Alma Mater Studiorum – Università di Bologna

School of Science
Department of Physics and Astronomy
Master Degree Programme in Astrophysics and Cosmology

Can intermediate-mass black hole binaries coalesce by the present day? Insights from hybrid modelling and *N*-body simulations

Graduation Thesis

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Academic year 2025-2026
Graduation date III

Abstract

The next generation of gravitational-wave (GW) observatories, Laser Interferometer Space Antenna (LISA) and the Einstein Telescope (ET), will jointly probe compact objects with masses ranging from a few to $\sim 10^7\,M_\odot$ over the frequency interval extending from 10^{-4} to 10^4 Hz. This unprecedented observational window makes it essential to identify promising GW sources, determine which of them may emit a detectable signal and characterize its physical properties. Among the most compelling candidates are intermediate-mass black hole (IMBH) binaries, whose inspiral GW emission would offer unique insights into this still poorly constrained population of massive black holes.

This thesis investigates the dynamical evolution of IMBH binaries embedded in stellar backgrounds, with particular emphasis on the coalescence timescale and the time evolution of the binary separation and eccentricity. To this end, both the Quinlan–Sesana hybrid model and Ketju direct N-body simulations are employed. While the former approach provides an efficient framework for reconstructing the evolution of black-hole binaries once the stellar background properties and the binary initial conditions are specified, the KETJU module, implemented into GADGET-4, treating black-holes and stars as point particles during close encounters through an algorithmically regularized integrator, accurately captures the orbital decay driven by stellar scattering with unprecedented spatial resolution.

The analysis explores different stellar backgrounds, including present-day nucleated dwarf galaxies, high redshift non-nucleated galaxies with enhanced stellar density and observationally motivated high-redshift stellar clump models. The obtained results confirm that IMBH binaries embedded in nucleated dwarf galaxies can merge within a few Myr, making them potentially detectable GW-loud sources even when pairing in the present-day universe. More remarkably, they show that even non-nucleated high-redshift dwarf galaxies and stellar clumps may drive binary coalescence within the Hubble time, identifying these systems as promising sources of detectable gravitational waves, provided the IMBH binary formed at sufficiently high redshift. Overall, this thesis project identifies a broader range of stellar environments capable of hosting IMBH binaries coalescing within the present-day universe, including high-redshift, non-nucleated systems as additional promising sources for future GW observatories. Further investigations incorporating a wider range of galaxy properties, binary parameters, and dynamical processes will be necessary to assess the contribution of these systems to the expected GW source population detectabale by LISA or ET.

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Introduction

Massive black holes (MBHs) play a fundamental role in the formation and evolution of galaxies. Observations have established that they are ubiquitously found at the centers of nearby galaxies (Kormendy and Richstone 1995), while tight empirical correlations between their masses and those of hosting galactic bulges suggest a close co-evolution of black holes and galaxies (Magorrian et al. 1998). Besides, MBHs are belived to be common in galaxy centres at all cosmic epochs; a piece of evidence for their existence, from low to high redshift, comes from the transient activity of galactic nuclei (AGN) over the hosts lifetime (see, e.g., Jahnke 2025). Black holes (BHs) span a wide range of masses ($\sim 10 - 10^9\,M_\odot$) and various formation and accretion models have been proposed based on their mass. At the two extremes of the BH mass spectrum lie stellar-mass black holes and supermassive black holes (SMBHs). Stellar evolution theory predicts that stellar-mass BHs may be the remnants of massive stars with either Zero Age Main Sequence mass $M_{ZAMS} \lesssim 140\,M_\odot$ or $M_{ZAMS} \gtrsim 260\,M_\odot$; otherwise, they undergo fully disruption and end their life as Pair Instability Super Novae, leaving no remnant behind (Farmer et al. 2019). In recent years, the properties of this population have been significantly constrained through the detection of gravitational waves from merging compact binaries (Abbott et al. 2016), as well as by advances in microlensing techniques capable of identifying isolated BH candidates (Bennett et al. 2002). On the opposite side ($\sim 10^6 - 10^9\,M_\odot$), instead, Pulsar timing arrays has already probed the upper mass range of SMBHs population (Taylor 2021), whose most accepted formation model interpret them as the final stage of a secular accretion onto initially lighter black hole seeds (see, e.g., Peterson 1997).

While substantial progress has been made in detecting and characterizing both stellar-mass BHs and SMBHs, the observational confirmation and detailed investigation of intermediate-mass black holes (IMBHs), lying in the mass range $\sim 10^2$–$10^5, M_\odot$, remain significantly more challenging. Although an increasing number of candidates have been identified in the local Universe, particularly in dwarf galaxies, through accretion signatures (see, e.g., Reines et al. 2013), their sphere of influence—for istance, defined as the sphere enclosing a stellar mass equal to twice that of the central BH (Sesana and F. M. Khan 2015), within which the BH gravitational potential dominates—is typically too small for reliable stellar or gas dynamical measurements, especially at higher redshift. Nevertheless, the discovery of high-redshift quasars (see, e.g., Bañados et al. 2018) points to the existence of BH seeds with masses comparable to those of IMBHs already in the early Universe, strengthening the idea that even the nuclei of less massive galaxies harbor MBHs. Therefore, it is common practice to infer the central BH mass in low-mass galaxies by extrapolating the empirical correlation between BH mass and the stellar mass of the host galaxy bulge, even for quiescent galactic nuclei (see, e.g., Schutte et al. 2019; Maiolino et al. 2024).

Figure 1 illustrates three distinct theoretical pathways for IMBH formation, highlighting the number of BHs expected to form through each channel, their corresponding mass ranges, and the redshift range over which each formation mechanism is expected to be most efficient. Each formation channel proceeds as follows (Greene et al. 2020):

Pop III. Sufficiently massive primordial stars, commonly referred to as Population III (Pop III) stars, may possess enough gravitational binding energy to withstand the energy released during the photodisintegration-induced collapse of their cores, ultimately leading to the formation of a IMBH remnant (Fryer et al. 2001).

Directe collapse. At high redshift, metal-free gas clouds may undergo nearly monolithic gravitational collapse without significant fragmentation, in contrast to what is typically observed in the present-day Universe, owing to the inefficient cooling of neutral hydrogen, and lead to the formation of IMBHs (see, e.g., Loeb and Rasio 1994; Lodato and Natarajan 2006).

Gravitational runaway. Runaway stellar collisions in high-density clusters may trigger a sequence of stellar mergers, progressively producing more massive stars that can undergo further collisions. The final product of this repeated mergers process may eventually collapse into an IMBH (see, e.g., Rodriguez et al. 2019; Devecchi and M. Volonteri 2009).

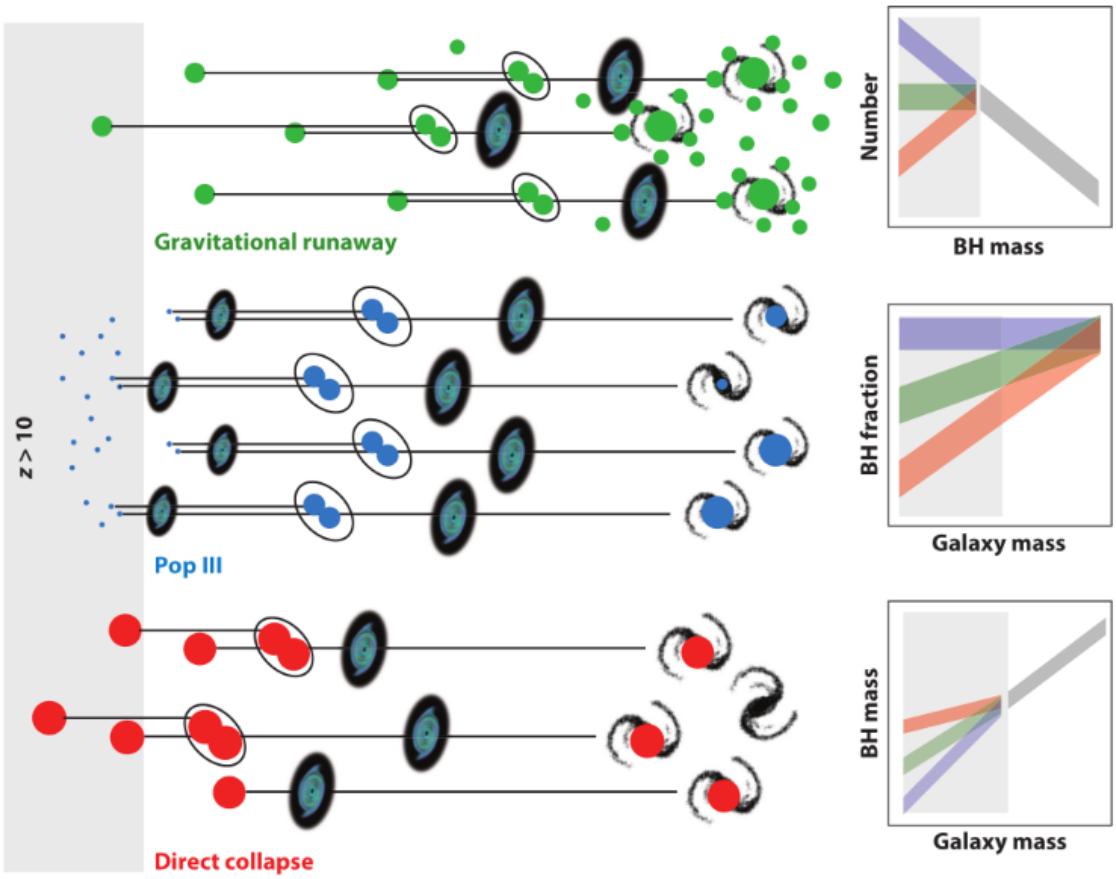


Figure 1: Schematic representation of different IMBH seeding channels, from their formation and subsequent growth through mergers and accretion to the present-day universe: Pop III stars (blue), direct collapse (red), and gravitational runaway (green). The corresponding present-day BH mass functions, occupation fractions, and BH–galaxy scaling relations are shown on the right. Figure taken from Greene et al. (2020).

Within the framework of the hierarchical growth scenario of galaxies (White and Rees 1978), the formation of BH binaries, and thus IMBH binaries, is expected to occur naturally as a consequence of mergers between their host stellar systems (Maiolino et al. 2024). These systems are among the most powerful sources of gravitational waves (GWs) in the Universe (Maggiore 2007), making them prime targets for current and future GW observatories. In particular,

the upcoming Laser Interferometer Space Antenna (LISA), a joint ESA/NASA space mission, will probe the low-frequency GW regime, from $\sim 10^{-4}$ to $\sim 10^{-1}$ Hz (Amaro-Seoane, Audley, et al. 2017; Amaro-Seoane, Andrews, et al. 2023), providing access to compact-object binaries with masses of $\sim 10^4 - 10^7, M_\odot$ and enabling the detection of their mergers out to redshift $z \sim 20$ (Colpi et al. 2024). At higher frequencies, the ground-based Einstein Telescope (ET) will complement LISA by extending the accessible frequency and mass ranges towards lower-mass systems, being expected to detect IMBH binaries with masses of $\sim 10^2 - 10^3, M_\odot$ out to redshifts of a few (Abac et al. 2026). Therefore, current and future GW observatories may provide a unique opportunity to unveil the population of IMBHs through their gravitational-wave emission, which may be detectable even when their electromagnetic signatures are too faint to be observed.

In light of this perspective, this thesis investigates the time evolution of the main orbital parameters of IMBH binaries, with focus on the semimajor axis a and eccentricity e, and examines how IMBH masses, stellar background properties, and the binary initial condition influence their dynamical evolution and coalescence timescale through a series of numerical experiments. To this end, the evolution of IMBH binaries is studied using two complementary approaches: the numerical integration of the differential equations for the binary separation and eccentricity within the hybrid model developed by Quinlan (1996) and extended by Sesana (2010), and high-resolution, gas-free N-body simulations of mergers between IMBHs and their stellar hosts performed with the tree-regularized code KETJU (Mannerkoski, Rawlings, et al. 2023).

The suite of numerical experiments presented in F. M. Khan and Holley-Bockelmann (2021) and F. M. Khan, Javed, et al. (2024) on both nucleated and non-nucleated dwarf galaxies have highlighted the crucial role of the stellar background density, particularly in the vicinity of the binary black-hole (BHB) sphere of influence, in driving fast and efficient mergers. In particular, high-density stellar environments ensure an adequate supply of low-angular-momentum stars, thereby sustaining efficient hardening via stellar slinghot (see, e.g., Mikkola and Valtonen 1992), while in low-density environments, the insufficient replenishment of such orbits may lead to binary stalling, giving rise to the so-called "last-parsec problem" (Milosavljević 2003). The evidence that different stellar environments produce peculiar pathways to coalescence highlights the need to investigate the dynamical evolution of IMBH binaries across a representative range of realistic stellar hosts.

The first class considered here comprises dwarf galaxies, whose most commonly adopted definition relies on an upper limit of $M_V > -18$ in absolute (V)-band magnitude Grebel et al. (2003). The observation of these systems even at high redshifts (see, e.g., Nguyen et al. 2018; Kikuchihara et al. 2020) makes them plausible stellar hosts at the time of IMBH pairing, whose GW emission may be probed by present-day observatories. Although N-body simulations have shown that IMBH binaries evolving in present-day, non-nucleated dwarf galaxies have coalescence times exceeding the Hubble time, their high-z progenitors are expected to be more compact, potentially enabling more rapid coalescence. This effect, combined with the observed increase in the BH-to-bulge mass ratio towards higher redshifts, as shown in Figure 2, may partially compensate for the absence of a nuclear star cluster. We therefore investigate this possibility by constructing models of non-nucleated, high-redshift dwarf galaxies from present-day analogues, following the cosmic size-evolution relation derived by Ono et al. (2023).

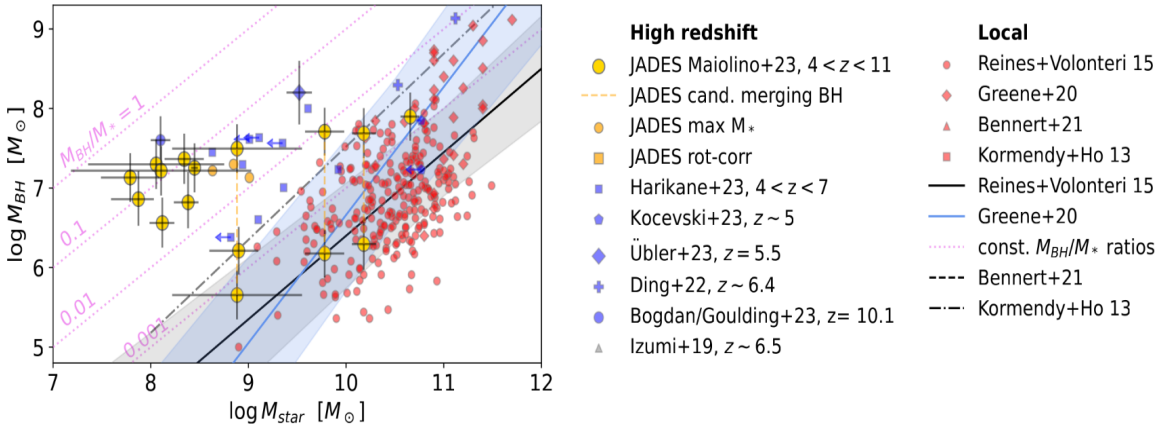


Figure 2: Black hole mass as a function of host galaxy stellar mass. Golden circles denote the results of JADES JWST survay, blue symbols other high-z JWST measurements, grey triangles high-z QSOs, and red symbols local galaxies. Dashed orange lines connect dual AGNs candidates. Figure taken from Maiolino et al. (2024).

In recent years, several claims have been made regarding the possible presence of IMBHs in Galactic and extragalactic globular clusters (GCs). Although these detections are not always conclusive, dynamical measurements and signatures of accretion in the X-ray band have provided indications of central IMBHs in a number of old stellar clusters (e.g., Gebhardt et al. 2005; H. Baumgardt 2017). At the same time, observations of galaxies at very high redshift are revealing the existence of massive stellar clumps. This suggests that star formation in the early universe may have been clump-dominated, with such stellar clumps potentially representing the progenitors of the more compact and less massive GCs observed in the local Universe, before undergoing substantial mass loss due to tidal interactions with their host galaxies. An example of this is constituted by the "Misty Moons" (Nakane et al. 2026) and the "Bullet arc" (Bradač et al. 2025), two strongly lensed galaxies at a photometric redshift of $z_{phot} \sim 11 - 12$ and spectroscopic redshift $z_{spec} \sim 11$ respectively, which are resolved into several massive stellar clumps. Figure 3 shows the size–mass relation of local star clusters, star-forming clumps, and dwarf galaxies, providing a direct comparison between the properties of local GCs and high-z stellar clumps.

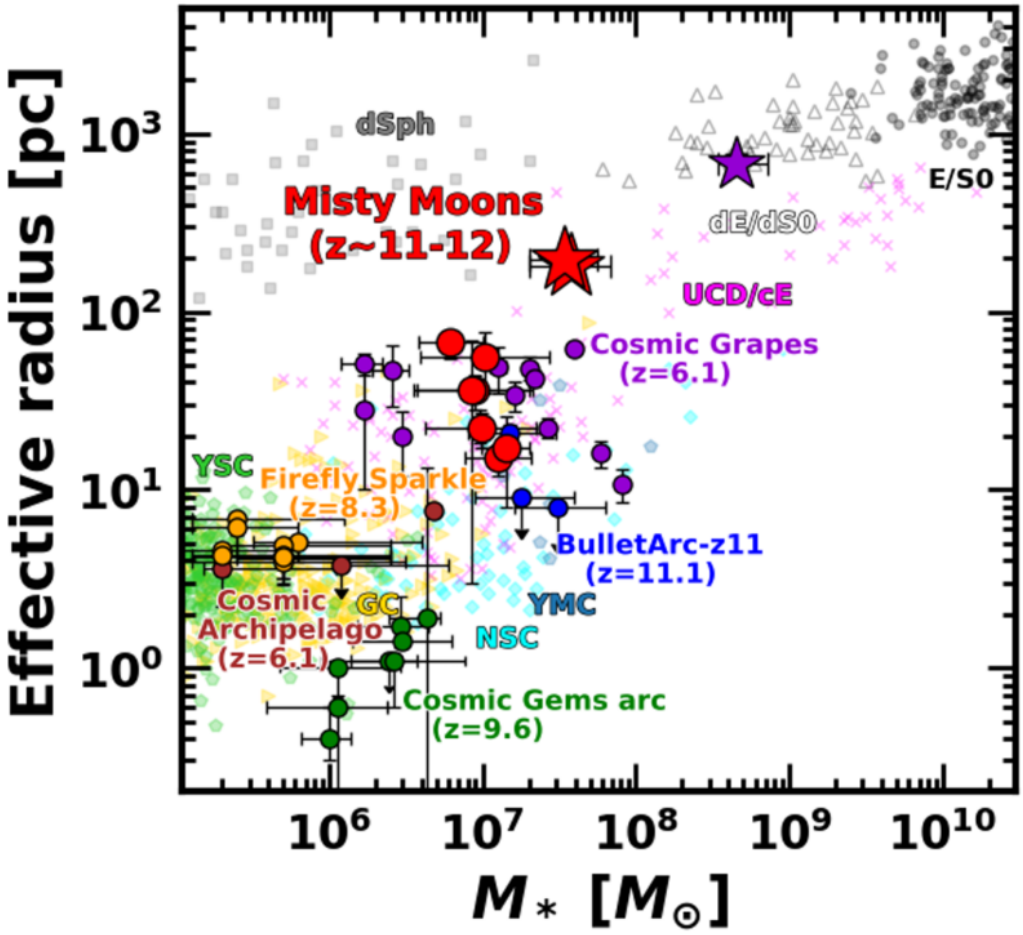


Figure 3: Size–mass relation for several stellar systems observed at different redshifts. Star symbols represent the complete systems, while circles denote individual clumps. Other symbols indicate local galaxies and star clusters, including Globular Clusters, Young Massive Star Clusters, and Nuclear Star Clusters, shown for comparison. Figure taken from Nakane et al. (2026).

Mindful of these considerations, we also investigate the dynamical evolution of IMBH binaries formed through the merger of stellar backgrounds whose masses and effective radii mimic those of the Misty Moons stellar clumps, hosting IMBHs with masses consistent with the stellar mass according to the empirical scaling relation shown in Figure 2.
The structure of this thesis is outlined as follows.
In chapter 1, we provide a brief overview of the main physical mechanisms promoting the shrinking of BH binaries, including violent relaxation, dynamical friction, stellar scattering by low-angular-momentum stars, and GW emission.
In chapter 2, we introduce the self-consistent hybrid model developed by Quinlan (1996) and Sesana (2010). We then present the evolutionary tracks obtained for the adopted sample of host galaxies and explore the relevant parameter space, highlighting how different environmental properties and initial conditions affect the binary evolution.
In chapter 3 we report the main features of the codes used to generate equilibrium N-body initial conditions and carry out the numerical simulations. The chapter then presents the results obtained by combining the evolution computed with KETJU, which follows the BH pair from early merger to the stellar-scattering-dominated regime, with the hybrid model used to track its subsequent evolution towards final coalescence.
Finally, the main results of this work are summarized, and possible future improvements are reported in the concluding chapter.

1

Physical Mechanisms of Black-Hole Binaries Hardening

In the "bottom-up" scenario, in which low-mass structures form first and progressively more massive systems assemble through hierarchical clustering and mergers, massive black hole binaries are expected to arise from mergers involving stellar systems or galaxies hosting such compact objects. This chapter reviews the main physical mechanisms governing the hardening of MBH binaries, from the initial merger of their stellar hosts to their final coalescence. We first discuss violent relaxation, which efficiently converts the orbital energy of the merging systems into disordered kinetic energy within the newly formed remnant. We then consider dynamical friction, which acts on massive components such as MBHs and causes them to lose orbital energy and sink toward the centre of the merger remnant, ultimately leading to the formation of a gravitationally bound binary. The subsequent evolution is driven by three-body interactions with low-angular-momentum stars. Therefore we discuss the gravitational slingshot mechanism, through which stellar ejections extract energy and angular momentum from the binary and promote its hardening, as well as the role of loss-cone refilling in sustaining this process. Finally, we derive the key equations governing the orbital evolution of the binary once gravitational-wave emission becomes the dominant mechanism driving its decay.

1.1 Galaxy Merging and Violent Relaxation

When two galaxies approach each other with sufficiently low relative velocity and small impact parameter, the encounter may result in a galaxy merger (see, e.g., A. Toomre and J. Toomre 1972; Cimatti et al. 2019). Galaxy mergers can be broadly classified according to the physical properties of the systems involved. Of particular relevance here are dissipationless, or dry, mergers, which involve gas-poor galaxies and are characterized by negligible radiative energy losses, so that the total energy of the system is approximately conserved (Khochfar and Silk 2009). Numerical N-body simulations have shown that collisions between galaxies of comparable mass, commonly referred to as "major mergers", generally result in a single, virialized remnant (see, e.g. Barnes and Hernquist 1992; Hilz et al. 2012). The formation of this remnant proceeds through a series of relaxation processes driven by the rapidly varying gravitational potential during the encounter, which redistributes the energy and angular momentum of the system and drives the initially out-of-equilibrium configuration toward a new equilibrium state (see, e.g. Lynden-Bell 1967; William C. Saslaw 1970).
In the scientific literature, several approaches have been proposed to define and derive the timescale T_r of violent relaxation. Here, we follow the approach of Lynden-Bell (1967) and define it as the characteristic oscillation period of the newly formed galaxy around the equilibrium configuration established by the virial condition

$$2K + W = 0, \tag{1.1}$$

holding in its barycentric inertial reference system S_{CM}, with K and W standing for the total kinetic and gravitational potential energy of the galaxy remnant, respectively. Therefore, we start by characterizing the equilibrium state attained by the system following the relaxation process.
Let us consider two galaxies, each with mass M_i and total energy $E_i = K_i + W_i$. In an ideal dissipationless merger, the total mass and energy are conserved, so that the resulting remnant has mass $M = M_1 + M_2$ and energy $E = E_1 + E_2 + E_{\rm orb}$, where E_{orb} accounts for the orbital energy of the pair, including both its ordered kinetic and mutual gravitational contributions. Following violent relaxation on a timescale T_r, the remnant settles into an equilibrium configuration that can be characterized by the virial velocity dispersion σ_v and gravitational radius r_g, defined by

$$\sigma_v^2 = \frac{2K}{M}, \qquad r_g = -\frac{GM^2}{W}, \tag{1.2}$$

which provide proxies for the characteristic stellar velocity dispersion and size of the galaxy in equilibrium, thus representing key information on the final stage reached by the merging galaxies.
By combining the virial condition (1.1)—which applies both to the progenitor galaxies prior to the interaction and to the remnant after the violent relaxation timescale T_r—with energy conservation, one obtains

$$\sigma_v^2 = \frac{M_1\sigma_{v,1}^2 + M_2\sigma_{v,2}^2 - 2E_{\rm orb}}{M_1 + M_2} \tag{1.3}$$

and

$$\frac{(M_1 + M_2)^2}{r_g} = \frac{M_1^2}{r_{g,1}} + \frac{M_2^2}{r_{g,2}} - \frac{2E_{\rm orb}}{G}, \tag{1.4}$$

where $\sigma_{v,i}$ and $r_{g,i}$ denote the virial velocity dispersion and the gravitational radius, respectively, of the colliding systems (Cimatti et al. 2019). From Equations (1.3) and (1.4), it follows that parabolic encounters between identical progenitors, defined by $E_{orb} = 0$, which are the

configuration considered in this thesis to carry out N-body numerical experiments as shown in chapter 3, give rise to a galaxy remnant having

$$M = 2M_1 = 2M_2, \qquad r_g = 2r_{g,1} = 2r_{g,2}, \qquad \sigma_v = \sigma_{v,1} = \sigma_{v,2}. \tag{1.5}$$

Therefore, before the merger to occur, each progenitor galaxy, which is assumed to be made of N stars of masses m_i and velocities $\mathbf{v_i}$, is individually virialized and satisfies the virial equilibrium condition

$$2K_i + W_i = 2\sum_{i=1}^{N} \frac{1}{2} m_i v_i^2 - \frac{1}{2}\sum_{i=1}^{N}\sum_{j\neq i} \frac{G m_i m_j}{|\mathbf{x_i} - \mathbf{x_j}|} = 0. \tag{1.6}$$

Neverthless, this condition is temporarily violated once the collision begins, as the newly formed remnant is driven far from equilibrium by the coherent motion inherited from the relative orbital motion of the progenitors. Consequently, during violent relaxation, Equation (1.1) is no longer satisfied at every instant; rather, the system obeys the time-dependent virial theorem,

$$\frac{1}{2}\ddot{I} = 2K + W, \tag{1.7}$$

where I denotes the polar moment of inertia (Ciotti 2021).
To quantitatively characterize the level of virialization of any system evolving towards equilibrium, it is usefull to introduce the so-called virial ratio

$$Q \equiv \frac{2K}{|W|}. \tag{1.8}$$

Figure 1.1 shows the results of N-body simulations of dissipationless galaxy mergers (see Carlo Nipoti et al. 2007), focusing on the time evolution of the virial ratio Q. The oscillations of Q have been reconstructed both within the framework of Modified Newtonian Dynamics (MOND) and under Newtonian gravity. In the former case, the standard Poisson equation is replaced by the non-relativistic field equation (Bekenstein and Milgrom 1984)

$$\nabla \cdot \left[\mu\left(\frac{\|\nabla\phi\|}{a_0} \right) \nabla\phi \right] = 4\pi G\rho, \tag{1.9}$$

where ϕ denotes the MOND gravitational potential, ρ is the mass density, $a_0 \simeq 1.2\times10^{-10}\,\mathrm{m/s^2}$ is the characteristic acceleration scale of MOND, and $\mu(y) = y/\sqrt{1+y^2}$ (Milgrom 1983).
Although the merging timescales are significantly longer in MOND than in Newtonian gravity with dark matter, in the Newtonian case the amplitude of the virial ratio oscillations rapidly decreases with time as Q approaches its equilibrium value, $Q = 1$, owing to the progressive conversion of ordered kinetic energy into random stellar motions. As illustrated by the green curve, the system approaches virial equilibrium through a series of progressively damped oscillations, with most of the virialization process occurring in the vicinity of the equilibrium value. We will take advantage of this observed behaviour for estimating the timescale T_r over which violent relaxation takes place.

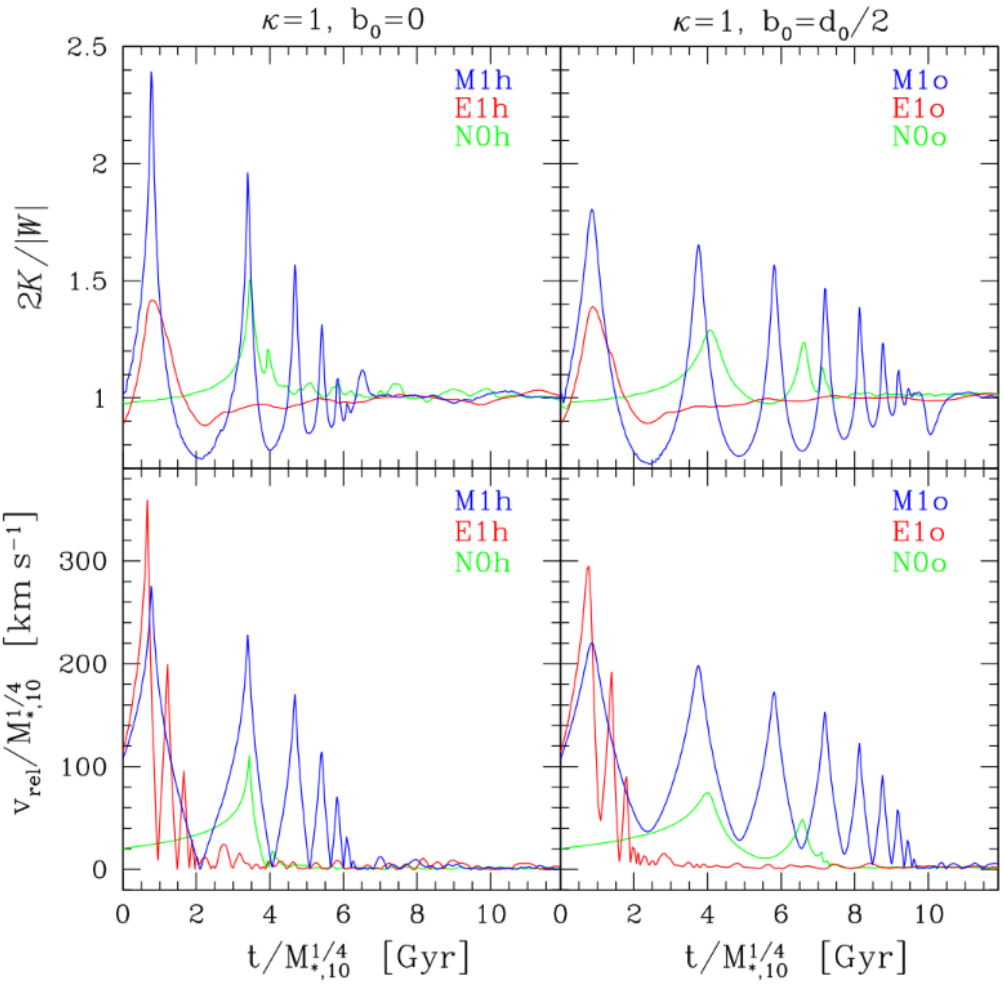


Figure 1.1: Time evolution of the virial ratio (top panels) and of the barycentric relative speed (bottom panels) for the two sets of galaxy merger simulations: head-on (left panels) and off-centre (right panels). Blue, red, and green curves refer to MOND, equivalent Newtonian, and purely baryonic Newtonian simulations, respectively. Figure taken from Carlo Nipoti et al. (2007).

To this end, it is useful to start from the physical mechanism underlying violent relaxation and consider the evolution of the energy ϵ_* of an individual star moving in a time-dependent gravitational potential $\psi(\mathbf{x}, t)$. We first show that the stellar energy can change only as a consequence of an explicit time dependence of the gravitational potential. Thus, if the potential is chosen to vanish at infinity, the energy ϵ_* of a star with mass m_* and velocity $\mathbf{v}$ is given by

$$\epsilon_* = \frac{1}{2} m_* v^2 + m_* \psi(\mathbf{x}, t). \tag{1.10}$$

Differentiating both sides of Equation (1.10) with respect to time yields

$$\frac{d\epsilon_*}{dt} = m_* \left[\mathbf{v} \cdot \dot{\mathbf{v}} + \nabla\psi \cdot \mathbf{v} + \frac{\partial \psi}{\partial t} \right]. \tag{1.11}$$

For a star moving under the gravitational potential ψ, its acceleration satisfies $\dot{\mathbf{v}} = -\nabla\psi$ and the previous equation reduces to

$$\frac{d\epsilon}{dt} = \frac{\partial \psi}{\partial t}, \tag{1.12}$$

where $\epsilon = \epsilon_*/m_*$ is the stellar energy per unit mass. Equation (1.12) shows that individual stars continuously gain and lose orbital energy as a consequence of the rapidly varying gravitational

field. This naturally motivates defining the violent relaxation timescale T_r as the timescale over which the energy change of a typical star becomes comparable to its initial energy (Lynden-Bell 1967), i.e.

$$T_r = \left\langle \frac{(d\epsilon/dt)^2}{\epsilon^2} \right\rangle^{-1/2} = \left\langle \frac{(\partial\psi/\partial t)^2}{\epsilon^2} \right\rangle^{-1/2}. \tag{1.13}$$

As discussed above, when the remnant galaxy is approaching the equilibrium condition, $\ddot{I}$ si small but non-zero and a mean star experiences changes in both kinetic and gravitational energies in a coupled manner, approximately satisfying

$$\frac{1}{2} m_* v^2 \sim -\frac{1}{4} m_* \psi, \tag{1.14}$$

as dictated by Equation (1.6).
By substituting Equation (1.14) into Equation (1.10), we can express the specific stellar energy solely in terms of the gravitational potential and Equation (1.13) turns into

$$T_r \equiv \frac{3}{4} \left\langle \frac{\psi^2}{\dot{\psi}^2} \right\rangle^{1/2}. \tag{1.15}$$

The dimensional analysis suggests that the polar moment of inertia can be expressed in terms of the total mass M and the square of the instantaneous characteristic radius $R(t)$ as

$$I(t) = \lambda^2 M R^2(t), \tag{1.16}$$

where λ^2 is a dimensionless structural factor of order unity that encodes the instantaneous mass distribution and is determined by the geometrical configuration of the masses m_i.
Let us consider small radial perturbations of the system about its virial equilibrium configuration. In this regime, the instantaneous characteristic radius of the system can be expressed as a small deviation from its equilibrium value, which we identify with the gravitational radius. Therefore, to the first order,

$$R(t) \approx r_g + \delta R, \tag{1.17}$$

with $|\delta R| \ll r_g$ representing the small radial displacement from equilibrium. In this regime, linearizing Equation (1.1) with respect to the perturbation gives

$$2\lambda^2 r_g \ddot{\delta R} - \frac{GM}{r_g^2} \delta R, \tag{1.18}$$

which we recognize to be the equation of a simple harmonic oscillator, with angular frequency

$$\omega = \left(\frac{GM}{2\lambda^2 r_g^3} \right)^{1/2}. \tag{1.19}$$

It is worth of noticing that, introducing the galaxy mean density $\bar{\rho} = 3M/(4\pi r_g^3)$, the latter quantity can be expressed as

$$\omega = (2\pi G \bar{\rho})^{1/2}. \tag{1.20}$$

Since the gravitational potential scales as $\psi \sim 1/r$, its time variation is governed by the same characteristic timescale as the radial oscillations, i.e. $\sim \omega^{-1}$. Plugging this scaling into the definition of the violent relaxation timescale in Equation (1.15) returns

$$T_r \simeq \frac{3}{4\omega} = \frac{3}{\sqrt{32\pi G \bar{\rho}}}. \tag{1.21}$$

Thus, the violent relaxation timescale is directly related to the characteristic dynamical timescale of the newly formed system and is expected to be of the order of its free-fall timescale $t_{ff} \propto 1/\sqrt{G\bar{\rho}}$.
To obtain an order-of-magnitude estimate of this timescale for the merger of two identical non-nucleated dwarf galaxies, we may consider the non-nucleated galaxy model built by F. M. Khan, Javed, et al. (2024) from a sample of non-nucleated dwarf galaxies of the Fornax cluster. For a galaxy of stellar mass $M_* = 10^8 \times M_\odot$, effective radius $R_e = 0.56\,\mathrm{kpc}$ and Sersic index $n = 0.81$ (see Section 2.3), the violent relaxation timescale is expected to be of the order of $T_r = 21.2\,Myr$. This estimate has been obtained from Equation (1.21) by considering the average stellar density of the mentioned reference galaxy within its effective radius. This result demonstrates that this mechanism can redistribute the orbital energy of the progenitor galaxies on a much shorter timescale than that associated with two-body relaxation t_{2b}. The latter will be introduced in § 1.2 as a key process responsible for replenishing the so-called binary "loss cone".

1.2 Two-Body Relaxation and Dynamical Friction

Galaxies are multi-component systems in dynamical equilibrium, where different stellar populations, gaseous phases, dust, and dark matter coexist (Cimatti et al. 2019). These environments host an enormous number of stars, which makes it impossible to track the orbit of each element. Neverthless, by abandoning the idea of characterizing the time evolution of each degree of freedom in the phase-space (PhS), a probabilistic description of the stellar population is still possible in terms of the one-particle distribution function (DF), namely $f(\mathbf{x}, \mathbf{v}, t)$. This function can be interpreted as the probability density of finding a star at a given point $\mathbf{w} = (\mathbf{x}, \mathbf{v})$ in PhS at time t. It is conventionally normalized so that

$$\rho(\mathbf{x}, t) = \int_{\mathbb{R}^3} d^3\mathbf{v}\, f(\mathbf{x}, \mathbf{v}, t), \tag{1.22}$$

where $\rho(\mathbf{x}, t)$ is the stellar density providing a smooth approximation to the actual, discrete distribution of N stars.
In the collisionless approximation, DF is conserved along the orbits of the gravitational potential $\Phi(\mathbf{x}, t)$ associated with the stellar density $\rho(\mathbf{x}, t)$. In this regime, it thus obeys the so-called Collisionless Boltzmann Equation (CBE),

$$\frac{df}{dt} \equiv \frac{\partial f}{\partial t} + \frac{d\mathbf{x}}{dt} \cdot \frac{\partial f}{\partial \mathbf{x}} + \frac{d\mathbf{v}}{dt} \cdot \frac{\partial f}{\partial \mathbf{v}} = 0, \tag{1.23}$$

that will be discussed more in detail in § 3.1.
Equation (1.23) is expected to hold as long as the effects of repeated gravitational encounters between stars remain negligible over the age of the system. Although individual encounters are typically weak, their cumulative effect can progressively redistribute energy and angular momentum among stars, causing their trajectories to deviate from those drawn by the smooth potential Φ and DF to diffuse in PhS (Binney and Tremaine 2008).
The characteristic timescale over which these cumulative encounters significantly alter stellar trajectories is the two-body relaxation time t_{2b}, which provides a natural criterion for assessing the validity of the collisionless approximation. Systems whose age is shorter than their two-body relaxation time, i.e $t_{age} < t_{2b}$, are conventionally referred to as collisionless; otherwise, they will be referred to as collisional (Ciotti 2021).
The approach of Chandrasekhar (1943) for estimating t_{2b}, that we will be following, relies on the impulsive approximation. It consists of two main assumptions:

1. Stellar encounters are treated as independent events, so that correlations between consecutive encounters and simultaneous multiple interactions are neglected.

2. The gravitational influence of the remaining $N-2$ stars is neglected during each encounter, reducing the problem to a sequence of isolated hyperbolic two-body interactions.

Let us work under the simplifying assumption of a spatially homogeneous distribution of field stars with isotropic velocity distribution. The DF $f(\mathbf{x_f}, \mathbf{v_f})$ of such system can be expressed as the product of the position-independent number density n_f and probability density function $g(v_f)$ in the velocity space,

$$f(\mathbf{x_f}, \mathbf{v_f}) = n_f g(v_f). \tag{1.24}$$

Two-body encounters between a test particle of mass m_t and a field star of mass m_f induce small changes in an orbital property P of the test particle. Although each interaction produces only a small perturbation, the cumulative effect of many encounters causes P to diffuse. This process is quantified by the diffusion coefficient $D(P)$, which measures the rate at which P changes per unit time. Owing to the impulsive approximation, in this framework it can be written as

$$D(P) \equiv 2\pi n_f \int_{\mathbb{R}^3} d^3\mathbf{v_f} \|\mathbf{v_t} - \mathbf{v_f}\| g(v_f) \int_0^\infty db \langle P \rangle b, \tag{1.25}$$

where $\langle P \rangle$ denotes the angle-averaged value of the orbital property P, and b is the impact parameter of the hyperbolic encounter (Ciotti 2021).
Since significant deflections of the original orbits are expected beyond t_{2b}, corresponding to the onset of the collisional regime, a useful indicator for determining t_{2b} may be the orbital property $P = \Delta\mathbf{v}_{t\perp}$, that is the component of the test-particle velocity change perpendicular to the initial relative velocity with respect to the field particle. However, under the assumption of isotropy, the first-order diffusion coefficients of the velocity components vanish (Binney and Tremaine 2008). Consequently, the leading non-vanishing contribution is represented by the second-order diffusion coefficient, hence, making the quantity $P = \|\Delta\mathbf{v}_{t\perp}\|^2$ the relevant orbital parameter for our analysis. By taking advantage of the small-angle scattering approximation (see, e.g., Chandrasekhar 1943),

$$\|\Delta\mathbf{v}_{t\perp}\| \simeq \frac{2Gm_f}{vb}, \tag{1.26}$$

with v denoting the modulus of the relative velocity between m_t and m_f at infinity, and Equation (1.25) becomes

$$D(\|\Delta\mathbf{v}_{t\perp}\|^2) \simeq 8\pi G^2 m_f^2 n_f \int_{\mathbb{R}^3} d^3\mathbf{v_f} \frac{g(v_f)}{v} \int_0^\infty \frac{db}{b}. \tag{1.27}$$

The integral in Equation (1.27) exhibits two logarithmic divergences. The former, known as ultraviolet (UV) divergence, arises in the limit $b \to 0$ and results from extending the small-angle scattering approximation to encounters with sufficiently small impact parameters, where it breaks down. The latter, the infrared (IR) divergence, arises in the limit $b \to \infty$ and is a consequence of modelling the stellar system as infinite and homogeneous. The UV divergence is avoided by replacing the small-angle approximated solution in Equation (1.26) with the exact two-body scattering solution; mathematically, this operation is equivalento to impose a lower cutoff to the impact parameter equals $b_{\pi/2}(v)$, corresponding to the impact parameter associated with a deflection angle of $\pi/2$. The IR divergence cannot be eliminated and thus we have to set a fiducial upper cutoff $b_{\max}$.
The resulting diffusion coefficient can then be written as

$$D(\|\Delta\mathbf{v}_{t\perp}\|^2) \sim 8\pi G^2 n_f m_f^2 \int_{\mathbb{R}^3} \frac{g(v_f) \ln\Lambda}{v} d^3\mathbf{v}_f, \tag{1.28}$$

where $\ln(\Lambda) = \ln(b_{max}/b_{\pi/2})$, is the so-called Coulomb logarithm. Although it depends on test-field particle relative velocity through $b_{\pi/2}(v)$, by defining the velocity-weighted Coulomb logarithm $\ln(\bar{\Lambda})$ so that

$$\int_{\mathbb{R}^3} \frac{g(v_f)\ln\Lambda}{\|\mathbf{v}_t - \mathbf{v}_f\|}\, d^3\mathbf{v}_f \equiv \Psi(v_t)\ln\bar{\Lambda}, \tag{1.29}$$

with

$$\Psi(v_t) = \int_{\mathcal{R}^3} \frac{g(v_f)}{\|\mathbf{v_t} - \mathbf{v_f}\|} d^3\mathbf{v_f}, \tag{1.30}$$

being the Rosenbluth potential, Equation (1.28) turns into

$$D(\|\Delta\mathbf{v}_{t\perp}\|^2) \sim 8\pi G^2 n_f m_f^2 \Psi(v_t)\ln\bar{\Lambda}. \tag{1.31}$$

One of the major simplifications introduced by the assumption of isotropy is that $\Psi(v_t)$ can be expressed as

$$\Psi(v_t) = \int_{\mathbb{R}^3} \frac{g(v_f)}{\|\mathbf{v}_t - \mathbf{v}_f\|}\, d^3\mathbf{v}_f \equiv \frac{\Xi(v_t)}{v_t} + 4\pi\int_{v_t}^{\infty} g(v_f) v_f\, dv_f, \tag{1.32}$$

where

$$\Xi(v_t) = 4\pi\int_0^{v_t} g(v_f) v_f^2\, dv_f \tag{1.33}$$

is normalized to unity as a consequence of the normalization of velocity-space distribution function $g(v_f)$.
Finally, if one defines the two-body relaxation time t_{2b} as the ratio between the orbital parameter $P = \|\Delta\mathbf{v}_{t\perp}\|^2$ and the associated diffusion coefficient, it turns out that

$$t_{2b} \equiv \frac{v_t^2(-\infty)}{D(\Delta E_{t\perp})} \sim \frac{v_t^3}{8\pi G^2 n_f m_f^2 ln\bar{\Lambda}}, \tag{1.34}$$

holding in the high velocity limit from Equation (1.32).
Mindful of T_r estimate obtained in § 1.1 for a representative non-nucleated dwarf galaxy, we now compare it with the corresponding two-body relaxation time, t_{2b}, of the same system. An order-of-magnitude estimate of t_{2b} is given by

$$t_{2b} \approx t_{cross}\frac{N}{12\log N}, \tag{1.35}$$

where N denotes the number of stars and $t_{cross} = 2\sigma_v/R_e$ (Ciotti 2021). Assuming equal-mass stars with $m_* = 1\,M_\odot$, Equation (1.35) yields a two-body relaxation time of the order of tens of Gyr, thus significantly exceeding the associated value of T_r. Therefore, merging galaxies settle into a new equilibrium configuration—corresponding to the newly formed galaxy—on a timescale T_r. Once this equilibrium has been reached and the gravitational potential becomes approximately time-independent, however, changes in the original stellar orbits occur instead on the much longer two-body relaxation timescale t_{2b}. This timescale governs the replenishment of low-angular-momentum orbits, as it will be discussed in § 1.4. We now continue by deriving the dynamical friction time scale by following the same approach used above.
While two-body encounters redistribute the kinetic energy of a test particle, increasing its velocity component perpendicular to the initial direction of motion, they also produce a systematic loss of kinetic energy along the direction of motion. In this sense, stellar encounters lead to heating in the transverse direction and cooling along the direction of motion (see, e.g., Chandrasekhar 1943; Binney and Tremaine 2008). This net loss of kinetic energy, commonly referred to as dynamical friction, causes massive objects such as an intruding secondary BH of mass M_2 to lose orbital energy and progressively sink toward the centre of the host stellar system. N-

body simulations performed by Baumgardt et al. (2006) have shown that dynamical friction efficiently drives the secondary BH down to a characteristic separation a_0, defined as the radius enclosing a stellar mass equal to twice the mass of M_2, that is

$$M_*(< a_0) = 2M_2. \tag{1.36}$$

To quantify the effect of this physical mechanism, we now derive an estimate of the dynamical friction timescale, t_{df}. This can be obtained by taking the ratio between the modulus of the initial velocity $\mathbf{v_t}$ of a test particle of mass m_t, moving through a stellar field composed of particles of mass m_f, and the diffusion coefficient associated with the orbital property $P = \Delta\mathbf{v_t}$. Using the Newton's law and the definition diffusion coefficient yield

$$\frac{d\mathbf{v_t}}{dt} = D[\Delta\mathbf{v_t}] = -4\pi G^2 m_t m_f \ln\Lambda \int d^3\mathbf{v_f} f(\mathbf{v_f}) \frac{\mathbf{v_t} - \mathbf{v_f}}{|\mathbf{v_t} - \mathbf{v_f}|^3}. \tag{1.37}$$

Assuming an isotropic velocity field, Equation (1.37) simplifies to

$$\frac{d\mathbf{v_t}}{dt} = -16\pi^2 G^2 m_t m_f \ln\Lambda \left[\int_0^{v_t} dv_f\, v_f^2 f(v_f)\right] \frac{\mathbf{v_t}}{v_t^3}, \tag{1.38}$$

known as Chandrasekhar's dynamical friction formula (Chandrasekhar 1943).
Equation (1.38) shows that the effect of dynamical friction is to systematically decreases the test particle's velocity through an acceleration directed antiparallel to its velocity vector.
By following an analogous derivation to that used to estimate the two-body relaxation timescale t_{2b} in Equation (1.34), the dynamical friction timescale t_{df} can be expressed as

$$t_{df} \equiv \frac{\|\mathbf{v_t}(-\infty)\|}{\|D(\Delta\mathbf{v_{t\|}})\|} \sim \frac{v_t^3}{4\pi G^2 n_f m_f (m_f + m_t) \ln(\bar{\Lambda}) \Xi(v_t)}, \tag{1.39}$$

We conclude this Section by deriving, under physically motivated assumptions, an analytic expression for the dynamical friction timescale t_{df} of an intruding secondary BH of mass m_2 and velocity $\mathbf{v_2}$, which sinks towards the center of a stellar background hosting a primary BH of mass m_1.
Let us assume the stellar distribution to possess a Maxwellian velocity distribution and a power-law density profile

$$\rho(r) = \rho_0 \left(\frac{r}{r_0}\right)^{-\gamma}. \tag{1.40}$$

We begin by noting that for a Maxwellian velocity distribution, the integral over the stellar velocity distribution appearing in Equation (1.38) can be evaluated analytically, leading to

$$\frac{d\mathbf{v_2}}{dt} = -4\pi \ln\Lambda \left[erf(X) - \frac{2X}{\sqrt{\pi}} e^{-X^2}\right] \frac{G^2 m_2 \rho(r)}{v_2^3} \mathbf{v_2}, \tag{1.41}$$

where

$$erf(X) = \frac{2}{\sqrt{\pi}} \int_0^X dt\, e^{-t^2} \tag{1.42}$$

is the so-called error function. Here, $X \equiv v_c/(\sqrt{2}\sigma)$ is a dimensionless parameter that combines the circular velocity v_c of the test particle with the velocity dispersion σ of the stellar background.
For an intruder BH moving along quasi-circular orbits within the influence radius r_{inf} of the primary BH, hereafter defined by

$$M_*(< r_{inf}) = 2(m_1 + m_2), \tag{1.43}$$

the angular momentum and the dynamical friction force acting on it are

$$\tilde{L} = m_2\sqrt{Gm_1 r} \tag{1.44}$$

and

$$\tilde{F} = m_2\frac{dv_2}{dt} = -C(\Lambda, X)\frac{G^2 m_2^2 \rho(r)}{v_c^2} \tag{1.45}$$

respectively. Here, $C(\Lambda, X)$ is the dimensionless factor implicitly defined through the Equations (1.41) and (1.45).
At this point, we can exploit the fact that, within the primary BH sphere of influence, the circular velocity can be approximated as

$$v_c^2 = \frac{Gm_1}{r}. \tag{1.46}$$

Under these assumptions, the second cardinal equation of dynamics takes the form

$$\frac{d\tilde{L}}{dt} = \tilde{F}r, \tag{1.47}$$

and leads to the differential equation for $r = r(t)$ upon substituting the expressions given in Equations (1.45), (1.44) and (1.46) (see, e.g., Baumgardt et al. 2006; Gualandris et al. 2022):

$$r^{-5/2}\frac{dr}{dt} = -2C(\Lambda, X(r))\frac{\sqrt{G}m_t}{m_f^{3/2}}\rho(r). \tag{1.48}$$

Equation (1.48) admits an analytic solution for an isotropic power-law density profile. Indeed, in this case

$$X = \sqrt{\gamma - 1} \tag{1.49}$$

and is therefore independent of radius (Madigan et al. 2011).
Integrating the differential equation subject to the initial condition $r(t = 0) = r_0 < r_{inf}$, we obtain

$$r(t) = \begin{cases} r_0\left[1 - 2C(\Lambda, X)\frac{\sqrt{G}m_2}{m_1^{3/2}}\left(\gamma - \frac{3}{2}\right)\rho_0 r_0^{3/2} t\right]^{\frac{1}{\gamma - 3/2}} & \text{if } 1 < \gamma \leq 3 \wedge \gamma \neq \frac{3}{2} \\ r_0 \exp\left[-2C(\Lambda, X)\frac{\sqrt{G}m_2}{m_1^{3/2}}\rho_0 r_0^{3/2} t\right] & \text{if } \gamma = \frac{3}{2} \end{cases} \tag{1.50}$$

Equation (1.50) applies to an isotropic, power-law stellar background, or more generally to a stellar distribution that approaches such a density profile within the sphere of influence of the central BH.
Lastly, setting the reference radius r_0 and density ρ_0 to their respective values at the influence radius r_{inf}, and defining $t_{\rm fric}$ such that $r(t_{\rm fric}) = 0$ (Baumgardt et al. 2006), we obtain

$$t_{\rm fric} = \begin{cases} \dfrac{m_1^{3/2}}{2C(\Lambda, X)\sqrt{G}\, m_2\, \rho_{\rm inf} r_{\rm inf}^{\gamma}}\dfrac{r_{\rm inf}^{\gamma - 3/2}}{\gamma - \frac{3}{2}}, & \text{if } \gamma \neq \frac{3}{2}, \\ +\infty, & \text{if } \gamma = \frac{3}{2}. \end{cases} \tag{1.51}$$

Although Equations (1.51) provide an estimate of the dynamical friction timescale from r_{inf} to the centre, as mentioned above, this mechanism becomes inefficient once the binary separation decreases below $a_0 < r_{inf}$. From this point onward, the further hardening of the binary is primarily driven by individual encounters with low-angular-momentum stars that are unbound to the binary and can extract energy and angular momentum from it. This shrinking mechanism is addressed in § 1.3.

1.3 Black Hole Binary–Star Three-Body Scattering

As discussed in § 1.1 and § 1.2, during a dry, major merger, the remnant evolves towards equilibrium on the violent relaxation timescale T_r, while the MBHs initially located at the centers of the progenitor galaxies sink towards the center of the remnant through dynamical friction on a timescale t_{fric}. As their separation decreases, the MBHs eventually cross each other sphere of influence and form a gravitationally bound pair at a separation of order a_0, defined in Equation (1.36). At this stage, the binary separation is sufficiently small that distant stars mainly perturb the binary center of mass, with negligible effect on its orbital energy (Quinlan 1996). Further hardening is therefore driven predominantly by low angular momentum stars passing in the vicinity of the binary and extracting both energy and angular momentum from it. Each time such a star—belonging to the so-called binary "loss cone", as we will discuss in more detail in § 1.4—passes sufficiently close to the binary, a three-body scattering takes place. Under suitable conditions, this interaction produces hyper-velocity stars (see, e.g., Quinlan 1996; Sesana, Haardt, et al. 2006) through the so-called stellar slingshot mechanism (Mikkola and Valtonen 1992). In this section, we review the main physical aspects of three-body scattering and the stellar slingshot mechanism, which are key processes driving the hardening of massive black-hole binaries (MBHBs). To this end, we begin by considering the so-called "two-body problem" and reviewing some of its key properties, which will be useful in understanding the physics of the stellar slingshot.

By adopting the same notation as in the previous Section, let us consider a test particle and a field particle, with masses m_t and m_f, respectively, interacting solely through gravity. Introducing the total mass of the system, $M = m_t + m_f$, and the reduced mass, $\mu = m_t m_f / M$, the total energy in the mass centre frame S_{CM} is given by

$$E = \frac{1}{2}\mu \|\mathbf{v}\|^2 - \frac{G m_f m_t}{r}, \tag{1.52}$$

where $r(t) \equiv \|\mathbf{x_t} - \mathbf{x_f}\|$ and $\mathbf{v}(t) \equiv \mathbf{v_t} - \mathbf{v_f}$ denote particles relative separation and velocity.

For two unbound particles moving along hyperbolic orbits in S_{CM}, energy conservation, as expressed by Equation (1.52), evaluated in the asymptotic limit $r(-\infty) = r(+\infty) \to +\infty$, yields

$$\|\mathbf{v}(-\infty)\| = \|\mathbf{v}(+\infty)\|. \tag{1.53}$$

Although it is not conserved along the entire orbit, Equation (1.53) shows that the magnitude of the relative velocity—i.e., the velocity of the test particle as measured by an observer comoving with the field particle—is the same at the two asymptotic limits. Therefore, the encounter results only in a rotation of the velocity vector $\mathbf{v}$ and the corresponding velocity change $\Delta\mathbf{v} = \mathbf{v}(+\infty) - \mathbf{v}(-\infty)$ can be conveniently decomposed as the sum of the components perpendicular and parallel to the initial relative velocity $\mathbf{v}(-\infty)$ as follows:

$$\mathbf{v}(+\infty) = \mathbf{v}(-\infty) + \Delta\mathbf{v}_{\parallel} + \Delta\mathbf{v}_{\perp}. \tag{1.54}$$

Substituting the expression for $\mathbf{v}(+\infty)$ provided by Equation (1.54) into Equation (1.53) yields

$$\|\Delta\mathbf{v}_{\parallel}\|^2 + \|\Delta\mathbf{v}_{\perp}\|^2 + \langle \Delta\mathbf{v}_{\parallel}, \mathbf{v}(-\infty)\rangle = 0. \tag{1.55}$$

A first inspection of Equation (1.55) reveals that a necessary condition for it to hold is that the scalar product be negative. This observation will be crucial for assessing whether a MBHB–star interaction results in binary hardening or softening (Ciotti 2021).

The identity derived above impose a constraint on the relative velocity of the interacting particles in a specific inertial frame, which is S_{CM}. However, the interaction can be described in any inertial frame S_0. Indeed, while Equation (1.53) shows that both the total energy and

the individual mechanical energies of the particles are conserved between the two asymptotic limits, the close encounter generally induces changes in the individual energies E_t and E_f in S_0. Therefore, between the two configurations at infinite relative separation, the test particle undergoes a net variation in mechanical energy,

$$\Delta E_t = \frac{1}{2} m_t [|\mathbf{v_t}(+\infty)|^2 - |\mathbf{v_t}(-\infty)|^2] = -\Delta E_f, \tag{1.56}$$

which is exactly compensated by the corresponding variation in the mechanical energy of the field particle, such that

$$\Delta E = \Delta E_t + \Delta E_f = 0. \tag{1.57}$$

By combinig the velocity composition law

$$\mathbf{v_t}(t) = \mathbf{v_{CM}} + \frac{\mu}{m_t} \mathbf{v}(t) \tag{1.58}$$

with the conservation of the center-of-mass velocity vector $\mathbf{v_{CM}}$, which holds for isolated systems, Equation (1.56) can be written as (Ciotti 2021)

$$\Delta E_t = \mu \langle \Delta \mathbf{v}, \mathbf{v_{CM}} \rangle. \tag{1.59}$$

We are now in a position to address three-body scattering involving a gravitationally bound BH binary and an incoming star. The dynamics of this encounter can be understood by considering the limiting case of a strongly asymmetric binary in which $M_1 \gg M_2 \gg m_*$ so that both the star and the secondary BH, having masses m_* and M_2, can travel along nearly Keplerian orbits around M_1 with slightly different radii and, consequently, possess slightly different Keplerian velocities. In this scenario, the secondary BH acts as the field particle, while the star plays the role of the test particle. The primary BH serves only to establish the relative velocity between the two bodies. Consequently, the center of mass motion of this gravitationally unbound pair is essentially determined by the orbit of M_2 around M_1. Therefore, the approximation

$$\mathbf{x_{CM}} \simeq \mathbf{x_f}, \qquad \mathbf{v_{CM}} \simeq \mathbf{v_f}, \tag{1.60}$$

applies and the center of mass will move locally at costant velocity if the observation time is short compared to the orbital period, thus making negligible its acceleration.
Hereafter, by analogy with the previous notation, we identify $M_2 \equiv m_f$ and $m_* \equiv m_t$.

Binary hardening. When the star is outside the secondary BH orbit, the Keplerian velocity profile imposes $\|\mathbf{v_f}(-\infty)\| = \|\mathbf{v_{CM}}\| > \|\mathbf{v_t}(-\infty)\|$. Therefore, at point where m_f surpasses m_t and they are both located on the same radius, $\mathbf{v}(-\infty)$ is nearly antiparallel to $\mathbf{v_{CM}}$ and $\Delta \mathbf{v}_\parallel$ can be read as the projected vector along $\mathbf{v_{CM}}$ rather than $\mathbf{v}(-\infty)$. Taking advantage of Equation (1.55), it follows that

$$0 < \langle \Delta \mathbf{v}_\parallel, \mathbf{v_{CM}} \rangle = \langle \Delta \mathbf{v}, \mathbf{v_{CM}} \rangle = \Delta E_t / \mu. \tag{1.61}$$

Equation (1.61) shows that in the reference system centered on M_1 the star gains orbital energy and is scattered onto a wider orbit. On the other hand, the secondary BH experiences an equal and opposite change in orbital energy, leading to a reduction of the binary separation and, consequently, to binary hardening.
It is worth noticing that if $\Delta E_t > |E_t(-\infty)|$, which occurs at very high circular velocity of the binary BHs, stellar ejection occurs. In this regime "hypervelocity stars" (HVSs) travelling along nearly radial orbits are produced (see, e.g., Quinlan 1996; Sesana, Haardt, et al. 2006).

Binary softening. Contrary to the previous case, binary softening takes place when the star is inside the secondary BH orbit. Now $\mathbf{v}(-\infty)$ is nearly parallel to $\mathbf{v_{CM}}$ at the crossover point and by repeating the same arguments used above it follows

$$E_t/\mu = \langle \Delta\mathbf{v}, \mathbf{v_{CM}} \rangle = \langle \Delta\mathbf{v}_{\parallel}, \mathbf{v_{CM}} \rangle < 0. \tag{1.62}$$

On one hand the star loses energy to the binary, it is pushed inward and may undergo tidal disruption due to primary BH (Lightman and Shapiro 1977); on the other hand the binary gains energy and its semimajor axis increases, resulting in binary softening.

Although our discussion has so far focused on a specific three-body scattering configuration, the results can be readily generalized by observing that the distinction between binary hardening and softening is determined by the relative magnitude of the BH binary's orbital velocity and the velocity of the incoming low angular momentum star, hereafter "loss-cone" star.
The binary BH embedded in an isotropic stellar background with costant velocity dispersion σ is expected to undergo three-body scatterings with loss-cone stars that will approach the galactic centre where the binary lives with velocity of the order of σ. Because the specific binding energy of a Keplerian with reduced mass μ and semimajor axis a is

$$E_b = \frac{G\mu}{2a} \tag{1.63}$$

and the specific energy of the loss-cone star is of the order of σ^2, the velocity criterion can be conveniently reformulated in terms of the specific energy. When $E_b \ll \sigma^2$, the binary is said to be soft, and repeated stellar encounters tend to further decrease its binding energy. On the other hand, when $E_b \gg \sigma^2$, the binary is hard, and stellar encounters, on average, increase its binding energy. This behavior is well summarized by Heggie's law (see, e.g., D. C. Heggie 1975; Douglas C. Heggie and Hut 1993).

Heggie's law. – *On average, hard binaries get harder and soft binaries get softer.*

By extrapolation of the two limiting cases, one can define the "hardening radius" a_h as the binary separation at which binary passes from the soft to the hard regime. Throughout this thesis we adopt the definition proposed by Quinlan (1996), unless otherwise specified:

$$a_h = \frac{GM_2}{4\sigma^2}. \tag{1.64}$$

At this stage, it should be clear that stellar scattering plays a fundamental role in driving the binary toward the regime in which GWs emission dominates its evolution. Nevertheless, one may wonder how a binary can evolve from the soft to the hard regime. The key point is that this transition is driven primarily by dynamical friction against the stellar background, rather than by individual stellar encounters, as shown in § 1.2.
The discussion then proceeds in § 1.4, where we examine how these loss-cone orbits are populated and subsequently replenished.

1.4 Loss Cone Problem

In § 1.3 we have been concerned in describing the phisics of three-body scattering, pointing out its relevance to the orbital decay of the BH binary. In the this Section, instead, we will be explaining which stars can effectively exchange energy and angular momentum with the binary, the rate at which these close encounters take place and how all of this enters the self consistent model for MBHBs evolution developed by Quinlan (1996) and Sesana (2010) that

will be presented in Chapter 2.
Let $a(t)$ and M be the binary semimajor axis and total mass, respectively . Numerical experiments demonstrated that only stars intersecting the binary orbit can efficiently extract energy and angular momentum from the binary when ejected at higher velocity (Mikkola and Valtonen 1992). This class of stars of the stellar background populates the so-called "binary loss cone" and comprises both low-energy stars and high-energy, low angular momentum stars. Under the assumption of Keplerian orbits, the former class have small mean radius, i.e. $\langle r \rangle < a$, and specific energy $E_* < -GM/a$, while latter spend most of their orbital period $P(E_*)$ far from the binary but a close encounter can occur at the pericenter r_P whenever $r_P < a$.
The size of the binary loss cone, J_{lc}, in specific angular momentum space, hereafter J-space, is a function of E_* and can be determined from the maximum and minimum angular momentum values allowed at a given specific energy.
At the configuration of closest approach, the star radial velocity vanishes and the specific energy in terms of r_P is

$$E_* = \frac{J^2}{2r_P^2} - \frac{GM}{r_P}. \tag{1.65}$$

From Equation (1.65) it follows that there exists a value of J below which the star enters the binary loss cone. Hereafter, we refere to it as $J_{lc}(E_*)$ and is given by

$$J_{lc}(E_*) \equiv \left[2\left(E_* + \frac{GM}{a}\right)\right]^{1/2} a, \tag{1.66}$$

which can be approximated to $\sim \sqrt{2GMa}$ as the vast majority of stars orbit far from the binary and have a specific angular energy $|E_*| \ll GM/a$.
Besides, stars of fixed specific energy E_*, other than a minimum of the specific angular momentum $J_{min} = J_{lc}(E_*)$, possess a maximum set by the chain of inequalities

$$E_* \geq \frac{J^2}{2r^2} - \frac{GM}{r} \equiv V_{eff}(r) \geq V_{eff,min} = -\frac{G^2M^2}{2J^2}, \tag{1.67}$$

holding because of the non-negative behavior of the radial kinetic term. This constrain poses an upper limit to J, that is

$$J_{max}(E_*) = \frac{GM}{\sqrt{2|E_*|}}, \tag{1.68}$$

corresponding to the specific angular momentum of a circular orbit of specific energy E_* (Capozziello and De Laurentis 2008).
Once the binary has depleted, on a dynamical timescale $P(E_*)$, the population of stars that are bound to it and intersect its orbit—most of which, hence, reside within the binar sphere of influence—further hardening through stellar slingshot interactions depends on the rate at which the binary loss cone is replenished by stellar diffusion in both energy and angular momentum space, allowing new stars to enter the loss cone (Lightman and Shapiro 1977). This diffusion arises from stellar encounters, addressed in § 1.2, that have been proven to be efficient in exchanging angular momentum and results in a random walk (RW) in J-space (see, e.g., Milosavljević 2003; Merritt 2013). Determining the rate at which this process occurs is essential for understanding how it is incorporated into the self-consistent hybrid model developed by Quinlan (1996) and Sesana (2010), illustrated in Chapter 2.
To this end, let us define j_2 the rms angular momentum transferred in one orbital period and $J_{max}(E_*)$ the same quantity associated with the timescale t_{2b}. Only stars with $J_{lc} \leq J \leq J_{lc}+j_2$ can be scattered into the loss cone over one orbital period and the kind of loss cone replenishing mechanism, along with the associated timescale, are set by the dimensionless quantity

$$q(E_*) \equiv \frac{j_2^2}{J_{min}^2}. \tag{1.69}$$

Taking Figure 1.2 as a reference, $q(E_*)$ enables us to distinguish between two extreme scenarios:

Empty Loss Cone. This condition is represented by picture on the left in Figure 1.2(b). When $q(E_*) \ll 1$ the RW step size is very small compared to the loss cone size, making the process ascribable to a diffusion problem with timescale t_{2b}. Therefore, stars are removed from the loss cone faster than the rate at which it is fed, hence, remeining nearly empty.

Full Loss Cone. This condition is represented by the picture on the right in Figure 1.2(b). The loss cone is smaller than the RW step size and even high J stars can be scattered into the loss cone by many, weak stellar encounters taking place during one robital period. Thus, the rate at which loss cone stars are expelled is lower than the rate of refilling, causing the loss cone to remain almost full.

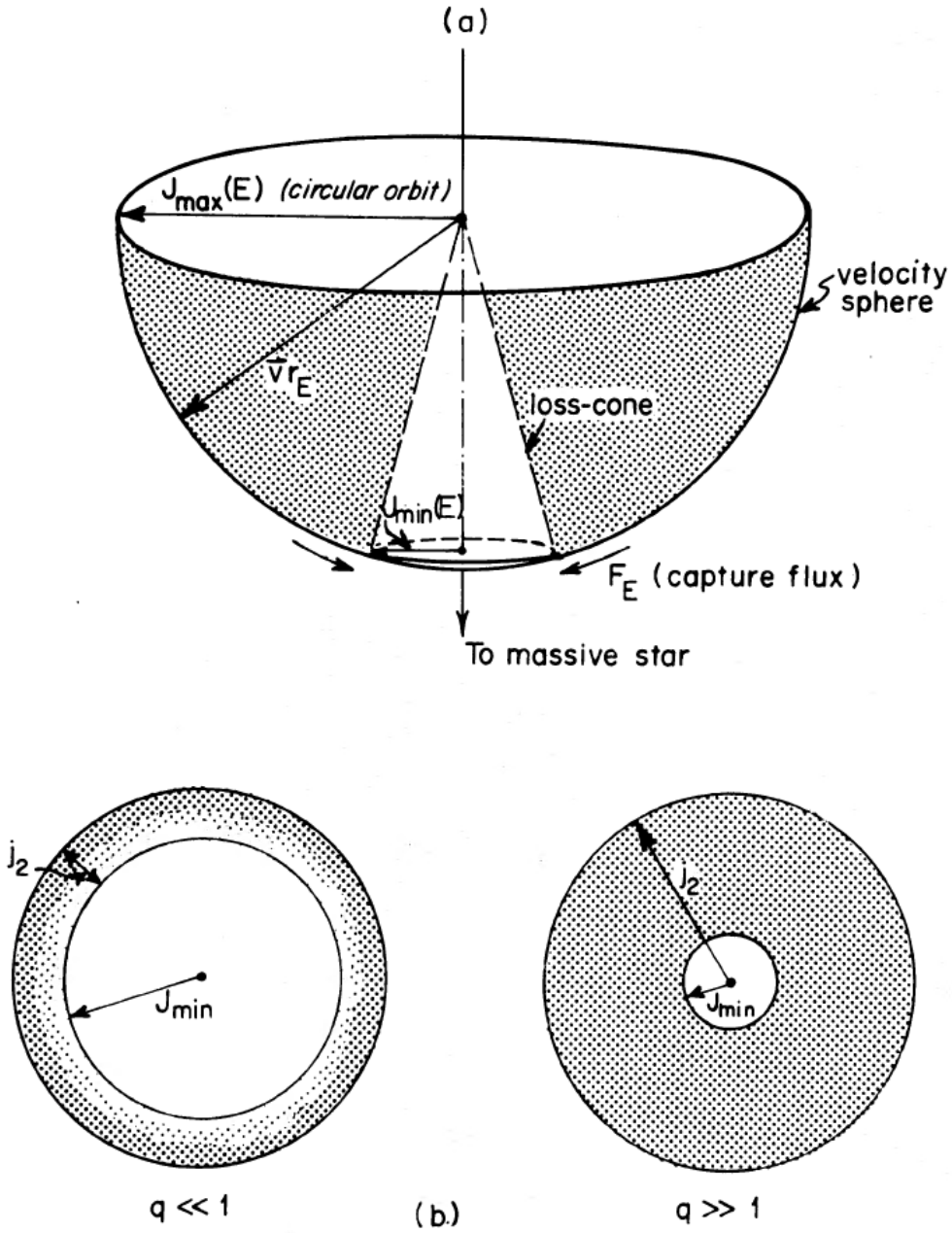


Figure 1.2: Panel (a): Velocity distribution of stars with fixed energy $E_* \equiv E$ and angular momentum J, with the latter varying accordingly to Equation (1.67). The vertical axis points towards the compact object of mass M.
Panel (b): The velocity sphere viewed from below. On the left the empty loss cone scenario. On the right the full loss cone regime. Figure taken from Lightman and Shapiro (1977).

By employng the same dimensional analysis carried out by Lightman and Shapiro (1977), we expect that the stellar capture rate per unit energy $\Gamma_E(E_*)$ could be written as

$$\Gamma_E(E_*) \equiv \frac{d\Gamma}{dE_*} \approx \epsilon |E_*|^{-1} F_{max}(E_*), \tag{1.70}$$

where $F_{max}(E_*) \sim N(E_*)/P(E_*)$ denotes the maximum rate of loss cone replenishing from stars with specific energy E_* and ϵ is a correction factor helping us to account for the fraction of stars within the loss cone.
Owing to the 2-D nature of the loss-cone problem in the J-space, ϵ is expected to be an area

correction factor, thus scaling as

$$\epsilon \sim \frac{J_{lc}^2}{(J_{lc}+j_2)^2}. \tag{1.71}$$

Therefore, in the full loss cone regime, where $J_{lc} \ll j_2 \approx J_{max}$, one easily finds that $\epsilon \sim J_{lc}^2/J_{max}^2 \sim a/r$ and, by substitution into Equation (1.70),

$$\Gamma_E(E_*) \approx |E_*|^{-1} \frac{a}{r} \frac{N(E_*)}{P(E_*)}. \tag{1.72}$$

Equation (1.72) allows us to finally discuss how loss-cone refilling is taken into account in the Quinlan–Sesana hybrid model.
In his scattering experiments, Quinlan (1996) quantified the binary dimensionless hardening rate H and eccentricity growth rate K through the flux of stars

$$\Gamma_Q = \left(\frac{\rho}{m_*}\right) \Sigma v \tag{1.73}$$

across the binary effective cross section Σ, where ρ is the stellar mass density, m_* is the stellar mass, and v is the stellar velocity.
Let b denote the impact parameter of a star initially unbound to the binary of mass M and undergoing a three-body interaction with it. The corresponding interaction cross section can then be defined as

$$\Sigma \equiv \pi b_{\max}^2, \tag{1.74}$$

where $b_{\max}$ is the maximum impact parameter for which stellar capture occurs. If we assume this to take place whenever $r_p < a$, mindful of the gravitational focusing formula,

$$b^2 = \frac{2GMr_P}{v^2}, \tag{1.75}$$

relating impact parameter and pericenter distance, then we can pose $b_{max} = b(a)$ and express Γ_Q in Equation (1.73) as

$$\Gamma_Q = 2\pi \left(\frac{M}{m_*}\right) \frac{G\rho a}{v}, \tag{1.76}$$

with ρ standing for the mass density of the homogeneous stellar background.
On the other hand, Sesana (2010) estimated the rate Γ in Equation (1.70) under the assumption of a full loss cone, adopting an isothermal stellar sphere of total mass M_*, influence radius $r_{\rm inf}$ and costant stellar velocity dispersion σ.
If all stars have the same mass m_*, the total number of stars inside r is simply proportional to the stellar mass enclosed within this sphere, that is

$$N(<r) = \frac{M}{m_*} \frac{r}{r_{inf}}, \tag{1.77}$$

while, under the assumption of Keplerian orbits bound to the binary, the energy of each star depends on the semimajor axis only through the relation

$$E_*(r) = -\frac{GMm_*}{2r}. \tag{1.78}$$

In this regime, by exploiting the relation $dE_* = |E_*| d\ln r$, we can rewrite Equation (1.72) as (see, e.g., Perets, Hopman, et al. 2007; Perets and Alexander 2008)

$$\frac{d\Gamma}{d\log r} = \frac{a}{r} \frac{N(<r)}{P(r)}. \tag{1.79}$$

Finally, approximating the orbital period as $P(r) \sim r/\sigma$ and substituting the estimates for $N(< r)$ and $P(r)$ derived above into Equation (1.79), the integration yields

$$\Gamma = 2\pi \frac{M}{m_*} \frac{G\rho_{\rm inf} a}{\sigma}. \tag{1.80}$$

A comparison between Equations (1.73) and (1.80) shows that, by identifying the velocity v of an unbound star at infinity with the velocity dispersion σ of the isothermal sphere, the prescriptions adopted for H and K effectively describe a system with a full loss cone inside r_{inf}. However, introducing the dimensionless factor LCN allows the hybrid model to account for different loss-cone regimes by simply replacing ρ_{inf} with $LCN \cdot \rho_{\rm inf}$. Since this factor cannot be constrained a priori, we adopt $LCN = 1$ throughout the analysis presented in chapter 2-3 unless otherwise specified.

1.5 Gravitational Radiation from Black Hole Binaries

This Section reviews the physics of gravitational-wave emission from a binary system within the framework of linearized general relativity, following the treatment presented by Maggiore (2007), and examines their role as the final mechanism driving a binary black hole system towards progressively smaller separations and, ultimately, coalescence.
The Section is organized into three Subsections. The first focuses on the linearized Einstein equations in the harmonic gauge. The second discusses the generation of GWs in the quadrupole approximation. Finally, the third derives the expected GW signal from a binary system and examines the associated radiation-reaction effects on its orbital evolution.

1.5.1 Linearized Gravitational Field Equation

The Einstein equations are the fundamental field equations of general relativity (GR). These equations read

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = \frac{8\pi G}{c^4} T_{\mu\nu}, \tag{1.81}$$

where $R_{\mu\nu}$, $g_{\mu\nu}$, and $T_{\mu\nu}$ denote the components of the Ricci, metric, and energy-momentum tensors, respectively, while R is the Ricci scalar, obtained by contracting the Ricci tensor.
Equations (1.81) constitute ten coupled, non-linear, second-order partial differential equations for the metric tensor. Neverthless, owing to the four Bianchi identities, they are not sufficient to uniquely set the ten indipendent components of the metric tensor$g_{\mu\nu}$, hence, resulting in six independent equations only. These four degrees of freedom are intimately connected to gauge symmetries, reflecting the arbitrariness of the four coordinates x^μ used to represent and solve the tensor Equations (1.81).
To examine how gravitational radiation arises in linearized GR within the quadrupole approximation, we first consider the weak-field limit, in which the metric tensor deviates only slightly from the Minkowski metric $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$, which describes the flat spacetime. We then derive the corresponding form of the Einstein equations in this limit. We therefore assume

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x), \tag{1.82}$$

with $|h_{\mu\nu}| \ll 1$ being a symmetric tensor representing a small perturbation of the flat spacetime. We note that, at linear order, the inverse metric tensor can be approximated as

$$g^{\mu\nu} \approx \eta^{\mu\nu} - h^{\mu\nu}. \tag{1.83}$$

Linearizing Equations (1.81) requires performing this operation on the various terms appearing on the left-hand side of it. This side encodes the geometry and curvature of spacetime and contains all terms that are functions of the metric and its derivatives. This becomes clear once we introduce the affine connections

$$\Gamma^{\lambda}_{\mu\nu} = \frac{1}{2} g^{\lambda\rho}(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}) \tag{1.84}$$

and the Riemann tensor

$$R_{\lambda\mu}{}^{\rho}{}_{\nu} = \partial_\lambda \Gamma^{\rho}_{\mu\nu} - \partial_\mu \Gamma^{\rho}_{\lambda\nu} + \Gamma^{\rho}_{\lambda\sigma}\Gamma^{\sigma}_{\mu\nu} - \Gamma^{\rho}_{\mu\sigma}\Gamma^{\sigma}_{\lambda\nu}. \tag{1.85}$$

Indeed, by definition

$$R_{\mu\nu} = R_{\sigma\mu}{}^{\sigma}{}_{\nu} \tag{1.86}$$

and

$$R = g^{\alpha\beta} R_{\alpha\beta} = R^{\sigma}_{\sigma}. \tag{1.87}$$

Bearing in mind that, owing to Equation (1.82), the metric derivatives reduce to derivatives of the perturbation $h_{\mu\nu}$ and it is thus straightforward to prove that, to the first order in the metric perturbation, these terms reduce to

$$R_{\mu\nu} = -\frac{1}{2}(\Box h_{\mu\nu} + \partial_\mu \partial_\nu h - \partial_\lambda \partial_\mu h^{\lambda}_{\nu} - \partial_\nu \partial_\lambda h^{\lambda}_{\mu}) \tag{1.88}$$

and

$$R = \partial_\mu \partial_\nu h^{\mu\nu} - \Box h, \tag{1.89}$$

respectively, whith h denoting the trace of $h_{\mu\nu}$.
The linearized Eistein equations are greatly simplified if they are expressed in terms of the so-called "trace reverse" of $h_{\mu\nu}$, defined as

$$\bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} h. \tag{1.90}$$

In this way, writing the original perturbation $h_{\mu\nu}$ as a function of the associated trace reverse $\bar{h}_{\mu\nu}$ and plugging the terms of Equations (1.88) and (1.89) into Equation (1.81), the field equations, finally, take the form

$$\Box \bar{h}_{\mu\nu} + \eta_{\mu\nu}\partial_\alpha \partial_\beta \bar{h}^{\alpha\beta} - \partial_\mu \partial_\lambda \bar{h}^{\lambda}_{\nu} - \partial_\nu \partial_\lambda \bar{h}^{\lambda}_{\mu} = -\frac{16\pi G}{c^4} T_{\mu\nu}. \tag{1.91}$$

Equations (1.91) are already linearized in the metric perturbation, however they can be further simplified if we exploit the gauge freedom holding for the Einstein equations. This reflects the possibility of an infintesimal coordinate transformation of the kind

$$x^{\mu} \to x'^{\mu} = x^{\mu} + \xi^{\mu}(x), \tag{1.92}$$

with $\xi^{\mu}(x)$ denotingfour arbitrary functions of the spacetime.
As tensors obey to precise law of transformations under change of coordinates, the coordinates transformation in Equation (1.92) induces a variation of the trace reverse $\bar{h}_{\mu\nu}$ components, specifically:

$$\bar{h}'_{\mu\nu} = \frac{\partial x^{\alpha}}{\partial x'^{\mu}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} \bar{h}_{\alpha\beta} \approx \bar{h}_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu + \eta_{\mu\nu}\partial_\sigma \xi^{\sigma}. \tag{1.93}$$

A direct inspection of Equations (1.91) and (1.93) shows that, among the possible transformations, choosing the four functions ξ^{μ} such that they satisfy the so-called De Donder gauge

condition, i.e. $\Box\xi^\mu = \partial_\sigma \bar{h}^{\sigma\mu}$, leads to $\partial_\sigma \bar{h}'^{\sigma\mu} = 0$. Dropping the prime, Equation (1.91) then simplifies to

$$\begin{cases} \Box \bar{h}_{\mu\nu} = -\dfrac{16\pi G}{c^4} T_{\mu\nu} \\ \partial_\mu \bar{h}^{\mu\nu} = 0. \end{cases} \tag{1.94}$$

We are now ready to address the problem of determining the general solution of the linearized Einstein's equations coupled with the De Donder gauge condition.

1.5.2 Generation and Propagation of Gravitational Waves

To study the formation of GWs we initially focus on the solution of the problem defined in Equation (1.94). Let us analyze first how GWs propagate in vacuo.
Outside the GWs source, $T_{\mu\nu}$ vanishes and, owing to the linearity of the d'Alembert operator, the solution can be expressed as superposition of plane waves without loss of generality, that is

$$\bar{h}_{\mu\nu}(x) = \int d^3k (A_{\mu\nu}(\mathbf{k}) e^{ik_\rho x^\rho} + c.c.), \tag{1.95}$$

with "c.c" being a short for complex conjugate of the first term.
It is worth noting that not all plane waves satisfy the linearized field equation in vacuo along with the gauge condition. However, linearity enables us to study the properties of a single plane wave separately; consequently, inserting the Ansatz $\bar{h}_{\mu\nu} \sim A_{\mu\nu} e^{ik_\rho x^\rho}$ into Equations (1.94) yields the following two constraints:

1. $(k_\rho k^\rho = 0)$: the four-vector $k^\mu = (\omega/c, \mathbf{k})$ is null, indicating that gravitational waves propagate at the speed of light c in vacuum.

2. $(k_\mu A^{\mu\nu} = 0)$: for GWs propagating along the x^3 direction, with $k^\mu = (k, 0, 0, k)$, this condition implies $A^{0\nu} = A^{3\nu}$.

At this stage, the amplitude tensor $A^{\mu\nu}$ possesses six indipendet components and can be written as

$$A^{\mu\nu} = \begin{pmatrix} A^{00} & A^{10} & A^{20} & A^{00} \\ A^{10} & A^{11} & A^{12} & A^{10} \\ A^{20} & A^{12} & A^{22} & A^{20} \\ A^{00} & A^{10} & A^{20} & A^{00} \end{pmatrix}. \tag{1.96}$$

Out of the six polarizations shown in Equation (1.96), four are unphysical because they vanish when the residual gauge freedom associated with four arbitrary harmonic functions ξ^μ is exploited. Indeed, one is free to adopt functions proportional to the plane wave solutions of the vacuo problem, e.g. $\xi^\mu = \epsilon^\mu e^{ik_\rho x^\rho}$, preserving the validity of the De Donder gauge, and to set their amplitudes ϵ^μ so that

$$A^{\mu\nu} = A^{\mu\nu}_{TT} \equiv \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_x & 0 \\ 0 & h_x & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{1.97}$$

The coordinate system where $A^{\mu\nu}$ is in the form of Equation (1.97), with two independent polarizations only, and defined by

$$\partial_\mu h^{\mu\nu} = 0, \quad h^{0\mu} = 0, \quad h^i_i = 0, \tag{1.98}$$

is referred to as transverse, traceless, or "TT", gauge[1].
We now focus on the solutions of Equations (1.94) in presence of the source term $T_{\mu\nu} \neq 0$. It can be read as the source term of the gravitational field equations and turns out to be

$$T^{\mu\nu} = (\rho + p/c^2)u^\mu u^\nu + pg^{\mu\nu} \tag{1.99}$$

for a perfect fluid of density ρ, pressure p and four-velocity v^μ.
Solutions for linear partial differential equations can be obtained by means of the Green's function method. By definition, the Green's function $G(x-x')$ is the solution of the associated equation in case of a pont-like source, i.e.

$$\Box_x G(x-x') = \delta^{(4)}(x-x'), \tag{1.100}$$

with $\delta^{(4)}(x-x')$ being the four-dimensional Dirac function.
It is relevant to note that, apart from the boundary conditions of the problem, $G(x-x')$ depends only on the linear operator. Therefore, from the radiation problem in electromagnetism, we already know that the proper solution is the retarded Green's function

$$G(x-x') = -\frac{1}{4\pi|\mathbf{x}-\mathbf{x}'|}\delta^{(4)}(ct_{ret} - ct'), \tag{1.101}$$

where

$$t_{ret} = t - \frac{|\mathbf{x}-\mathbf{x}'|}{c} \tag{1.102}$$

is the time in past at which the radiation was emitted at $\mathbf{x}'$ for it to be observed at $\mathbf{x}$ at time t. Once the Green's function is found, the general solution will be the convolution between it and source term:

$$\bar{h}_{\mu\nu}(t,\mathbf{x}) = \frac{4G}{c^4}\int d^3\mathbf{x}'\,\frac{T_{\mu\nu}(t_{ret},\mathbf{x}')}{|\mathbf{x}-\mathbf{x}'|}. \tag{1.103}$$

One of the greatest simplifications to this integral consists of the so-called "Compact-source approximation". It applies outside the source whenever its size is negligible with respect to the source-detector separation. In this regime, we can retain the leading term of the multipole expansion only, thus obtaining

$$h_{ij}^{TT}(t,\mathbf{x}) = \frac{4G}{c^4 r}\int d^3\mathbf{x}'\,T_{ij}(t_{ret},\mathbf{x}'). \tag{1.104}$$

Furthermore, far from the source, the conservation of the Energy-Momentum tensor, namely $\nabla_\mu T^{\mu\nu} \approx \partial_\mu T^{\mu\nu} = 0$, yields

$$h_{ij}^{TT}(t,\boldsymbol{x}) = \frac{2G}{rc^4}\ddot{Q}_{ij}^{TT}(t-r/c) \tag{1.105}$$

at lowest order in the low-velocity expansion. The term Q_{ij}^{TT} introduced in Equation (1.105)[2] is the representation of the quadrupole moment in the class of coordinate systems satisfying the TT gauge conditions.
Defining the quadrupole moment of $T^{00}/c^2 = \rho$ as

$$M^{ij} \equiv \frac{1}{c^2}\int d^3x T^{00}(t,\boldsymbol{x})x^i x^j, \tag{1.106}$$

[1] In the TT gauge $\bar{h}_{ij}^{TT} = h_{ij}^{TT}$.
[2] The components with at least one time index are excluded from our analysis because they are constant and do not contribute to GWs formation.

it is thus possible to write

$$Q^{ij} \equiv M^{ij} - \frac{1}{3}\delta^{ij} M_{kk} = \int d^3\mathbf{x}\, \rho(t,\mathbf{x}) \left(x^i x^j - \frac{1}{3} r^2 \delta^{ij} \right). \tag{1.107}$$

Equation (1.105) shows that, unlike electromagnetic radiation, at lowest order the gravitational radiation arises from changes in the quadrupole moment.
We conclude by reporting other important observables such as the energy and angular momentum carried by GWs, needed to estimate the back-reaction and related to the time derivatives of quadrupole moment tensor defined in Equation (1.107).
The conservation of energy and angular momentum requires that the rates at which GWs carry these quantities away from the source match the corresponding rates of orbital energy and angular momentum loss. The former are measured at a large distance r from the source and at time t, while the latter must be evaluated at the appropriate retarded time.
By neglecting the contributions from possible spins of the two black holes, it can be proved that the energy and angular momentum transported by GWs per unit time are

$$P_{quad} = \frac{dE_{gw}}{dt} = \frac{G}{5c^5} \langle \dddot{Q}_{ij} \dddot{Q}_{ij} \rangle = -\frac{dE_{source}}{dt}, \tag{1.108}$$

or equivalently

$$P_{quad} = \frac{G}{5c^5} \left\langle \dddot{M}_{ij} \dddot{M}_{ij} - \frac{1}{3} (\dddot{M}_{kk})^2 \right\rangle \tag{1.109}$$

in terms of M_{ij}, and

$$\frac{dL^i_{gw}}{dt} = \frac{2G}{5c^5} \epsilon^{ijk} \langle \ddot{Q}_{jl} \ddot{Q}_{kl} \rangle = -\frac{dL^i_{source}}{dt}. \tag{1.110}$$

respectively. Equations (1.108) and (1.110) will be of fundamental importance for estimating the back-reaction induced by GWs emission on the black hole binary.

1.5.3 Gravitational Waves from Black Hole Binaries and Back-Reaction

We now focus on the gravitational radiation emitted by two masses m_1 and m_2 moving along elliptic Keplerian orbit and the associated back-reaction.
Let $m = m_1 + m_2$ and $\mu = m_1 m_2/m$ be the total mass and the reduced mass, respectively. Whether the orbit is bound or unbound, the motion of the two bodies is confined to a plane by the conservation of the total angular momentum $\mathbf{L}$. It is therefore convenient to choose this plane as the x-y plane, perpendicular to $\mathbf{L}$, and introduce polar coordinates (r, ψ) centered on the center of mass. In this reference frame, denoted by S_{CM},

$$L_z = \mu r^2 \dot{\psi} \tag{1.111}$$

and, owing to the Köenig theorem, the total binary energy,

$$E = \frac{1}{2}\mu \dot{r}^2 + \frac{L^2}{2\mu r^2} - \frac{Gm\mu}{r}, \tag{1.112}$$

exhibits only the term of kinetic energy corresponding to the relative motion of the system, as in S_{CM} the center of mass kinetic energy vanishes.
The integration of Equation (1.112) requires the introduction of the auxiliary variable $w(\psi) = 1/r(\psi)$ (Capozziello and De Laurentis 2008). Plugging it into the energy conservation identity, Equation (1.112) becomes

$$w'^2 + w^2 - \frac{2Gm_1 m_2 \mu}{L^2} w = \frac{2\mu E}{L^2} \tag{1.113}$$

where $w' \equiv dw/d\psi$.
Differentiating both sides of the last identity with respect to ψ leads to a new differential equation, which is

$$w' \left(w'' + w - \frac{Gm_1 m_2 \mu}{L^2} \right) = 0. \tag{1.114}$$

Apart from the solution $w = const$, corresponding to a circular binary, the problem is formally identical to that of a harmonic oscillator. Therefore, upon solving for $r(\psi)$, it is found

$$r(\psi) = \frac{R}{1 + e\cos\psi}, \tag{1.115}$$

with $R = L^2/(Gm\mu^2)$ denoting the so-called the semi-latus rectum.[3]
Being interested in the GW emission process by a BH binary, which is by definition a gravitationally bound system, from this point onward it will be assumed $E < 0$. In this energy range, the trajectory is an ellipse having semiaxes

$$a = \frac{R}{1 - e^2} \qquad\qquad b = \frac{R}{\sqrt{1 - e^2}}, \tag{1.116}$$

with

$$e^2 = 1 - \frac{R}{a} = 1 + \frac{2EL^2}{G^2 m^2 \mu^3}. \tag{1.117}$$

denoting the eccentricity $0 \leq e < 1$ of the binary.
Inserting Equation (1.117) into the first identity of Equation (1.116) yields the semi-major axis a in terms of E, i.e.

$$a = -\frac{Gm\mu}{2E}, \tag{1.118}$$

implying that fixing the energy means setting the orbit size.
The functions $r(t)$ and $\psi(t)$ can be obtained by direct integration of Equations (1.111) and (1.112); in the parametric form they can be written as

$$r(t) = a[1 - e\cos u(t)] \qquad\qquad \psi(t) = \frac{\cos u(t) - e}{1 - e\cos u(t)}, \tag{1.119}$$

involving the so-called "eccentric anomaly" $u(t)$ satisfying

$$u(t) - e\sin u(t) = \left(\frac{Gm}{a^3} \right)^{1/2} t \equiv \omega_0 t. \tag{1.120}$$

Once the motion of the Keplerian binary has been characterised, we have all the necessary ingredients to compute the rates at which energy and angular momentum are carried away by GWs.
In the reference system S_{CM}, the mass density distribution $\rho(\mathbf{x}, t)$ of two point-like masses is

$$\rho(\mathbf{x}, t) = \mu\delta(x - r\cos\psi)\delta(y - r\sin\psi)\delta(z). \tag{1.121}$$

Therefore, according to the prescription provided by Equation (1.106), it is immediate to obtain

$$M_{ij} = \mu r^2(t) \begin{bmatrix} \cos^2(\psi) & \sin(\psi)\cos(\psi) \\ \sin(\psi)\cos(\psi) & \sin^2(\psi) \end{bmatrix}. \tag{1.122}$$

[3]The polar axis is aligned with the major axis $2a$, so that R corresponds to the distance between the origin and the point of the conic at $\psi = \pi/2$.

The time derivatives of M_{ij}, required to compute the energy and angular momentum transfer rates via GW emission, are more easily computed when r is expressed as a function of ψ. This is accomplished by replacing this variable with the general solution of the two-body problem provided by Equation (1.115), thus finding

$$\dddot{M}_{11} = \beta(1 + e\cos\psi)^2 \left[2\sin 2\psi + 3e\sin\psi\cos^2\psi\right], \tag{1.123}$$

$$\dddot{M}_{22} = \beta(1 + e\cos\psi)^2 \left[-2\sin 2\psi - e\sin\psi(1 + 3\cos^2\psi)\right], \tag{1.124}$$

and

$$\dddot{M}_{12} = \beta(1 + e\cos\psi)^2 \left[-2\cos 2\psi + e\cos\psi(1 - 3\cos^2\psi)\right], \tag{1.125}$$

where the factor β is defined by

$$\beta^2 \equiv \frac{4G^3\mu^2 m^3}{a^5(1-e^2)^5}. \tag{1.126}$$

Plugging Equations (1.123)-(1.124)-(1.125) into Equation (1.108) yields

$$P(\psi) = \frac{8G^4}{15c^5}\frac{\mu^2 m^3}{a^5(1-e^2)^5}(1 + e\cos(\psi))^4[12(1 + e\cos(\psi))^2 + e^2\sin^2(\psi)]. \tag{1.127}$$

However, we recall that this istantaneous quantity is not constant over one orbital period for elliptic Keplerian orbits as in general M_{ij} will suffer from different rate of variation from pericenter to apocenter. Therefore, taking the temporal average over one orbital period, $T \equiv 2\pi/\omega_0$, returns

$$P \equiv \frac{1}{T}\int_0^T dt\, P(\psi) = \frac{\omega_0}{2\pi}\int_0^{2\pi} d\psi\, \frac{P(\psi)}{\dot{\psi}} = \frac{32G^4\mu^2 m^3}{5c^5 a^5}F(e), \tag{1.128}$$

with

$$F(e) = \frac{1}{(1-e^2)^{7/2}}\left(1 + \frac{73}{24}e^2 + \frac{37}{96}e^4\right) \tag{1.129}$$

being a function strongly depending on the eccentricity, which reduces to unity in case of circular binary.

Throughout the inspiral phase, the binary radiates not only energy but angular momentum as well, carried away by GWs, and undergoes secular changes in both semimajor axis and ellipticity until coalescence eventually occurs. Therefore, another key ingredient in evaluating the back-reaction on the binary evolution is the determination of the angular momentum loss rate. It was quantified in Equation (1.110), which now becomes

$$\frac{dL^i_{source}}{dt} = -\frac{2G}{5c^5}\epsilon^{ikl}\left\langle \dddot{M}_{ka}\dddot{M}_{la}\right\rangle \tag{1.130}$$

as the the terms of the form $\epsilon^{ijk}\delta_{jl}\dddot{Q}_{kl}$ vanish, since they correspond to the contraction of an antisymmetric tensor with a symmetric one.

Taking into account that, in S_{CM}, all components of kind M_{i3} are identically zero and that $L = L_z = L^3_{source}$, one can rewrite Equation (1.130) as

$$\frac{dL}{dt} = \frac{4G}{5c^5}\langle \ddot{M}_{12}(\dddot{M}_{11} - \dddot{M}_{22})\rangle. \tag{1.131}$$

The last identity provides dL/dt in terms of the second and third derivatives of the quadrupole moment tensor M_{ij}. Although the second derivative $\ddot{M}_{12}$ has not been computed explicitly yet, it follows straightforwardly from the same procedure used to derive $\dddot{M}_{ij}$ that

$$\ddot{M}_{12} = \frac{G\mu m}{a(1-e^2)}\sin\psi\left[-4(1 + e\cos\psi)^2\cos\psi + 2e\left(3\cos^2\psi - 1 + 2e\cos^3\psi\right)\right]. \tag{1.132}$$

Yet, upon substituting the expressions given in Equations (1.123), (1.124), and (1.132), respectively, we obtain the angular momentum loss, corresponding to the negative of the angular momentum carried away by gravitational radiation in the quadrupole approximation.
Taking the temporal average over one orbital period T and expressing it as an integral over the angular position ψ then yields

$$\frac{dL}{dt} = -\frac{32}{5}\frac{G^{7/2}\mu^2 m^{5/2}}{c^5 a^{7/2}}\frac{1}{(1-e^2)^2}\left(1+\frac{7}{8}e^2\right) \tag{1.133}$$

Finally, the secular evolution of the binary orbit can be obtained by relating the losses of orbital energy and angular momentum to the corresponding changes in the semi-major axis and eccentricity. In particular, applying the corresponding conservation laws to the combined system of the BH binary and the emitted GWs, and differentiating Equations (1.117) and (1.118) with respect to time, we get

$$\frac{da}{dt} = \frac{Gm\mu}{2E^2(a)}(-P) \tag{1.134}$$

and

$$2e\frac{de}{dt} = \frac{2}{G^2 m^2 \mu^3}\left[-PL^2(a,e) + 2E(a)L(a,e)\frac{dL}{dt}\right]. \tag{1.135}$$

Combining the resulting expressions and eliminating the intermediate quantities, Peters and Mathews (1963) derived

$$\frac{da}{dt} = -\frac{64}{5}\frac{G^3\mu m^2}{c^5 a^3}\frac{1}{(1-e^2)^{7/2}}\left(1+\frac{73}{24}e^2+\frac{37}{96}e^4\right) \tag{1.136}$$

and

$$\frac{de}{dt} = -\frac{304}{15}\frac{G^3\mu m^2}{c^5 a^4}\frac{e}{(1-e^2)^{5/2}}\left(1+\frac{121}{304}e^2\right). \tag{1.137}$$

Equations (1.136) and (1.137) show that GW emission becomes increasingly effective as the binary separation decreases, as reflected by the steep a^{-3} and a^{-4} dependences of the semi-major axis and eccentricity evolution, respectively. Therefore, at sufficiently small separations, GW emission is expected to become the dominant mechanism governing the binary evolution, superseding the effects of stellar scattering discussed in § 1.3. These terms were incorporated into the self-consistent hybrid model developed independently by Quinlan (1996) and Sesana (2010), which combines stellar dynamical interactions and GW emission to describe the evolution of massive black hole binaries across the different stages of their inspiral. A detailed discussion of this model, together with the characteristic transition lengthscales that ideally separate the regimes in which the different evolutionary mechanisms dominate, is presented in Chapter 2.
We conclude this Section, and chapter 2, deriving the expected coalescence timescale τ_{coal} in case GW emission alone, which will be useful in light of the N-body, numerical experiments presented in chapter 2.
Combining Equations 1.136 and 1.137 leads to a differential equation for $a = a(e)$, which reads

$$\frac{da}{de} = \frac{12}{19}\frac{1+\frac{73}{24}e^2+\frac{37}{96}e^4}{e(1-e^2)\left(1+\frac{121}{304}e^2\right)}. \tag{1.138}$$

It can be demonstrated that Equation (1.138) admits analytic solution of the kind

$$a(e) = a_0\frac{g(e)}{g(e_0)}, \tag{1.139}$$

where

$$g(e) \equiv \frac{e^{12/19}}{1-e^2}\left(1+\frac{121}{304}e^2\right)^{870/2299} \tag{1.140}$$

and (a_0, e_0) denotes the initial point of the binary evolutionary track. Therefore, given the initial binary conditions (a_0, e_0), the coalescence time is obtained by integrating Equation (1.138) over the eccentricity. The orbit circularization during this evolutionary stage imposes $e(\tau) = 0$ and yields

$$\tau(a_0, e_0) = \int_0^{\tau(a_0,e_0)} dt = \int_{e_0}^{0} de\,\frac{dt}{de} = \int_{e_0}^{0} de\,\frac{a^4(e)(1-e^2)^{5/2}}{e\left(1+\frac{121}{304}e^2\right)}. \tag{1.141}$$

Finally, by plugging Equation (1.139) and (1.140) into Equation (1.141) one finds the final expression

$$\tau_{\rm coal}(a_0, e_0) = \frac{12}{19}\frac{c^5}{G^3}\frac{a_0^4}{\mu m^2}\int_0^{e_0} de\,\frac{e^{29/19}\left(1+\frac{121}{304}e^2\right)^{1181/2299}}{(1-e^2)^{3/2}}. \tag{1.142}$$

Equation (1.142) shows that, for an equal-mass BH binary, owing to $\mu = m/2$, the coalescence time scales as

$$\tau_{coal} \propto \frac{a_0^4}{m^3}. \tag{1.143}$$

This scaling may provide a possible explanation for the numerical bottleneck encountered in N-body simulations of merger between two identical stellar systems, each containing an IMBH, presented in chapter 3

2

A Hybrid Approach to Black-Hole Binaries Evolution

In light of the forthcoming GW observatories, LISA and ET, investigating the properties of potentially GW-loud sources is of particular importance. Among these, IMBH binaries are especially compelling, and studying their dynamical evolution, together with their associated coalescence times, is essential for assessing their prospects as GW sources. This chapter aims to review the physical framework underlying the Quinlan-Sesana hybrid model for the dynamical evolution of black hole binaries and to present the main results of the numerical experiments performed using this approach. The chapter begins with a detailed description of the hybrid model, focusing on its main parameters and on the numerical integration of the differential equations governing the evolution of the binary semimajor axis and eccentricity. The model is then applied to IMBH binaries living within physically motivated stellar backgrounds. The second section is therefore devoted to numerical experiments in stellar-cusp environments. The second and third section presents the main results obtained by applying the model to a present-day, nucleated and non-nucleated dwarf galaxy, respectively. The chapter concludes with a study of IMBH binaries evolving within a family of reconstructed, high-redshift, non nucleated galaxies assessing for which combination of redshift of binary pairing and initial conditions it coalesce within the present time.

2.1 Quinlan-Sesana Hybrid Model Framework

As already discussed in Chapter 1, the coalescence time of a BH binary driven solely by GW emission depends strongly on several parameters, including the initial binary separation a_0 and eccentricity e_0. However, when evolving within a stellar background, the GW-driven regime is reached only after a sequence of earlier hardening mechanisms, primarily dynamical friction and three-body stellar scattering, which dominate the early stages of the binary shrinking. It is therefore essential to model the dynamical evolution of the binary by consistently accounting for the different hardening processes from its formation to final coalescence in order to obtain a reliable description of its evolutionary tracks, i.e. $a(t)$ and $e(t)$, and its coalescence time.
Although direct N-body simulations provide a self-consistent approach to carrying out this analysis, they remain computationally too demanding to efficiently explore the broad parameter space governing MBH binary evolution. This parameter space, for our purposes, includes the BH masses, the binary separation and eccentricity at pairing, and the properties of the stellar background. In this context, the self-consistent hybrid model developed by Quinlan (1996) and Sesana (2010) provides an efficient alternative, overcoming the computational limitations and long runtimes associated with direct N-body experiments while retaining the key physical processes driving the binary evolution.
Let us begin by introducing the Quinlan formalism, reviewing the restricted three-body scattering experiments from which the main model parameters were determined, and deriving the resulting coupled system of differential equations governing the evolution of the binary separation $a(t)$ and eccentricity $e(t)$. Adopting the original notation, we denote by m_1 and $m_2 < m_1$ the primary and secondary BH masses, respectively, by $M_{12} = m_1 + m_2$ the total binary mass, by $q = m_2/m_2$ the BHs mass ratio, and by $m_* \ll m_2$ the mass of an individual star, assuming all stars have the same mass. In his work, Quinlan (1996) studied the dynamical evolution of a BH binary embedded in a fixed, homogeneous background of unbound stars with density ρ and a Maxwellian velocity distribution characterised by a constant velocity dispersion σ.
The model relies on three dimensionless quantities, H, K, and J, which denote the dimensionless binary hardening rate, eccentricity growth rate, and stellar mass ejection rate, respectively. They are defined as follows:

$$H = \frac{\sigma}{G\rho}\frac{d}{dt}\left(\frac{1}{a}\right), \tag{2.1}$$

$$K = \frac{de}{d\ln(1/a)} \tag{2.2}$$

and

$$J = \frac{1}{M_{12}}\frac{dM_{ej}}{d\ln(1/a)}. \tag{2.3}$$

Let us focus on Equations (2.1) and (2.2). Both dimensionless rates account for the various, independent mechanisms involved during the binary evolution, so that, upon solving Equations (2.1) and (2.2) for the derivatives of the orbital parameters a and b, one can write

$$\begin{cases} \dfrac{da}{dt} = \displaystyle\sum_i \left.\frac{da}{dt}\right|_i = -\frac{a^2 G\rho}{\sigma}\sum_i H_i \\ \dfrac{de}{dt} = \displaystyle\sum_i \left.\frac{de}{dt}\right|_i = -\frac{aG\rho}{\sigma}\sum_i H_i K_i \end{cases} \tag{2.4}$$

or, equivalently,

$$\begin{cases} H_i = \dfrac{\sigma}{a^2 G\rho}\left.\dfrac{da}{dt}\right|_i \\ K_i = a\left.\dfrac{de}{dt}\right|_i \left(\displaystyle\sum_j \left.\frac{da}{dt}\right|_j\right)^{-1} \end{cases} \tag{2.5}$$

where the index i identifies a dynamical process (Sesana 2010).
As in Quinlan (1996) and Sesana, Haardt, et al. (2006), in our numerical experiments, we study the binary evolution due to close encounters with unbound, loss-cone stars and by GW emission. Therefore, we assume

$$\begin{cases} \dfrac{da}{dt} = \left.\dfrac{da}{dt}\right|_u + \left.\dfrac{da}{dt}\right|_{GW} = -\dfrac{a^2 G\rho}{\sigma}(H_u + H_{GW}) \\ \dfrac{de}{dt} = \left.\dfrac{de}{dt}\right|_u + \left.\dfrac{de}{dt}\right|_{GW} = -\dfrac{aG\rho}{\sigma}(H_u K_u + H_{GW} K_{GW}) \end{cases} \tag{2.6}$$

Equations (1.136) and (1.137) ruling the binary separation and eccentricity evolution under GW emission only and derived in § 1.5, are reported here to ensure consistency with the notation adopted in this chapter:

$$\left.\frac{da}{dt}\right|_{GW} = -\frac{64}{5}\frac{G^3\mu M_{12}^2}{c^5 a^3}\frac{1}{(1-e^2)^{7/2}}\left(1 + \frac{73}{24}e^2 + \frac{37}{96}e^4\right), \tag{2.7}$$

$$\left.\frac{de}{dt}\right|_{GW} = -\frac{304}{15}\frac{G^3\mu M_{12}^2}{c^5 a^4}\frac{e}{(1-e^2)^{5/2}}\left(1 + \frac{121}{304}e^2\right). \tag{2.8}$$

The terms in Equations (2.6) arasing from three-body stellar scattering, instead, were independently derived by Quinlan (1996) and Sesana, Haardt, et al. (2006) using two different approaches, as shown below. Indeed, with the aim of deriving the dimensionless rates, both works employed suites of three-body scattering experiments designed to characterize the response of a BH binary to repeated close encounters with stars over its lifetime. However, while Quinlan (1996) fixed the incoming stellar velocity v and varied the binary separation a, Sesana, Haardt, et al. (2006) adopted the opposite approach, varying v at fixed a. It is worth of noting that the two approaches are physically equivalent, but lead to different parametrizations of the functions H_u and K_u. We will be using the latter as, when numerically integrating Equations (2.6), the parametrization provided by Sesana, Haardt, et al. (2006) turns out to be the most convenient to be adopted, since it directly involves the quantities being integrated.
Let us now briefly review how H_u and K_u were originally estimated. The underlying idea is to infer the variations in the binary orbital energy and angular momentum from the corresponding changes experienced by the scattered star, namely ΔE_* and $\Delta\mathbf{L}_*$, which are equal in magnitude and opposite in sign as a consequence of energy and angular momentum conservation.
The first step is to define the dimensionless energy change

$$C \equiv \frac{a\Delta E_*}{G\mu}, \tag{2.9}$$

introduced by Hills (1983), and the dimensionless impact parameter

$$x \equiv \frac{b}{b_0}, \tag{2.10}$$

with $b_0^2 = 2GM_{12}a/v^2$.
Denoting by $\langle C\rangle$ the average over all angular variables, the "single-velocity" hardening rate

$H_1(v)$, derived from restricted three-body scattering experiments in which all stars have the same velocity at infinity, is given by

$$H_1(v) = \frac{1}{\pi b_0^2} \int db\,(2\pi b) 4\pi \langle C \rangle = 8\pi \int dx\,\langle C \rangle x \equiv 8\pi I_x(C), \tag{2.11}$$

where the operator $I_x(C)$ is implicitly defined by Equation (2.11). Finally, the hardening rate $H_u(\sigma)$ is obtained by averaging the single-velocity hardening rate $H_1(v)$ over the Maxwellian distribution of stellar velocities, which leads to

$$H_u(\sigma) = \int_0^\infty dv\, 4\pi v^2 H_1(v) \frac{\sigma}{v} f(v, \sigma), \tag{2.12}$$

where

$$f(v, \sigma) = \frac{1}{(2\pi\sigma^2)^{3/2}} e^{-\frac{v^2}{2\sigma^2}} \tag{2.13}$$

is the Maxwellian velocity distribution associated with the costant velocity dispersion σ.
Similarly, one finds K_u from the "single-velocity" dimensionless eccentricity growth rate K_1, having the same meaning of H_1. The quantity K_1 can be derived starting from the expression for the binary eccentricity

$$e^2 = 1 + \frac{2EL^2}{\mu (G M_{12})^2}, \tag{2.14}$$

which relates the eccentricity to the binary orbital energy E and angular momentum L.
Differentiating Equation (2.14) and expressing the resulting variations in terms of finite differences yields

$$\frac{\Delta e}{\Delta ln(1/a)} = \frac{1 - e^2}{2e} \left[-2 \left(\frac{E}{\Delta E} \right) \left(\frac{\Delta L_z}{L_z} \right) - 1 \right] = \frac{1 - e^2}{2e} (B - C), \tag{2.15}$$

where

$$B \equiv -\frac{M_{12}}{m_*} \frac{\Delta L_z}{L_z}. \tag{2.16}$$

is the dimensionless angular momentum change. Finally, by integrating over all possible impact parameters and angular variables and normalizing the result to B, the single-velocity eccentricity growth rate K_1 turns out to be

$$K_1 = \frac{\int dx\, x C(x) \left\langle \frac{\Delta e}{\Delta \ln(1/a)} \right\rangle}{\int dx\, x C(x)} = \frac{1 - e^2}{2e} \left[\frac{I_x(B - C)}{I_x(C)} \right]. \tag{2.17}$$

The best-fitting functions found by Sesana, Haardt, et al. (2006) from numerical experiments to H_u and K_u are

$$H_u(a) = A_H \left(1 + \frac{a}{a_{0,H}} \right)^{\gamma_H} \tag{2.18}$$

and

$$K_u(a, e) = A_K \left(1 + \frac{a}{a_{0,K}} \right)^{\gamma_K} + B_K, \tag{2.19}$$

with best-fitting parameters listed in Tables 2.1 and 2.2.

q	A_H	$a_{0,H}$	γ_H
1	14.55	3.48	-0.95
1/3	15.82	4.18	-0.90
1/9	17.17	3.59	-0.79
1/27.....	18.15	3.32	-0.77
1/81.....	18.81	3.87	-0.82
1/243.....	19.16	4.16	-0.86

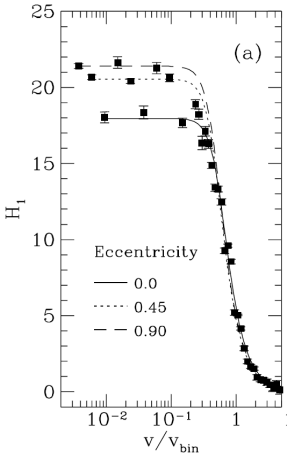
Table 2.1: Best-fit parameters for the dimensionless stellar hardening function H_u defined by Equation (2.18) as a function of the BH mass ratio q. The parameter $a_{0,H}$ is given in units of the binary hardening radius a_h. Values taken from Sesana, Haardt, et al. (2006).

e	A_K	$a_{0,K}$	γ_K	B_K
$q = 1$				
0.15	0.037	0.339	-3.335	-0.012
0.3	0.075	0.151	-1.548	-0.008
0.45	0.105	0.088	-0.893	-0.005
0.6	0.121	0.090	-0.895	-0.008
0.75	0.134	0.064	-0.544	-0.006
0.9	0.082	0.085	-0.663	-0.004
$q = 1/3$				
0.15	0.082	0.042	-0.168	-0.048
0.3	0.095	0.213	-1.152	-0.012
0.45	0.129	0.137	-0.655	-0.006
0.6	0.166	0.081	-0.546	-0.006
0.75	0.159	0.079	-0.497	-0.010
0.9	0.095	0.122	-0.716	-0.008
$q = 1/9$				
0.15	0.051	0.385	-0.891	-0.011
0.3	0.111	0.307	-1.107	-0.007
0.45	0.172	0.526	-1.174	-0.016
0.6	0.181	0.251	-1.169	-0.007
0.75	0.179	0.195	-0.846	-0.004
0.9	0.117	0.400	-1.170	-0.001
$q = 1/27$				
0.15	0.064	0.284	-1.206	0.021
0.3	0.143	1.033	-1.537	-0.021
0.45	0.212	0.722	-1.257	-0.022
0.6	0.216	0.430	-1.163	-0.014
0.75	0.173	0.771	-1.934	-0.014
0.9	0.129	0.329	-1.125	-0.020

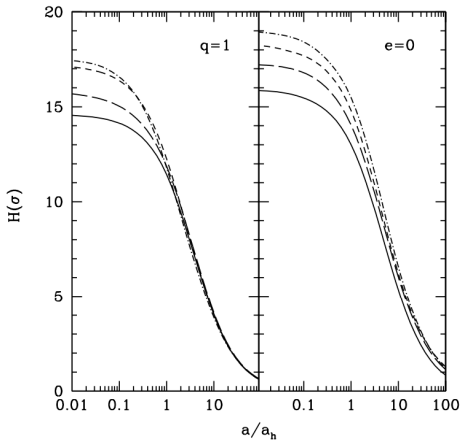
Table 2.2: Best-fit parameters for the dimensionless eccentricity growth rate function K_u defined by Equation (2.19) as a function of the BH mass ratio q and binary eccentricity e. The parameter $a_{0,K}$ is given in units of the binary hardening radius a_h. Values taken from Sesana, Haardt, et al. (2006).

The absence of a column dedicated to the eccentricity sampling in Table 2.1 reflects one of the main findings of the scattering experiments by Quinlan (1996) and Sesana, Haardt,

et al. (2006): the dimensionless hardening rate H_u is almost independent of the BH binary eccentricity. Besides, it was found that hard binaries, having separation $a \ll a_h$ by definition, hardens at almost costant rate. These results are displayed in Figure 2.1a.



(a) "Single-velocity" binary hardening rate H_1 (see Equation 2.11) as a function of the initial stellar velocity V, normalized to the binary associated circular velocity $V_{bin} = \sqrt{GM_{12}/a}$, for three different values of the BH binary eccentricities, namely $e = (0, 0.45, 0.90)$, at fixed BH binary masses ratio $q = 1$.
Figure taken from Quinlan (1996).



(b) Hardening rate H_u as a function of binary separation a, normalized to a_h for different eccentricities e (left) and mass ratios q (right). In both panels, solid, long-dashed, short-dashed, and dot-dashed lines correspond to $e = (0, 0.3, 0.6, 0.9)$ and $q = (1/3, 1/9, 1/27, 1/81)$, respectively. Figure taken from Sesana, Haardt, et al. (2006).

Figure 2.1: Comparison of the dimensionless stellar hardening rates obtained by Quinlan (1996) and Sesana, Haardt, et al. (2006) from indipendent three-body scattering experiments.

In contrast, the dimensionless eccentricity growth rate K_u depends on both the binary eccentricity e and the separation a. Following the procedure illustrated by Sesana (2010), at a given binary separation, $K_u(a, e)$ is obtained from Equation (2.19) by linearly interpolating in the eccentricity domain the best-fitting parameters reported in Table 2.2. The interpolation is supplemented by the boundary conditions $K(a, e = 0) = 0$, as theoretically predicted in the test-particle limit for circular orbits, since circular binaries remain circular, and $K(a, e = 1) = 0$.
We are now in a position to illustrate how the model can be applied to an arbitrary stellar background and how the corresponding evolution of the binary separation and eccentricity, $a(t)$ and $e(t)$, is obtained numerically. As reported by Quinlan (1996) and Sesana, Haardt, et al. (2006), the three-body scattering experiments underlying the hybrid model consider stars that are initially unbound to the binary and whose velocities follow a Maxwellian distribution. Since these calculations are performed in a specific stellar background, a prescription is required to extend their results to the diverse stellar distributions hosting MBHs. A useful calibration is provided by Sesana and F. M. Khan (2015), who showed that the hardening rate measured from direct N-body simulations of a MBHB embedded in a spherically symmetric and isotropic stellar system, characterized by a mass density $\rho(r)$ and velocity dispersion $\sigma(r)$, agrees to within approximately 30% with that predicted by the hybrid model for a homogeneous stellar background characterized by a constant density ρ_{inf} and velocity dispersion σ_{inf}, denoting the stellar mass density and velocity dispertion values at the binary influence radius R_{inf}, respectively. Indeed, once the binary has ejected most of the stars initially bound within its radius of influence, the loss cone is predominantly replenished by stars coming from just outside r_{inf}, thus having locally a nearly homogeneous spatial distribution with $\rho = \rho_{inf}$ and velocity dispersion $\sigma = \sigma_{inf}$.

In summary, once the BH masses and the properties of the stellar background are specified, the dynamical evolution of the binary is governed by the following system of coupled differential equations:

$$\begin{cases} \dfrac{da}{dt} = -\dfrac{a^2 G\rho_{inf}}{\sigma_{inf}} H_u(a) - \dfrac{64}{5}\dfrac{G^3\mu M_{12}^2}{c^5 a^3}\dfrac{1}{(1-e^2)^{7/2}}\left(1+\dfrac{73}{24}e^2+\dfrac{37}{96}e^4\right) \\ \dfrac{de}{dt} = -\dfrac{aG\rho_{inf}}{\sigma_{inf}} H_u(a)K_u(a,e) - \dfrac{304}{15}\dfrac{G^3\mu M_{12}^2}{c^5 a^4}\dfrac{e}{(1-e^2)^{5/2}}\left(1+\dfrac{121}{304}e^2\right) \end{cases} \tag{2.20}$$

which can be solved numerically for a given set of initial conditions (a_0, e_0).
The system of Equations (2.20) constitutes a set of first-order autonomous ordinary differential equations (ODEs), which can be written in the general form

$$\frac{d\mathbf{u}}{dt} = \mathbf{f}(\mathbf{u}), \tag{2.21}$$

where $\mathbf{u} = (a, e)$ is the vector of dynamical variables.
For the numerical integration of Equations (2.20), a fourth-order Runge–Kutta (RK4) scheme is employed. To ensure the stability of this method, the time step must satisfy an appropriate stability condition. For a system of N ODEs expressed in terms of the N-dimensional vector $\mathbf{u}$, the first-order stability analysis of the RK4 scheme yields the following criterion (Butcher 2016):

$$\Delta t \leq min\left\{\frac{2.79}{|\lambda_i|}\right\} \qquad \forall i = 1, N \tag{2.22}$$

where λ_i denotes the i-th eigenvalue of the Jacobian matrix $\mathbf{J} = \partial\mathbf{f}/\partial\mathbf{u}$.
For the system considered here, the Jacobian elements can be derived directly from Equations (2.20) and are listed below:

$$J_{11} = \left.\frac{da}{dt}\right|_u \left(\frac{2}{a} + \frac{\gamma}{a + a_0 a_h}\right) - \left.\frac{da}{dt}\right|_{gw}\left(\frac{3}{a}\right), \tag{2.23}$$

$$J_{12} = \left.\frac{da}{dt}\right|_{GW}\left(\frac{7e}{1-e^2} + \frac{73e/12 + 37e^3/24}{1 + 73e^2/24 + 37e^4/96}\right), \tag{2.24}$$

$$J_{21} = \left.\frac{de}{dt}\right|_u \left(\frac{1}{a} + \frac{\gamma_H}{a + a_{0,H} a_h} + \frac{\gamma_K}{a + a_{0,K} a_h}\right) - \left.\frac{de}{dt}\right|_{GW}\left(\frac{4}{a}\right), \tag{2.25}$$

$$J_{22} = \left.\frac{de}{dt}\right|_u \left(\frac{\partial \ln K}{\partial e}\right) + \left.\frac{de}{dt}\right|_{GW}\left(\frac{5e}{1-e^2} + \frac{1 + 363e^2/304}{e + 121e^3/304}\right). \tag{2.26}$$

These expressions allow the two eigenvalues, λ_1 and λ_2, of the Jacobian matrix to be evaluated at each point along the binary evolutionary track and the adaptive time-step to be calculated as

$$\Delta t \leq 2.79 \cdot \min\left\{\frac{1}{|\lambda_1|}; \frac{1}{|\lambda_2|}\right\}. \tag{2.27}$$

In the following Sections, will be presented the results of the numerical experiments performed with the hybrid model described above, considering different IMBH binaries masses, initial conditions and host stellar systems.

2.2 Evolution of an IMBH Binary in a Family of Stellar Cusp Profiles

Observations and direct N-body simulations suggest that galaxies hosting dynamically evolved BH binaries may undergo stellar scouring during binary hardening, significantly altering the original stellar distribution and leading to the formation of a shallower inner cusp (see, e.g., Milosavljević and Merritt 2001; Merritt and Milosavljević 2005). As the binary orbit shrinks within the galactic core, it transfers energy to the surrounding stars, progressively depleting the central stellar distribution and producing a stellar mass deficit in the innermost region of the host galaxy.
In light of these results, the first stellar background considered here for studying the dynamical evolution of an IMBH binary, from pairing to coalescence, is thus a power-law density profile within the binary influence radius, matched to a singular isothermal sphere (SIS) outside it (Sesana 2010). The stellar density $\rho(r)$ of this spherically symmetric distribution is given by

$$\begin{cases} \rho(r) = \rho_{inf}\left(\dfrac{r}{r_{inf}}\right)^{-\gamma} & \text{if} \quad r < r_{inf} \\ \rho(r) = \dfrac{\sigma^2}{2\pi G r^2} & \text{if} \quad r > r_{inf} \end{cases} \tag{2.28}$$

where γ is the exponent of the inner power-law stellar cusp and σ denotes the position-independent velocity dispersion of the SIS ($\gamma = 2$) profile.
Imposing the continuity of the stellar profile in Equation (2.28) at the binary influence radius r_{inf} returns the following analytic expressions for r_{inf} and ρ_{inf}:

$$r_{inf} = (3-\gamma)\frac{GM_{12}}{\sigma^2}, \tag{2.29}$$

$$\rho_{inf} = \frac{\sigma^2}{2\pi G r_{inf}^2}. \tag{2.30}$$

Adopting the notation introduced in the previous Section, we assume that σ is uniquely determined by the binary mass M_{12} through the $M-\sigma$ relation

$$\left(\frac{M_{12}}{10^6\, M_\odot}\right) = 0.84\left(\frac{\sigma}{70\,\mathrm{km/s}}\right)^4. \tag{2.31}$$

As already mentioned in § 1.2, dynamical friction exerted by the stellar background on an intruding secondary BH efficiently drags it towards the galactic center, where it eventually forms a gravitationally bound binary with the primary BH, with semimajor axis a_0. This separation is defined as the radius enclosing a stellar mass equal to the secondary BH mass m_2 and, for an adopted stellar cusp profile, can be expressed as

$$a_0 = (3-\gamma)\frac{GM_{12}}{\sigma^2}\left(\frac{q}{1+q}\right)^{1/(3-\gamma)} = r_{inf}\left(\frac{q}{1+q}\right)^{1/(3-\gamma)}. \tag{2.32}$$

In addition to a_0, it is useful to define a set of characteristic distances associated with different binary separations, which allows us to distinguish the various stages of binary evolution and identify the dominant physical mechanism governing each stage. At separation a_0, the binary is initially too "soft", in the sense of Heggie's law, for scattering with unbound stars to significantly contribute to its hardening. Therefore, once dynamical friction has become inefficient at a_0, the subsequent binary evolution is driven by interactions with bound stars orbiting within the

binary sphere of influence (Sesana, Haardt, et al. 2008). As the binary hardens, these stars are progressively ejected from the central region, carrying away the orbital energy extracted from the binary and causing its separation to decrease towards the hardening radius

$$a_h = \frac{1}{4(3-\gamma)} \left(\frac{q}{1+q} \right)^{(2-\gamma)/(3-\gamma)}, \tag{2.33}$$

below which the stellar-scattering term included in Equations (2.20) becomes dominant (Sesana 2010).
Finally, provided that a sufficient supply of loss-cone stars is available, the binary eventually enters the GW-dominated regime, in which GW emission drives the orbital decay towards final coalescence. For a stellar cups like stellar background, the transition occurs at the binary separation

$$a_{GW} = 4 \left[\frac{128\pi(3-\gamma)F(e)}{5H_u} \right]^{1/5} \frac{\sigma}{c} q^{-4/5} (1+q)^{3/5} a_h \tag{2.34}$$

defined by the condition $H_u = H_{GW}$ (Sesana 2010). Here $F(e)$ represents the same function of the eccentricity e defined in Equation (1.129).
In conclusion, it is worth of noting that for the equal-mass BH binaries, the initial binary separation a_0, given by Equation (2.32), and the hardening radius a_h, defined in Equation (2.33), are comparable. In particular, their separation in logarithmic space is small compared with the overall range of binary separations spanned during the subsequent evolution. Consequently, the binary is already close to the hard-binary regime at its birth, and the additional hardening associated with the erosion of the bound stellar cusp can safely be omitted from our analysis (Sesana and F. M. Khan 2015).
To investigate the role played by the central stellar density in the overall evolution of the binary, while keeping all other relevant conditions fixed, we consider three numerical experiments, one for each value of the stellar cusp slope $\gamma = (1, 1.5, 2)$. Using the hybrid model described above, we follow the evolutionary tracks of the same equal-mass IMBH binary, with $m_1 = m_2 = 10^5\, M_\odot$, initially placed at the same initial conditions (a_0, e_0) in all cases. Although the value of a_0 depends on the adopted cusp slope γ, as determined from Equation (2.32), we adopt a common initial configuration for all binaries, $(\bar{a}_0, 0.5)$, where $\bar{a}_0$ is the average of the a_0 values obtained for the different stellar cusps. All stellar cusps are instead characterized by the same velocity dispersion, $\sigma = 48.9\,\mathrm{km/s}$, owing to Equation (2.31). Combined with the different cusp slopes, this common value of σ yields the corresponding stellar densities at the influence radius, $\rho_{inf} = (1.71, 3.04, 6.83) \times 10^5\, M_\odot/\mathrm{pc}^3$ for $\gamma = (1, 1.5, 2)$, respectively.
Following the prescriptions given by Equations (2.29), (2.32), (2.33), and (2.34), the characteristic transition lengthscales are computed for each stellar background, or equivalently for each value of γ, and reported in Table 2.3.

γ	r_{inf} [pc]	a_0 [pc]	a_h [pc]	a_{GW} [pc]
1	0.72	0.51	0.016	0.97×10^{-4}
1.5	0.54	0.34	0.034	0.91×10^{-4}
2	0.36	0.18	0.045	0.84×10^{-4}

Table 2.3: Hybrid model parameters for three different stellar cusp slopes γ. Column 1: Stellar cusp slope. Column 2: Influence radius of the stellar cusp. Column 3: Initial binary separation. Column 4: Binary hardening radius. Column 5: Stellar scattering-GW hardening transition lengthscale.

A first inspection of Table 2.3 and Figure 2.2 shows that, for fixed BH masses, increasing γ leads to an increasingly concentrated stellar distribution and, consequently, to a higher stellar

density within the binary sphere of influence. As a result, the characteristic lengthscales r_{inf}, a_0, and a_{GW} progressively decrease with increasing γ. This reflects the fact that, in a steeper cusp, the stellar mass reaches twice the binary mass and twice the secondary BH mass at smaller radii, while the transition to the GW-dominated regime occurs at a smaller binary separation. Figure 2.2 shows the mass density profile of each stellar cups profile considered here, along with the corresponding influence radius.

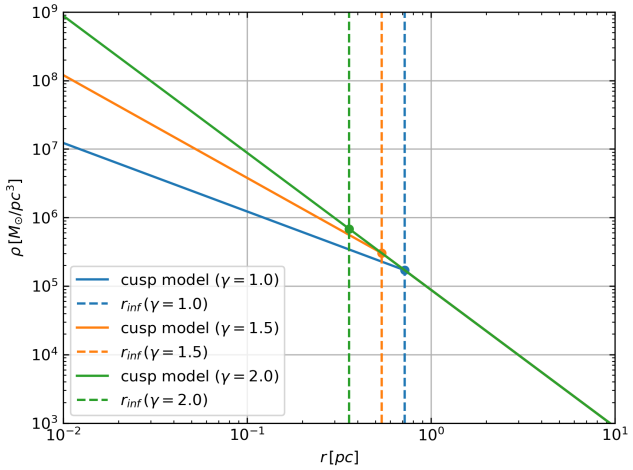


Figure 2.2: Stellar density profiles for three different stellar cusp slopes $\gamma = (1, 1.5, 2)$ (solid blue, orange, and green lines, respectively), with the corresponding influence radii r_{inf} marked by vertical dashed lines of the same colour. At radii larger than the largest influence radius, corresponding to $\gamma = 1$, the three curves converge to the same SIS profile, that is the stellar cusp with $\gamma = 2$.

As all three numerical experiments involve equal-mass binaries, the mass ratio is fixed to $q = 1$ for all values of γ. The dimensionless hardening rate H_u is therefore determined directly from Equation (2.18), using the hardening radii a_h listed in Table 2.3 for each cusp slope and the fitting parameters reported in the first row of Table 2.1. The resulting hardening rates for the three stellar cusps are shown in Figure 2.3.

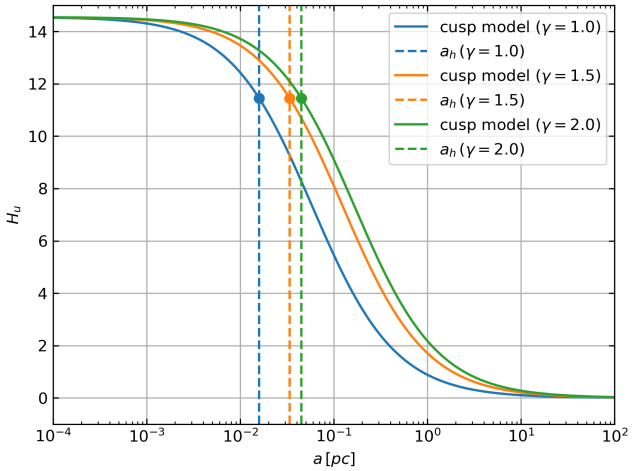
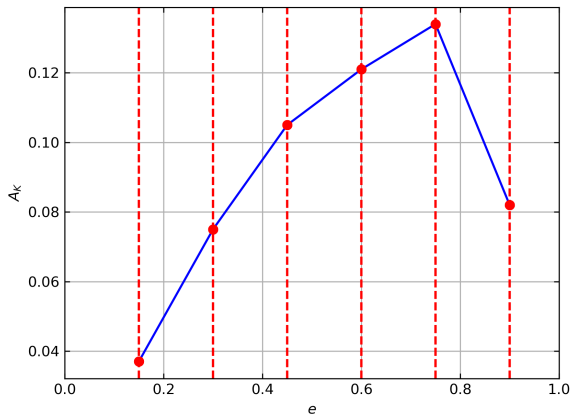


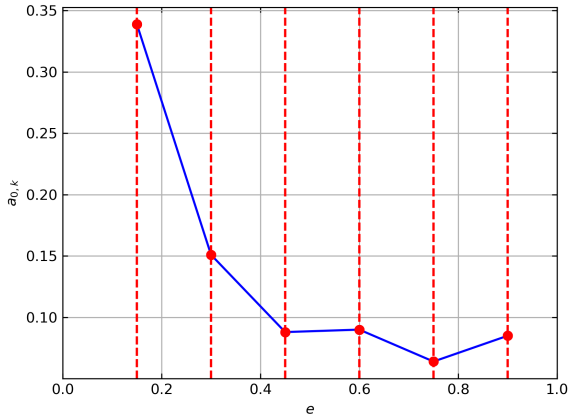
Figure 2.3: Dimensionless hardening rate H_u as a function of binary separation for three different stellar cusp slopes $\gamma = (1, 1.5, 2)$ (solid blue, orange, and green lines, respectively), with the corresponding hardening radii a_h marked by vertical dashed lines of the same colour.

The final quantity required to numerically integrate Equations (2.20) is the dimensionless

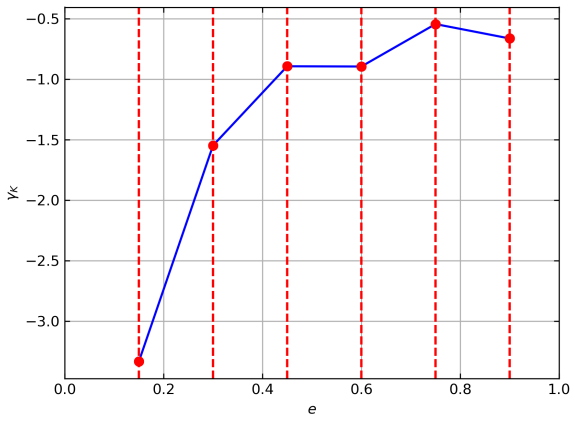
eccentricity growth function K_u. Unlike H_u, it depends not only on the binary semimajor axis but also on its eccentricity. This dependence on eccentricity is encoded in the best-fitting parameters of Equation (2.19), as reported in Table 2.2. Although the best-fitting parameters are provided only at a discrete set of eccentricities, namely $e = (0.15, 0.3, 0.45, 0.6, 0.75, 0.9)$, their values over the full interval $e \in [0, 1]$ can be obtained by linear interpolation in eccentricity, once the mass ratio q is specified (see, e.g., Sesana, Haardt, et al. 2006; Sesana 2010). Figure 2.4 (a)-(b)-(c)-(d) illustrates this procedure.



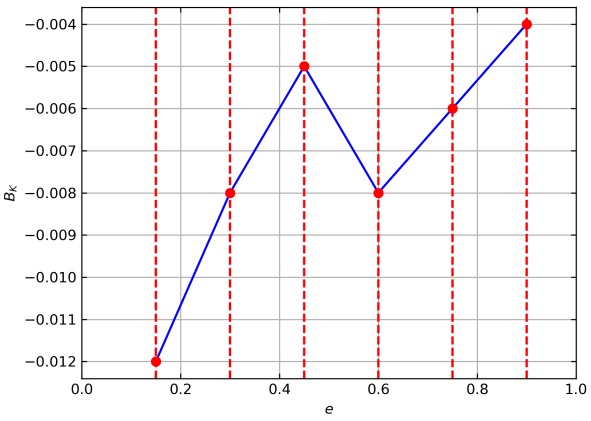
(a) Eccentricity growth rate parameter A_K (solid blue piecewise line) as a function of binary eccentricity e for binary masses ratio $q = 1$. Red dashed vertical lines mark the sampled eccentricities, with red dots corresponding to the values listed in Table 2.2.



(b) Eccentricity growth rate parameter $a_{0,K}$ (solid blue piecewise line), in a_h units, as a function of binary eccentricity e for binary masses ratio $q = 1$. Red dashed vertical lines mark the sampled eccentricities, with red dots corresponding to the values listed in Table 2.2.



(c) Eccentricity growth rate parameter γ_K (solid blue piecewise line) as a function of binary eccentricity e for binary masses ratio $q = 1$. Red dashed vertical lines mark the sampled eccentricities, with red dots corresponding to the values listed in Table 2.2.

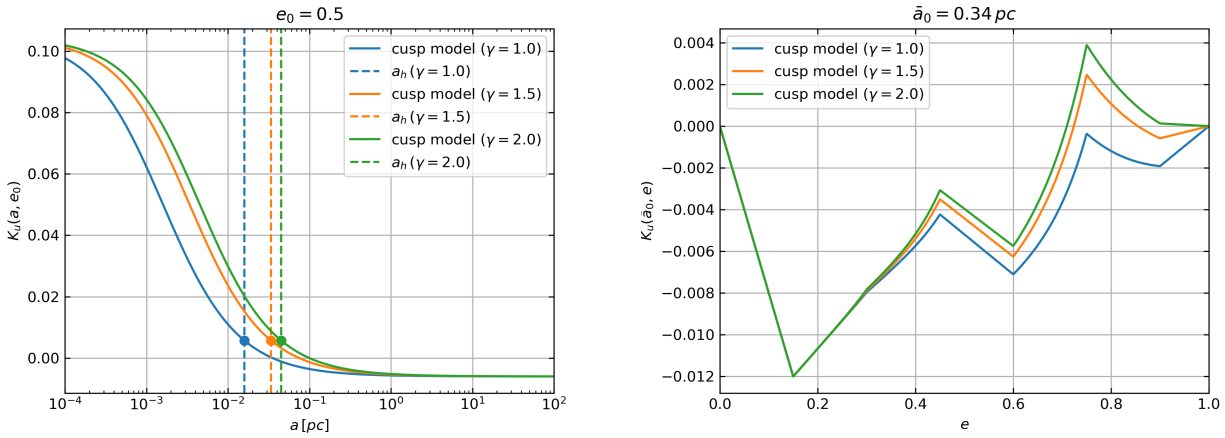


(d) Eccentricity growth rate parameter B_K (solid blue piecewise line) as a function of binary eccentricity e for binary masses ratio $q = 1$. Red dashed vertical lines mark the sampled eccentricities, with red dots corresponding to the values listed in Table 2.2.

Figure 2.4: Panels (a)–(b)–(c)–(d) show the best-fitting parameters A_K, $a_{0,K}$, γ_K, and B_K of the eccentricity growth-rate function K_u, respectively, as functions of the binary eccentricity e. Red points indicate the values obtained from three-body scattering experiments and listed in Table 2.2, while blue piecewise-linear curves show their linear interpolation. Vertical red dashed lines mark the sampled eccentricity values.

These interpolated best-fitting parameters can then be used to derive $K_u(a, e)$ and, in order to extend the interpolation to the boundaries of the physical eccentricity interval, two additional points are manually introduced at $e = 0$ and $e = 1$, so that $K_u(a, 0) = K_u(a, 1) = 0$.
Figure 2.5 (a)–(b) show $K_u(a) = K_u(a, e_0)$ and $K_u(e) = K_u(\bar{a}_0, e)$, respectively, where $(\bar{a}_0, e_0) = (0.34\text{pc}, 0.5)$ corresponds to the initial condition adopted in all three numerical simulations. As displyed, the derived eccentricity growth rate is smooth when evaluated at fixed e, while it is

piecewise linear when evaluated at fixed a.



(a) Dimensionless eccentricity growth rate $K_u(a, e)$, at fixed eccentricity $e_0 = 0.5$, as a function of binary separation for three different stellar cusp slopes $\gamma = (1, 1.5, 2)$ (solid blue, orange, and green lines, respectively), with the corresponding hardening radii a_h marked by vertical dashed lines of the same colour.

(b) Dimensionless eccentricity growth rate $K_u(a, e)$, at fixed binary semimajor axis $\bar{a}_0 = 0.34\,\mathrm{pc}$, as a function of binary eccentricity for three different stellar cusp slopes $\gamma = (1, 1.5, 2)$ (solid blue, orange, and green lines, respectively). The boundary conditions $K_u(a, 0) = K_u(a, 1) = 0$ have been included.

Figure 2.5: 1-D slices of the dimensionless eccentricity growth rate $K_u(a, e)$, at fixed eccentricity (a) and binary separation (b), for three different stellar cusps slopes $\gamma = (1, 1.5, 2)$.

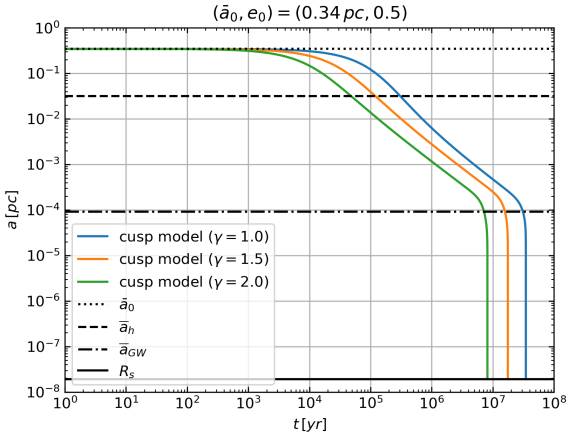
Having reported how the essential ingredients of the adopted hybrid model are have been derived, we can now specify the setup adopted for the numerical integrations. In all the experiments presented in this Section, the integration starts from the initial conditions $(\bar{a}_0, e_0) = (0.34\mathrm{pc}, 0.5)$ and is terminated when the binary separation reaches the Schwarzschild radius of the binary, defined as

$$R_s = \frac{2GM_{12}}{c^2}, \tag{2.35}$$

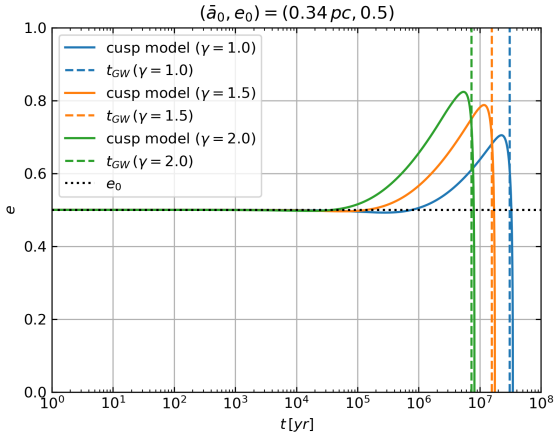
which, for the total binary mass considered here, corresponds to $R_s = 5.9 \times 10^5\,\mathrm{km}$.
The integration time step Δt is chosen to ensure the numerical stability of the RK4 scheme according to the criterion given in Equation (2.22). For all numerical runs, we adopt

$$\Delta t = \epsilon \cdot 2.79\, min\left\{\frac{1}{|\lambda_1|}; \frac{1}{|\lambda_2|}\right\} \tag{2.36}$$

where $\epsilon = 0.001$ is the chosen safety factor and λ_i denotes the i-th eigenvalue of the Jacobian matrix with elements defined in Equation (2.23)-(2.24)-(2.25)-(2.26).
Figure 2.6 (a)–(b) show the time evolution of the binary semimajor axis, $a(t)$, and eccentricity, $e(t)$, respectively, for the three stellar cusp slopes considered, $\gamma = (1, 1.5, 2)$.



(a) Binary semimajor axis time evolution for three different stellar cusp slopes $\gamma = (1, 1.5, 2)$ (solid blue, orange, and green lines, respectively).
Black horizontal lines indicate the mean values over the three stellar cusps of a_0 (dotted), a_h (dashed), a_{GW} (dash-dotted), and the binary R_s.



(b) Binary eccentricity evolution for three stellar cusp slopes, $\gamma = (1, 1.5, 2)$ (solid blue, orange, and green lines, respectively).
Vertical dashed lines of the corresponding colours mark the times t_{GW} at which $a = a_{GW}$ and GW-driven circularization becomes dominant.

Figure 2.6: Evolutionary tracks $(t, a(t))$ in panel (a) and $(t, e(t))$ in panel (b) for stellar cusp slopes $\gamma = (1, 1.5, 2)$ and initial binary conditions $(a_0, e_0) = (0.34, \mathrm{pc}, 0.5)$.

The first relevant result concerns the estimated coalescence timescales; they turns out to be $\tau_{coal} = (34.7, 17.6, 8.18)\,\mathrm{Myr}$. All three values are shorter than the Hubble time, $t_H = 13.8\,\mathrm{Gyr}$, indicating that, for the considered initial conditions, an equal-mass IMBH binary with $m_1 = m_2 = 10^5\,M_\odot$ may emit a potentially detectable GW signal in the LISA frequency band. Moreover, the coalescence time is found to decrease systematically with increasing cusp slope γ. This behaviour, shown in Figure 2.6 (a), can be attributed to the higher central stellar densities associated with steeper cusps, which provide a larger supply of stars interacting with the binary through loss-cone dynamics. The enhanced rate of stellar interactions accelerates the binary hardening, allowing the binary to reach the separation at which GW emission becomes the dominant mechanism driving the orbital decay at an earlier time, thereby reducing the overall coalescence timescale.
Besides, Figure 2.6 (b) shows that the binary eccentricity tends to increase during the phase dominated by stellar scattering. As the cusp becomes steeper, this dynamical regime becomes progressively shorter, allowing GW-driven circularization to take over at an earlier stage of the evolution. Consequently, the maximum eccentricity reached by the binary decreases with increasing γ.
Having established the key role of the stellar density in driving efficient binary hardening, the next Section extends the analysis to a nucleated, present-day galaxy model, whose high central density is expected to promote fast IMBH binaries coalescence.

2.3 Evolution of an IMBH Binary in a NGC 205-Like Stellar Background

The Local Group comprises several dozen galaxies, with the majority of its members being low-mass systems gravitationally associated with its two dominant galaxies, the Milky Way and M31. Among them, M 32, NGC 205, NGC 5102, and NGC 5206, four nucleated low-mass early-type galaxies, are of particular interest because their central regions can be resolved in detail

and their nuclear dynamics can be studied through high-resolution observations (Nguyen et al. 2018). Owing to the remarkably high density provided by their nuclear star clusters (NSCs), these galaxies represent ideal environments for efficient stellar-scattering-driven hardening of a BH binaries potentially living in their galactic centre.
In this Section, we thus investigate the evolution of an IMBH binary embedded in a NGC 205-like stellar background, as a representative present-day, nucleated dwarf galaxy ,with particular emphasis on the role played by the binary eccentrecy e_0 at pairing on the resulting evolutionary tracks and coalescence timescales.
NGC 205 is a nucleated dwarf elliptical galaxy orbiting M 31, located at RA $= 00^{\rm h}40^{\rm m}22.1^{\rm s}$ and Dec $= +41°41'07''$. Its distance has been determined through the tip of the red giant branch method, yielding a distance modulus of $(m-M)_0 = 24.58 \pm 0.07$, corresponding to a distance of approximately 824 ± 27 kpc (McConnachie et al. 2005). HST/STIS spectroscopy of the Ca II triplet region revealed no significant nuclear rotation and a nearly constant velocity dispersion of $\sim 20\,$km/s within the central $\sim 0.3''$ (~ 1 pc). Dynamical modelling of the nuclear region yielded a 3σ upper limit of $3.8 \times 10^4\,M_\odot$ for the mass of a possible central IMBH (see, e.g., Valluri et al. 2005; Geha et al. 2006).
The two-dimensional surface-brightness profile of NGC 205 can be decomposed into two distinct structural components, corresponding to the bulge and the nuclear star cluster. We model each component with a three-dimensional deprojected Sersic density profile (Sersic 1968) of the form (see, e.g., Prugniel and Simien 1997; Terzić and Graham 2005):

$$\rho(r) = \rho_0 \left(\frac{r}{R_e}\right)^{-p} e^{-b\left(\frac{r}{R_e}\right)^{1/n}}. \tag{2.37}$$

The stellar density profile given in Equation (2.37) is characterized by three structural parameters, (ρ_0, R_e, n), together with the model-dependent quantities $p(n)$ and $b(n)$. Here, R_e is the effective radius, enclosing half of the total luminosity of the stellar component, while n is the Sersic index controlling the shape of the profile. The normalization ρ_0 is determined by requiring the volume-integrated density to reproduce the total stellar mass of the component. The parameters p and b are functions of the Sersic index and can be approximated, for $0.5 < n < 10$, as $p \approx 1.0 - 0.6097/n + 0.05563/n^2$ and $b \approx 2n - 1/3 + 0.009876/n$ (Terzić and Graham 2005). Table 2.4 summarizes the main structural parameters of the bulge and NSC of NGC 205, which are required to specify the corresponding Sersic profiles.

Component		n	R_e [pc]	M_* [$10^7\,M_\odot$]
NSC	...	1.6	1.3	0.18
Bulge	...	1.4	516	97.2

Table 2.4: NGC 205 structural parameters of stellar bulge and NSC. Column 1: Galaxy component. Column 2: Sersic index. Column 3: Effective radius. Column 4: Stellar mass. Values taken from F. M. Khan and Holley-Bockelmann (2021).

Since Chapter 3 is devoted to studying the dynamical evolution of IMBH binaries embedded in Sersic-like stellar backgrounds through direct N-body simulations, we briefly examine here the properties of this profile, defined by Equation (2.37), that are relevant for constructing equilibrium N-body initial conditions and interpreting the results presented in the following Sections.
Given the mass density profile in Equation (2.37) for a spherically symmetric stellar component of total mass M_*, the mass enclosed within a sphere of radius r is obtained by integrating the

density profile over the corresponding volume:

$$M(<r) = 4\pi \int_0^r dx\, \rho(x) x^2. \tag{2.38}$$

By exploiting the standard substitution $Z \equiv b\left(\frac{r}{R_e}\right)^{1/n}$, the integral can be expressed in terms of the lower incomplete gamma function,

$$\gamma(a,x) = \int_0^x dt\, e^{-t} t^{a-1}, \tag{2.39}$$

as

$$M(<r) = 4\pi\rho_0 R_e^3 n b^{n(p-3)} \gamma\left[n(3-p), b\left(\frac{r}{R_e}\right)^{1/n}\right]. \tag{2.40}$$

Imposing the normalization condition $M(<r) \to M_*$ as $r \to +\infty$, and using the fact that the lower incomplete gamma function reduces to the complete gamma function,

$$\Gamma(a) = \int_0^\infty dt\, e^{-t} t^{a-1}, \tag{2.41}$$

in this limit, Equation (2.40) can be conveniently recast in terms of the total stellar mass as

$$M(<r) = M_* \frac{\gamma\left[n(3-p), b\left(\frac{r}{R_e}\right)^{1/n}\right]}{\Gamma[n(3-p)]} \tag{2.42}$$

and the normalizing density ρ_0 is unambiguously set by comparing Equations (2.40) and (2.42) to

$$\rho_0 = \frac{M_*}{4\pi R_e^3} \frac{b^{n(3-p)}}{n\,\Gamma[n(3-p)]}. \tag{2.43}$$

Yet, the gravitational potential generated by an arbitrary spherically symmetric distribution $\rho(r)$ can be written in the general form

$$\Phi(r) = \Phi(r_0) + \frac{GM(<r_0)}{r_0} - \frac{GM(<r)}{r} - 4\pi G \int_r^{r_0} dx\, \rho(x) x. \tag{2.44}$$

Although the stellar distribution is formally untruncated, its enclosed mass converges to the finite value M_* as $r \to +\infty$, allowing for the standard normalization $\Phi(\infty) = 0$ and $r_0 \to +\infty$. In this way, Equation (2.44) reduces to

$$\Phi(r) = -\frac{GM(<r)}{r} - 4\pi G \int_r^\infty dx\, \rho(x) x. \tag{2.45}$$

Upon substituting the Prugniel–Sérsic density profile reported in Equation (2.37) into Equation (2.45), the integral can be evaluated in terms of the upper incomplete gamma function,

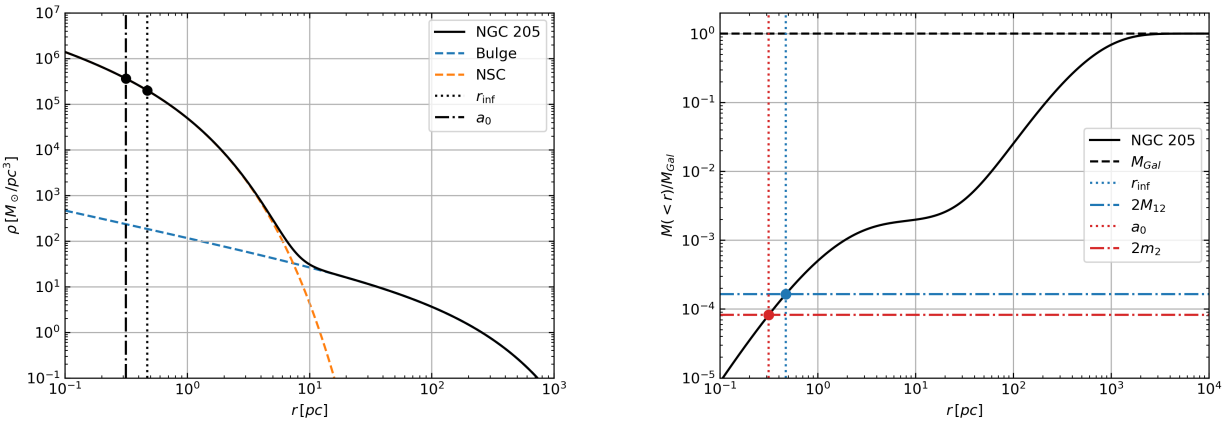
$$\Gamma(a,x) = \int_x^\infty dt, e^{-t} t^{a-1}, \tag{2.46}$$

yielding

$$\Phi(r) = -\frac{GM_*}{r}\left\{\frac{\gamma\left[n(3-p), Z\right]}{\Gamma\left[n(3-p)\right]} + Z^n \frac{\Gamma\left[n(2-p), Z\right]}{\Gamma\left[n(3-p)\right]}\right\}. \tag{2.47}$$

We can now turn to the suite of numerical experiments. Here, we present the evolutionary tracks obtained by evolving equal-mass IMBH binaries, with component masses $m_1 = m_2 =$

$4 \times 10^4\, M_\odot$[1], within a stellar environment reproducing the oservational features of NGC 205 (F. M. Khan and Holley-Bockelmann 2021). The IMBHs are initially placed at a separation a_0, as defined in Equation (2.32), and the simulations are performed for five different initial eccentricities, $e = (0.01, 0.25, 0.50, 0.75, 0.99)$.
At the end of § 2.1, we described the prescription proposed by Sesana and F. M. Khan (2015) for applying the self-consistent Quinlan–Sesana hybrid model to arbitrary spherically symmetric, isotropic stellar backgrounds. Other than the binary hardening radius a_h, it requires only the additional computation of the stellar mass density ρ_{inf} and velocity dispersion σ_{inf} at the binary influence radius r_{inf}, both set by the IMBH binary and stellar environment.
From Equation (2.43), the normalizing densities of the NGC 205 stellar bulge and NSC are found to be $\rho_{0,Bulge} = 2.95\, M_\odot/\mathrm{pc}^3$ and $\rho_{0,NSC} = 4.80 \times 10^5\, M_\odot/\mathrm{pc}^3$, respectively. These values, along with the effective radius R_e and the Sersic index n, fully specify the corresponding Sersic stellar mass density and cumulative mass profiles, from which those of NGC 205 can then be reconstructed. These profiles are shown in Figure 2.7 (a)-(b).



(a) Mass density profiles of the NGC 205 stellar components: bulge (blue dashed), NSC (orange dashed), and the total galaxy (black solid). Black vertical lines mark the binary influence radius (dotted) and the initial binary separation a_0 (dash-dotted).

(b) Cumulative stellar mass profile of NGC 205, normalized to the total stellar mass M_{Gal}. Vertical dotted lines mark the characteristic scales r_{inf} (blue) and a_0 (red), while horizontal dash-dotted lines of the same colours indicate the corresponding mass values $2M_{12}$ and $2m_2$, respectively.

Figure 2.7: NGC 205 mass density profile, together with those of its stellar components, (on left) and cumulative mass profile (on right).

As shown in Figure 2.7(b), the characteristic lengthscales r_{inf} and a_0 are determined from the cumulative mass profile by numerically solving equation

$$M(< r_{inf}) - 2M_{12} = 0 \tag{2.48}$$

and

$$M(< a_0) - 2m_2 = 0 \tag{2.49}$$

respectively. The former solution, i.e. r_{inf}, is then used to derive ρ_{inf} as highlighted in Figure 2.7 (a).
At this stage, the remaining quantities required to fully specify all the terms in Equations (2.20)

[1]This value is taken as potentially expected in a galaxy with the mass of NGC 205. It is not meant to represent specifically the case of NGC 205, for which, as we have pointed out above, the upper limit on the mass of the central IMBH is $3.8\, M_\odot$.

are the stellar velocity dispersion at the binary influence radius, σ_{inf}, and the hardening radius, a_h, which is given by (Quinlan 1996)

$$a_h = \frac{Gm_2}{4\sigma_{inf}}. \tag{2.50}$$

A more detailed discussion of the stellar velocity dispersion, along with its application to the construction of equilibrium N-body initial conditions, is deferred to Chapter 3. In the present context, we limit ourselves to using the radial velocity dispersion

$$\sigma^2(r) = \frac{1}{\rho(r)} \int_r^\infty ds \, \frac{GM(<s)}{s^2} \rho(s), \tag{2.51}$$

holding for spherically symmetric and isotropic stellar system.
By combining the previously reconstructed stellar mass density and cumulative mass profiles, Equation (2.51) returns the function $\sigma = \sigma(r)$, which is displayed in Figure 2.8 normalized to the circular speed $V_c(r) = \sqrt{GM(<r)/r}$.

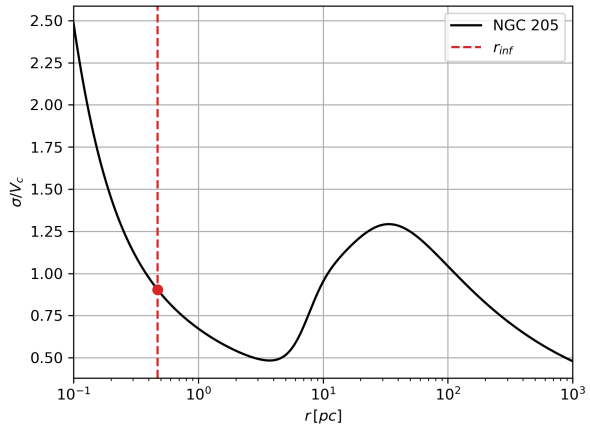


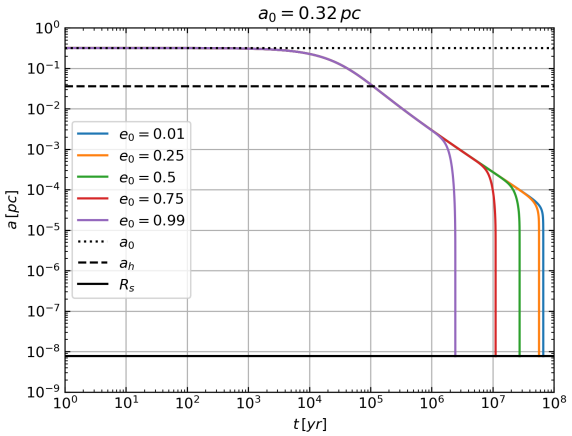
Figure 2.8: NGC 205 velocity dispersion profile (black solid line), normalized to the circular speed $V_c = \sqrt{GM(r)/r}$. The red vertical line marks the binary influence radius r_{inf}.

The main hybrid model parameters, obtained according the operations illustrated above and used for our numerical experiments, are conveniently listed in Table 2.5.

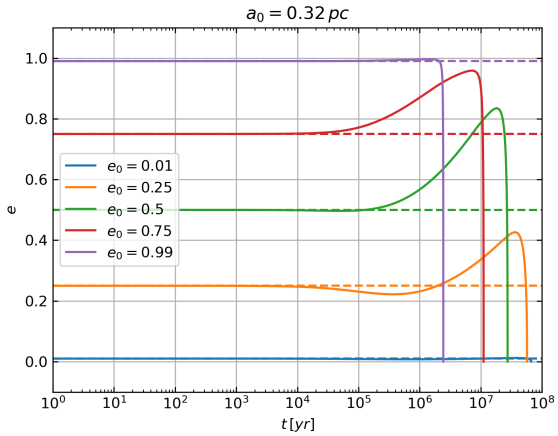
r_{inf} [pc]	a_0 [pc]	a_h [pc]	ρ_{inf} [$M_\odot$/pc^3]	σ_{inf} [km/s]
0.47	0.32	0.036	2.00	34.5

Table 2.5: NGC 205 hybrid model parameters. Column 1: Binary influence radius. Column 2: Binary initial separation. Column 3: Binary hardening radius. Column 4: Stellar mass density at binary influence radius. Column 5: Stellar velocity dispersion at binary influence radius.

Finally, the numerical integration of Equations (2.20), performed following the procedure described in § 2.2, yields the evolutionary tracks $(t, a(t))$ and $(t, e(t))$, along with the corresponding coalescence timescales for each numerical experiment. The results of this operation are shown in Figure 2.9.



(a) Binary semimajor axis time evolution for five different initial eccentricities $e_0 = (0.01, 0.25, 0.5, 0.75, 0.99)$ (blue, orange, green, red, and violet, blue solid lines, respectively). Black horizontal lines indicate the common binary initial separation (dotted), hardening radius (dashed) and the binary R_s (solid).



(b) Binary eccentricity time evolution for five, different initial eccentricities, $e_0 = (0.01, 0.25, 0.5, 0.75, 0.99)$ (blue, orange, green, red, and violet, blue solid lines, respectively). Horizontal dashed lines of the corresponding colours indicate the initial eccentricity of each the numerical experiment.

Figure 2.9: NGC 205 IMBH binary evolutionary tracks $(t, a(t))$ in panel (a) and $(t, e(t))$ in panel (b) for five initial eccentricities, $e_0 = (0.01, 0.25, 0.5, 0.75, 0.99)$ and binary initial separation $a_0 = 0.32\,pc$.

The obtained evolutionary tracks show a clear dependence of the coalescence timescale on the initial eccentricity of the IMBH binary. Specifically, we find $\tau_{coal} = (67.0, 57.1, 27.5, 11.1, 2.4)$ Myr for $e_0 = (0.01, 0.25, 0.50, 0.75, 0.99)$, respectively. The progressive decrease in τ_{coal} with increasing e_0 reflects the tendency of initially more eccentric binaries to reach higher eccentricities during the stellar hardening phase, causing the GW-driven regime to become dominant at an earlier stage of the evolution. This trend is consistent with the results of Sesana (2010), who reported the same dependence on the initial eccentricity in a series of numerical experiments based on the same hybrid model adopted in this analysis, considering BH binaries evolving in stellar cusp profiles as defined in Equation (2.28). Moreover, the NSC substantially increases the central stellar density, accelerating binary evolution relative to the case discussed in Section 2.2. Consequently, all resulting coalescence timescales $\tau_{\rm coal}$ are well below the Hubble time, $t_H \approx 14, \mathrm{Gyr}$.
We extend the analysis in the next Section by examining the evolutionary tracks of an IMBH binary embedded in a model of a present-day, non-nucleated dwarf galaxy and associated coalescence timescale.

2.4 Evolution of an IMBH Binary in a Non-Nucleated Present-Day Dwarf Galaxy

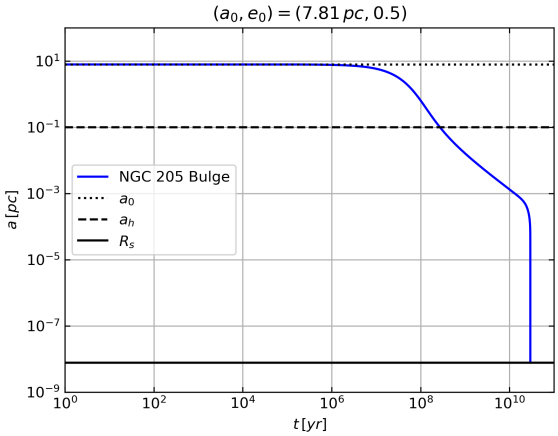
The study presented in the previous Section showed that the presence of a dense, stellar environment, such as a nuclear star cluster at the centre of a dwarf galaxy, makes more efficient the IMBH binary shrinking through close stellar encounters, leading to a shortening of the coalescence timescale to values of up to a few tens of Myr. This result confirms, as found by F. M. Khan and Holley-Bockelmann (2021), that nucleated dwarf galaxies may harbour GW-loud sources potentially detectable by LISA or ET, even if the pairing of the two IMBHs occurs in

the present-day Universe. Nevertheless, the same empirical relations (see, e.g., Magorrian et al. 1998; Maiolino et al. 2024) suggesting the presence of IMBHs at the centres of nucleated dwarf galaxies also provide the same prediction for non-nucleated galaxies, assuming a negligible NSC mass compared to the bulge mass. In this Section, we therefore investigate the dynamical evolution of the same IMBH binary studied in § 2.3 within a present-day, non-nucleated galaxy. To this end, without introducing a new galactic model, we take the galactic bulge of NGC 205 as a reference stellar background, whose structural parameters are listed in Table 2.4.
Once the stellar background is specified, the integration of the differential Equations (2.20) requires the specification of the binary initial conditions and the determination of a set of physical quantities that depend on both the binary mass and the properties of the hosting stellar system.
Repeating the same procedure described in § 2.2-2.3, we determine, in this order, the IMBH binary influence radius r_{inf}, the corresponding stellar density ρ_{inf} and stellar velocity dispersion σ_{inf}, and the binary hardening radius a_h. The relevant quantities entering the hybrid model for this numerical experiment are reported in Table 2.6.

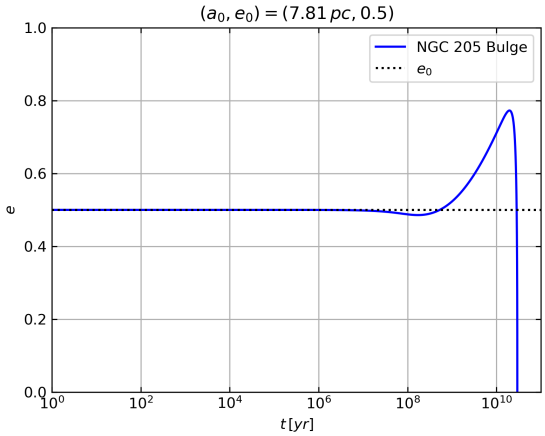
r_{inf} [pc]	a_0 [pc]	a_h [pc]	ρ_{inf} [$M_\odot$/pc^3]	σ_{inf} [km/s]
10.5	7.81	0.10	25.4	20.8

Table 2.6: NGC 205 stellar bulge hybrid model parameters. Column 1: Binary influence radius. Column 2: Binary initial separation. Column 3: Binary hardening radius. Column 4: Stellar bulge mass density at binary influence radius. Column 5: Stellar bulge velocity dispersion at binary influence radius.

For this numerical experiment, the hybrid-model differential equations are therefore solved by taking advantage of the same RK4 scheme and integration timestep adopted above. The solution is thus obtained numerically from the adopted initial conditions $(a_0, e_0) = (7.81\,\mathrm{pc}, 0.5)$, where a_0 is given by Equation (2.32), down to R_s.
Figure 2.10 (a)-(b) show the predicted evolution of the IMBH binary semimajor axis and eccentricity, respectively, for this setup.



(a) NGC 205-like stellar bulge without NSC: IMBH binary semimajor-axis evolution (blue, solid line). Black horizontal lines indicate the binary initial separation (dotted), the hardening radius (dashed), and the binary R_s (solid).



(b) NGC 205-like stellar bulge without NSC: IMBH binary eccentricity time evolution (blue, solid line).
The black, horizontal line indicate the adopted binary initial eccentricity $e_0 = 0.5$.

Figure 2.10: Evolutionary tracks $(t, a(t))$ in panel (a) and $(t, e(t))$ in panel (b) of an IMBH binary of masses $m_1 = m_2 = 4 \times 10^4\,M_\odot$ evolving within a NGC 205-like stellar bulge without NSC, starting from the initial condition $(a_0, e_0) = (7.81\,\mathrm{pc}, 0.5)$.

This numerical experiment shows that the lower stellar density at the IMBH binary influence radius weakens the stellar-scattering-driven phase and prolongs the binary evolution compared to the nucleated case analysed in § 2.3. As a result, the coalescence timescale is approximately three orders of magnitude longer than the corresponding predictions for the nucleated dwarf galaxies. In particular, as shown by the evolutionary tracks in Figure 2.10, we obtain $t_{coal} = 22.5\,\mathrm{Gyr}$, significantly exceeding the Hubble time, $t_H \approx 14\,\mathrm{Gyr}$, as pointed out by F. M. Khan, Javed, et al. (2024). Therefore, IMBH binaries pairing within the present-day universe and embedded in non-nucleated dwarf galaxies are predicted to not reach coalescence within the current age of the Universe. However, it is interesting to extend the analysis to non-nucleated galaxies models at higher redshift, which, at fixed stellar mass, are expected to be progressively more compact and thus characterised by higher stellar density and longer cosmic times available for the binary to evolve. This will be the main focus of the following Section.

2.5 Evolution of an IMBH Binary in a Family of Non-Nucleated High-Redshift Galaxies

While the analysis presented in § 2.4 appears to disfavour non-nucleated dwarf galaxies as potential hosts of GW sources detectable by LISA and ET, at least in the local Universe, such environments may still promote IMBH binaries coalescence within $z = 0$ when the effects of cosmic size evolution are taken into account. In particular, galaxies of a given mass are expected to have had smaller effective radii at higher redshifts, hence, potentially compensating for the absence of a NSC, when dealing with non-nucleated dwarfs. In this Section, we follow the cosmic size evolution empirical relation derived by Ono et al. (2023), according to which the effective radius $R_e = R_e(z)$ evolves with redshift as

$$R_e(z) = R_e(z = 0)(1 + z)^{-1.22}. \tag{2.52}$$

Therefore, we perform a suite of numerical experiments employing the Quinlan–Sesana hybrid model, in which the cosmic size evolution relation expressed in Equation (2.52) is applied building a family of galaxy models representing high-z dwarfs with stellar mass similar to the stellar bulge of NGC 205. Specifically, all simulations consider an equal-mass IMBH binary with $m_1 = m_2 = 4 \times 10^4\, M_\odot$, initially separated by $a_0 = 10$ pc and embedded in the corresponding Sérsic profile. The redshift range is sampled at 25 equally spaced values between $z = 0$ and $z = 12$, while the initial eccentricity is varied over the values $e_0 = (0.1, 0.3, 0.5, 0.7, 0.9)$, resulting in 125 indipendent runs.
For each galaxy model, corresponding to one sampled point of the reshift space, the effective radius $R_e(z)$ is thus computing according to Equation (2.52), while the normalizing mass density ρ_0 is derived from Equation (2.43) by assuming the stellar mass uquals to $M_{bulge} = 97.2 \times 10^7\, M_\odot$, as reported in Table 2.4.
Figure 2.11 compares the stellar mass density profiles, together with the corresponding influence radii, of a subsample of the reconstructed galaxy models with that of the NGC 205 NSC, which predominantly determined the influence radius of the overall galaxy as discussed in § 2.3. Only reconstructed galaxies at $z \approx 10$, or higher, seem to have influence radii, and associated stellar density, comparable to those of the nuclear star cluster.

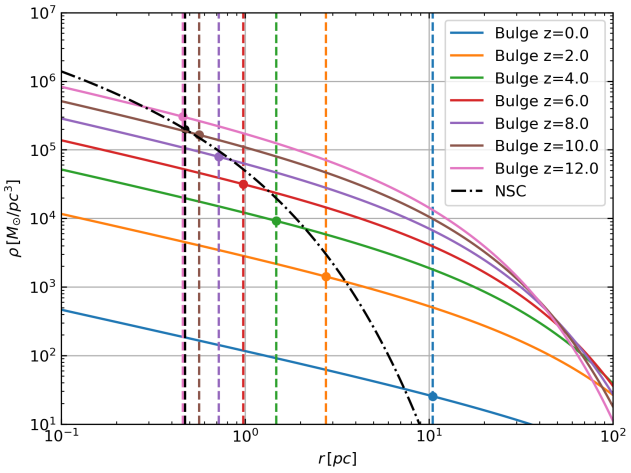


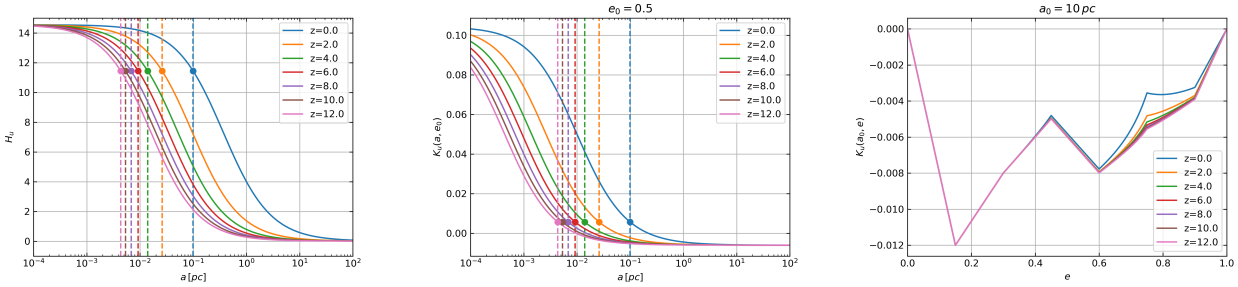
Figure 2.11: Mass density profiles of a representative subsample of galaxy models at $z = (0, 2, 4, 6, 8, 10, 12)$ (solid blue, orange, green, red, violet, brown, and pink lines, respectively), together with the NGC 205 NSC profile (black dash-dotted line). The corresponding influence radii are indicated by vertical dashed lines of the same colours.

Moving forward, repeating the same procedure extensively described in the previous Sections, for each density profile we determine the cumulative mass profile $M(< r)$ by using Equation (2.42), the influence radius r_{inf} by numerically solving Equation (2.48), the mass density at the binary influence radius ρ_{inf} by plugging r_{inf} into the stellar density profile, the velocity dispersion at the binary influence radius σ_{inf} by evaluating Equation (2.51) at r_{inf} and finally substituing it within Eqaution (2.50) to derive the hardening radius a_h.
In Table 2.7 are listed all the parameters mentioned above for each galaxy model.

z	R_e [pc]	ρ_0 [$M_\odot$/pc^3]	n	r_{inf} [pc]	ρ_{inf} [$M_\odot$/pc^3]	a_h [pc]	σ_{inf} [km/s]
0	516	2.95	1.4	10.52	2.54×10^1	0.100	20.8
0.5	314.6	1.30×10^1	1.4	6.41	1.12×10^2	0.061	26.6
1	221.5	3.73×10^1	1.4	4.51	3.21×10^2	0.043	31.7
1.5	168.7	8.43×10^1	1.4	3.44	7.27×10^2	0.033	36.3
2	135.1	1.64×10^2	1.4	2.75	1.42×10^3	0.026	40.6
2.5	111.9	2.89×10^2	1.4	2.28	2.49×10^3	0.022	44.6
3	95.1	4.71×10^2	1.4	1.94	4.06×10^3	0.018	48.4
3.5	82.4	7.25×10^2	1.4	1.68	6.25×10^3	0.016	52.0
4	72.4	1.07×10^3	1.4	1.48	9.20×10^3	0.014	55.4
4.5	64.5	1.51×10^3	1.4	1.31	1.30×10^4	0.012	58.8
5	58.0	2.08×10^3	1.4	1.18	1.79×10^4	0.011	62.0
5.5	52.6	2.78×10^3	1.4	1.07	2.40×10^4	0.010	65.1
6	48.0	3.65×10^3	1.4	0.98	3.15×10^4	0.009	68.1
6.5	44.2	4.70×10^3	1.4	0.90	4.05×10^4	0.009	71.0
7	40.8	5.95×10^3	1.4	0.83	5.14×10^4	0.008	73.9
7.5	37.9	7.43×10^3	1.4	0.77	6.41×10^4	0.007	76.6
8	35.4	9.16×10^3	1.4	0.72	7.90×10^4	0.007	79.4
8.5	33.1	1.12×10^4	1.4	0.67	9.63×10^4	0.006	82.0
9	31.1	1.35×10^4	1.4	0.63	1.16×10^5	0.006	84.6
9.5	29.3	1.61×10^4	1.4	0.60	1.39×10^5	0.006	87.2
10	27.7	1.91×10^4	1.4	0.56	1.65×10^5	0.005	89.7
10.5	26.2	2.25×10^4	1.4	0.53	1.94×10^5	0.005	92.2
11	24.9	2.63×10^4	1.4	0.51	2.26×10^5	0.005	94.6
11.5	23.7	3.05×10^4	1.4	0.48	2.63×10^5	0.005	97.0
12	22.6	3.52×10^4	1.4	0.46	3.04×10^5	0.004	99.3

Table 2.7: Structural parameters (left of the double vertical line) and hybrid-model parameters (right of the double vertical line) for 25 galaxy models identified with the corresponding redshift value. Column 1: Redshift. Column 2: Effective radius. Column 3: Sersic profile normalization density. Column 4: Sersic index. Column 5: Binary influence radius. Column 6: Stellar mass density at binary influence radius. Column 7: Binary hardening radius. Column 8: Stellar velocity dispersion at binary influence radius.

It is worth noting that the parameters $a_{0,H}$ and $a_{0,K}$ entering the parametrization of H_u and K_u in Equations (2.18) and (2.19) are expressed in units of the binary hardening radius. Therefore, despite the primary and secondary IMBH masses are fixed across all the galaxies considered, the shapes of $H_u(a)$ and $K_u(a, e)$ vary from one galaxy to another. This is because the galaxies have different velocity dispersions at the binary influence radius, and consequently different hardening radii. This behaviour is clearly illustrated in Figure 2.12(a)-(b)-(c).



(a) Dimensionless stellar hardening rate H_u of a representative subsample of galaxy models at $z = (0, 2, 4, 6, 8, 10, 12)$ (solid blue, orange, green, red, violet, brown, and pink lines, respectively). The corresponding hardening radii are indicated by vertical dashed lines of the same colours.

(b) Dimensionless eccentricity growth rate $K_u(a, 0.5)$ of a representative subsample of galaxy models at $z = (0, 2, 4, 6, 8, 10, 12)$ (solid blue, orange, green, red, violet, brown, and pink lines, respectively). The corresponding hardening radii are indicated by vertical dashed lines of the same colours.

(c) Dimensionless eccentricity growth rate $K_u(a, e)$, at fixed $a = 10$ pc, of a representative subsample of galaxy models at $z = (0, 2, 4, 6, 8, 10, 12)$ (solid blue, orange, green, red, violet, brown, and pink lines, respectively). The boundary conditions $K_u(a, 0) = K_u(a, 1) = 0$ have been included.

Figure 2.12: Dimensionless stellar hardening and eccentricity growth rates for a representative subsample of galaxy models at $z = (0, 2, 4, 6, 8, 10, 12)$. Panels (a), (b), and (c) show $H_u(a)$, $K_u(a, 0.5)$, and $K_u(10\,\mathrm{pc}, e)$, respectively.

Finally, Figure 2.13 shows the evolutionary tracks $(t, a(t))$ and $(t, e(t))$ for the same representative subsample of galaxies considered above, exploring the full range of initial eccentricities adopted in this analysis, $e_0 = (0.1, 0.3, 0.5, 0.7, 0.9)$, all starting at initial separation $a_0 = 10\,\mathrm{pc}$.

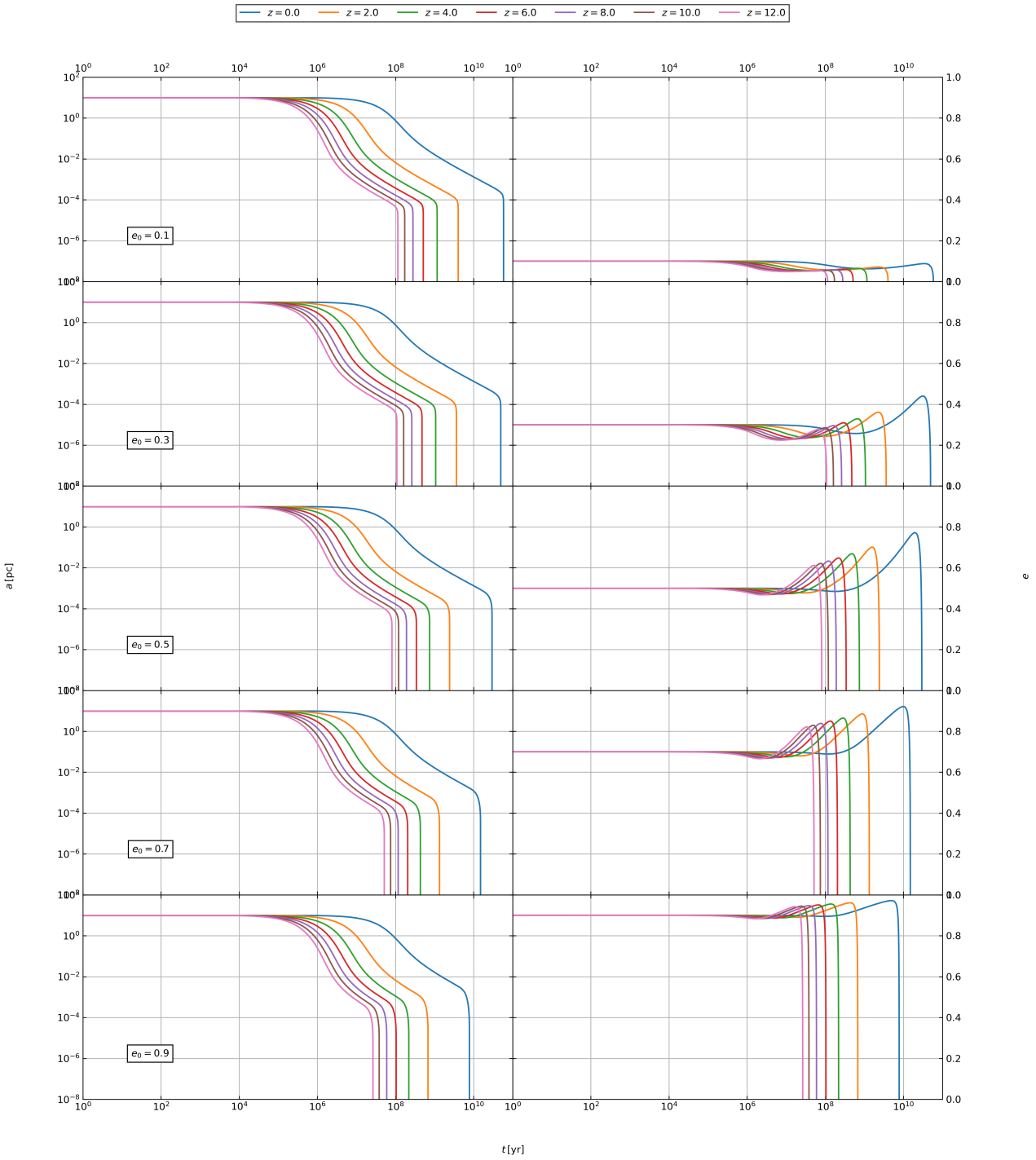


Figure 2.13: Evolutionary tracks of IMBH binaries evolving within a representative subsample of galaxy models. The left and right columns show the binary semimajor axis $a(t)$ and eccentricity $e(t)$, respectively, with each pair of panels in a given row corresponding to the evolution of the same galaxy model. The five rows, from top to bottom, correspond to initial eccentricities $e_0 = (0.1, 0.3, 0.5, 0.7, 0.9)$, indicated by the panels on the left. Coloured curves correspond to galaxy models at $z = (0, 2, 4, 6, 8, 10, 12)$ (blue, orange, green, red, violet, brown, and pink, respectively).

The 125 runs performed to explore the redshift and initial eccentricity parameter spaces provide, in addition to an equal number of pairs of evolutionary tracks $(t, a(t))$ and $(t, e(t))$, an estimate of the coalescence time t_{coal} for each sampled configuration. Here, t_{coal} is defined as the time elapsed from the beginning of the numerical integration $t_0 = 0$, corresponding to the

initial binary conditions (a_0, e_0), to the IMBH binary coalescence, assumed to occur at $a = R_s$. Whether a given galaxy model and set of binary initial conditions can host a GW-loud source detectable by LISA or ET in the present-day Universe depends not only on the estimated t_{coal}, but also on the redshift of the IMBH pairing, which is same used to build the hosting galaxy model.

In standard ΛCDM cosmology (see, e.g. Coles and Lucchin 1995), within which the Friedmann–Lemaître–Robertson–Walker (FLRW) metric provides the appropriate description of the large-scale homogeneous and isotropic Universe, the Hubble function $H(z)$ describes the expansion rate of the Universe as a function of redshift and can be written as

$$H(z) = H_0\sqrt{\Omega_k(1+z)^2\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + +\Omega_\Lambda}, \tag{2.53}$$

where H_0 is the present-day Hubble constant, Ω_k, Ω_m, Ω_r, and Ω_Λ are the present-day dimensionless density parameters associated with matter, radiation, spatial curvature, and dark energy, respectively. In the FLRW cosmological model, the cosmic time $t(z)$ corresponding to a given redshift z can be obtained from the Hubble function through the standard redshift–time relation

$$t(z) = \int_z^\infty \frac{dz'}{(1+z')H(z')}. \tag{2.54}$$

Adopting the cosmological parameters from Planck Collaboration et al. (2020), namely $H_0 = 67.4, \mathrm{km}, \mathrm{s}^{-1}, \mathrm{Mpc}^{-1}$ and $\Omega_m = 0.315$, and assuming a spatially flat ΛCDM cosmology, we compute the cosmic time $t(z)$ for each galaxy model using Equation (2.54). The radiation contribution is neglected at the redshifts considered, and $\Omega_\Lambda = 1 - \Omega_m = 0.685$, yielding a present-day age of the Universe of $t_{\mathrm{age}} = t(z = 0) = 13.8, \mathrm{Gyr}$. These estimates allow us to convert the coalescence times obtained from the numerical integrations, measured from the initial epoch of each model, into the corresponding cosmic times.

Both the coalescence times and the corresponding cosmic times of binary coalescence are reported in Table 2.8 (a) and (b), respectively, for each of the 125 runs.

z \ e_0	0.1	0.3	0.5	0.7	0.9
0	58.3	49.5	29.6	15.0	7.79
0.5	21.7	18.9	11.8	6.14	3.17
1	10.8	9.53	6.13	3.26	1.67
1.5	6.28	5.60	3.68	2.00	1.02
2	4.03	3.62	2.42	1.34	0.68
2.5	2.77	2.51	1.70	0.95	0.48
3	2.00	1.82	1.25	0.71	0.36
3.5	1.51	1.38	0.95	0.55	0.28
4	1.17	1.07	0.75	0.43	0.22
4.5	0.92	0.85	0.60	0.35	0.18
5	0.75	0.69	0.49	0.29	0.15
5.5	0.62	0.57	0.41	0.24	0.12
6	0.52	0.48	0.34	0.21	0.10
6.5	0.44	0.40	0.29	0.18	0.09
7	0.37	0.35	0.25	0.15	0.08
7.5	0.32	0.30	0.22	0.13	0.07
8	0.28	0.26	0.19	0.12	0.06
8.5	0.25	0.23	0.17	0.10	0.05
9	0.22	0.20	0.15	0.09	0.05
9.5	0.19	0.18	0.13	0.08	0.04
10	0.17	0.16	0.12	0.08	0.04
10.5	0.15	0.15	0.11	0.07	0.03
11	0.14	0.13	0.10	0.06	0.03
11.5	0.13	0.12	0.09	0.06	0.03
12	0.12	0.11	0.08	0.05	0.03

(a) Coalescence timescales, in Gyr, for the galaxy models evolved to different redshifts z and for different initial binary eccentricities e_0. Each entry gives the time elapsed from the beginning of the simulation, $t_0 = 0$, to binary coalescence.

z \ e_0	0.1	0.3	0.5	0.7	0.9
0	72.1	63.3	43.4	28.8	21.6
0.5	30.3	27.5	20.4	14.7	11.8
1	16.7	15.4	12.0	9.11	7.52
1.5	10.5	9.87	7.95	6.27	5.29
2	7.31	6.90	5.70	4.61	3.96
2.5	5.38	5.12	4.31	3.57	3.10
3	4.15	3.97	3.39	2.85	2.50
3.5	3.30	3.17	2.75	2.35	2.08
4	2.70	2.61	2.29	1.97	1.76
4.5	2.26	2.18	1.93	1.68	1.51
5	1.92	1.86	1.66	1.46	1.32
5.5	1.66	1.61	1.45	1.28	1.16
6	1.45	1.41	1.27	1.14	1.03
6.5	1.27	1.24	1.13	1.01	0.93
7	1.13	1.11	1.01	0.91	0.84
7.5	1.02	1.00	0.91	0.83	0.76
8	0.92	0.90	0.83	0.76	0.70
8.5	0.83	0.82	0.76	0.69	0.64
9	0.76	0.75	0.69	0.64	0.59
9.5	0.70	0.69	0.64	0.59	0.55
10	0.64	0.63	0.59	0.55	0.51
10.5	0.60	0.59	0.55	0.51	0.48
11	0.55	0.55	0.51	0.48	0.45
11.5	0.52	0.51	0.48	0.45	0.42
12	0.48	0.48	0.45	0.42	0.39

(b) Cosmic coalescence times, in Gyr, for the galaxy models evolved to different redshifts z and for different initial binary eccentricities e_0. Each entry gives the cosmic time at which the binary reaches coalescence, measured from the initial time $t_0 = t(z)$.

Table 2.8: Coalescence times of the IMBH binaries for the 25 galaxy models as a function of redshift z and initial binary eccentricity e_0. Panel (a) reports the coalescence timescale measured from $t_0 = 0$, while panel (b) gives the corresponding cosmic time of coalescence.

Figures 2.14 and 2.15 show how the cosmic time and redshift of coalescence, respectively, depend on the binary pairing redshift and initial eccentricity, highlighting the combinations of these parameters for which coalescence occurs within the present age of the Universe ($t_H \approx 13.8\,\mathrm{Gyr}$) and before the present epoch ($z = 0$).

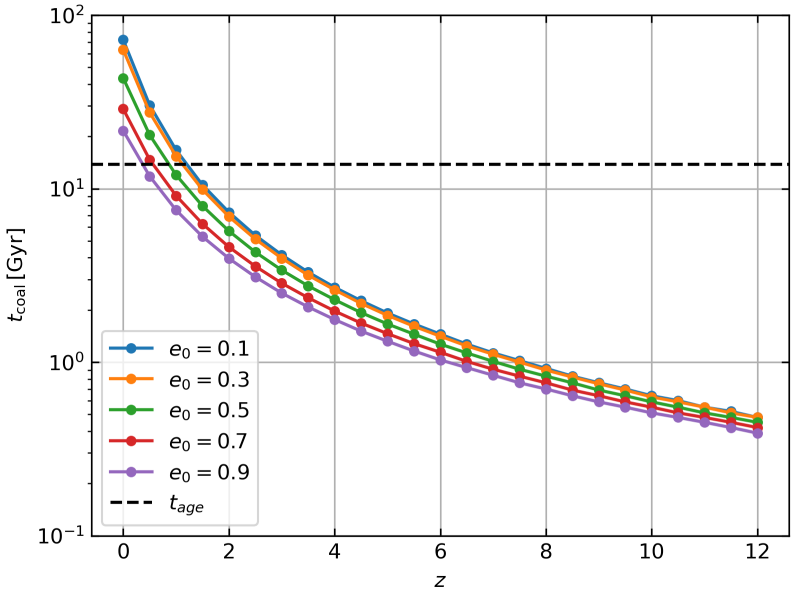


Figure 2.14: Cosmic coalescence time as a function of the redshift of IMBH binary pairing for the 125 simulations. The solid blue, orange, green, red, and violet piecewise lines connect, respectively, the points corresponding to $e_0 = (0.1, 0.3, 0.5, 0.7, 0.9)$ across the redshift sampling. The horizontal black dashed line marks the age of the universe.

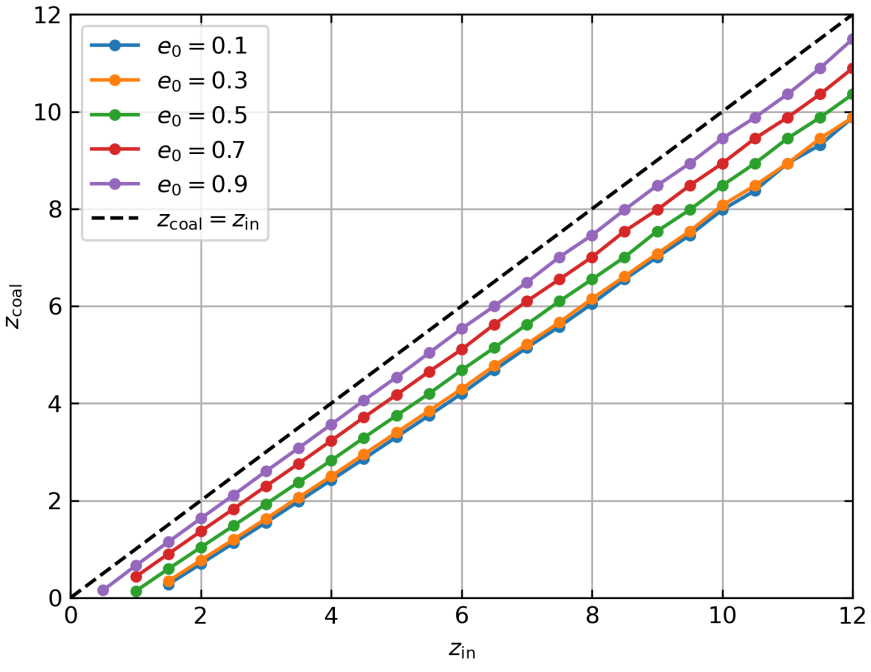


Figure 2.15: Redshift of IMBH binary coalescence as a function of the redshift at binary formation for the 125 simulations. The solid blue, orange, green, red, and violet lines connect the points corresponding to $e_0 = (0.1, 0.3, 0.5, 0.7, 0.9)$, respectively, across the sampled redshifts. The black dashed line marks the bisector of the first quadrant.

In this chapter, we have investigated the self-consistent hybrid model developed by Quinlan (1996) and Sesana, Haardt, et al. (2006) to describe the dynamical evolution of massive black hole binaries within a fixed stellar background. Taking advantage of the versatility of

this framework, we have performed a suite of numerical experiments aimed at identifying the main features of IMBH binary evolutionary tracks and systematically exploring the relevant parameter space. The collection of experiments presented in § 2.2-2.3-2.4-2.5, together with the analysis of the corresponding evolutionary tracks $(t, a(t))$ and $(t, e(t))$, allows us to draw the following conclusions:

- Stellar environments characterised by enhanced central densities, such as nucleated galaxies, promote more efficient dynamical hardening of IMBH binaries, leading to a faster orbital decay.
- Eccentric IMBH binaries generally undergo further eccentricity growth during the stellar-scattering hardening phase. As a consequence, they can enter the GW-dominated regime at larger eccentricities and reach coalescence on shorter timescales.
- Initially circular binaries tend to preserve their low eccentricity throughout both the stellar-driven and GW-driven regimes.
- In sufficiently low-density stellar environments, such as present-day non-nucleated dwarfs,stellar-driven hardening can be too inefficient to bring the binary into the GW-dominated regime within a Hubble time t_H, preventing its coalescence within the present age of the Universe and limiting its detectability by LISA or ET.
- At higher redshifts, non-nucleated dwarf galaxies are sufficiently compact that, even in the absence of a nuclear star cluster, IMBH binaries hosted at their centres can undergo rapid coalescence within the present age of the Universe.

The results obtained in this chapter motivate the next step of our investigation, namely the transition from the semi-analytical hybrid description to direct N-body simulations. This approach will allow us to follow the evolution of the IMBH binary and its surrounding stellar system self-consistently, overcoming some of the main limitations inherent to the adopted hybrid model. In particular, direct N-body simulations will make it possible to follow the mutual dynamical evolution of the binary and the stellar background, rather than prescribing the latter as a fixed external potential, and will open the possibility of investigating the dynamical evolution and eventual merger of the stellar systems themselves.

3

Evolution of IMBH Binaries with KETJU N-Body Simulations

The numerical experiments based on the Quinlan-Sesana hybrid model, presented in Chapter 2, have shown that, once an IMBH binary forms, its coalescence can occur within a Hubble time even in non-nucleated, high-redshift galaxies. These results motivate a further investigation through self-consistent, N-body simulations, which allow the dynamical evolution of the stellar system and the black holes to be followed from the stages preceding binary formation through the subsequent evolution of the binary. This approach also makes it possible to relax some of the simplifying assumptions adopted in the hybrid model, in particular the assumption of a fully populated loss cone and the evolution within a fixed, spherically symmetric stellar background. In this context, the N-body code KETJU provides a particularly suitable framework, as the interactions between the black holes and surrounding stars are resolved without gravitational softening.
The chapter is structured as follows. Section 1 introduces the equilibrium theory of N-body systems and the procedure used to construct equilibrium initial conditions with the software OpOpGadget. Section 2 presents the results of high-resolution, N-body simulations of mergers between two identical systems, each consisting of an NGC 205-like nuclear star cluster hosting an IMBH. The final section presents a suite of N-body simulations investigating how the initial orbital angular momentum of encounters between two high-redshift stellar clumps, each hosting a central IMBH, influences the subsequent dynamical evolution of the black-hole binary, from the pairing and to the final coalescence.

3.1 Dynamical Equilibrium of Isotropic and Spherically Symmetric Stellar Systems

This Section has two main purposes. First, it reviews the equilibrium theory of spherically symmetric, isotropic, collisionless stellar systems. Second, it introduces the main features of OpOpGadget, the software used to generate the N-body initial conditions adopted in this thesis.
The section is divided into three subsections. The first outlines the fundamental equations governing collisionless stellar systems, including the collisionless Boltzmann equation, the Jeans equations, and the Eddington inversion formula. The second describes the numerical methods implemented in OpOpGadget, focusing on the algorithm used to generate N-body initial conditions and the analysis tools provided by the code. The final subsection presents the analysis carried out to assess the dynamical equilibrium of the OpOpGadget initial conditions when evolved with the N-body code KETJU.

3.1.1 Equilibrium Theory of Stellar Systems

The first step towards performing N-body experiments of merging stellar systems hosting IMBHs is the generation of reliable N-body initial conditions that capture the main properties of the system in isolation, when at equilibrium, and reproduce its relevant observables, such as the density profile and stellar velocity dispersion, consistently with those of the corresponding idealized model. This, in turn, requires a well-defined framework for describing the dynamical state of the system.
As discussed in § 1.2, provided that the stellar system is considered on timescales shorter than its two-body relaxation time t_{2b}, collisional effects are negligible, and the stars can be treated as collisionless particles moving in a smooth gravitational potential generated by a continuous mass density distribution $\rho(\mathbf{x}, t)$, smooth proxy of the actual stellar distribution. In this regime, when dealing with stars that are all identical to one another, the dynamical state of the stellar component can be described by a unique one-particle distribution function (DF) $f(\mathbf{x}, \mathbf{v}, t)$ of phase-space (PhS), while the corresponding gravitational potential $\Phi(\mathbf{x}, t)$ is self-consistently determined by the mass distribution $\rho(\mathbf{x}, t)$ through the Poisson equation

$$\nabla^2 \Phi(\mathbf{x}, t) = 4\pi G \rho(\mathbf{x}, t). \tag{3.1}$$

For collisionless systems of this type, the corresponding DFs satisfy the so-called collisionless Boltzmann equation (CBE), which follows from the normalization of the DF and, consequently, from the conservation of the integrated probability.
Building on the discussion in § 1.2, we begin by introducing the normalized DF in phase space. If we indicate with $\mathbf{w} = (\mathbf{q}, \mathbf{p})$ an arbitrary set of canonical PhS coordinates, the DF of a system with total mass M can be normalized such that

$$\int_{\mathbb{R}^6} d\mathbf{w}\, f(\mathbf{w}) = M. \tag{3.2}$$

The CBE follows from requiring the DF to be conserved along the trajectories generated by the potential $\Phi(\mathbf{x}, t)$ associated with the density distribution $\rho(\mathbf{x}, t)$. Taking the total derivative with respect to the time of Equation (3.2) yields

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + \dot{\mathbf{w}} \cdot \frac{\partial f}{\partial \mathbf{w}} = \frac{\partial f}{\partial t} + [f, H] = 0, \tag{3.3}$$

where D/Dt has been used to denote the Lagrangian derivative, the square bracket indicates the Poisson bracket

$$[f, H] = \sum_i \left(\frac{\partial f}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial H}{\partial q_i} \right), \tag{3.4}$$

and $H(\mathbf{q}, \mathbf{p})$ is the Hamiltonian function of the system (see, e.g. Binney and Tremaine 2008). The arbitrariness of the canonical coordinates $\mathbf{w}$ enables us to specialize Equation (1.23) to the inertial Cartesian coordinates $(\mathbf{x}, \mathbf{v})$. With this choice, the Hamiltonian per unit mass becomes

$$H(\mathbf{x}, \mathbf{v}, t) = \frac{1}{2} v^2 + \Phi(\mathbf{x}, t) \tag{3.5}$$

and the Equation (3.3) turns into

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \frac{\partial f}{\partial \mathbf{x}} - \frac{\partial \Phi}{\partial \mathbf{x}} \cdot \frac{\partial f}{\partial \mathbf{v}} = 0. \tag{3.6}$$

Equation (3.6) gives the CBE for a collisionless system in Cartesian coordinates, valid on timescales shorter than the corresponding t_{2b} and expresses the conservation of the phase-space probability density along the orbits generated by the potential $\Phi(\mathbf{x}, t)$. However, it is too general for our purposes, as we are interested in studying the equilibrium of stellar systems, and in particular of spherically symmetric, equilibrium models. It is therefore useful to introduce the following definition: a system composed of N particles is said to be in equilibrium, or stationary, when any quantity $q(\mathbf{x}, t)$ describing the system, including its distribution function (DF), is time-independent, such that $\partial q/\partial t = 0$. The DF of such systems satisfies the so-called stationary CBE

$$\frac{Df}{Dt} = \dot{\mathbf{w}} \cdot \frac{\partial f}{\partial \mathbf{w}} = [f, H] = 0, \tag{3.7}$$

obtained from Equation (3.3) by setting $\partial f/\partial t = 0$.
The solutions to Equation (3.7) are referred to as integrals of motion (Ciotti 2021). Specifically, the function $I(\mathbf{x}, \mathbf{v})$ is an integral of motion for the potential $\Phi(\mathbf{x}, \mathbf{v})$ if it satisfies

$$\frac{D}{Dt}[I(\mathbf{x}(t), \mathbf{v}(t))] = 0 \tag{3.8}$$

along any orbit of the potential $\Phi(\mathbf{x})$ and a relevant example is represented by the Hamiltonian itself of the system, given that

$$[H, H] = 0. \tag{3.9}$$

The comparison between Equation (3.7) and (3.8) reveals that solving the steady-state CBE is equivalent to being an integral of motion; this leads to the formulation of the Jeans theorem (see, e.g., Lynden-Bell 1962; Binney and Tremaine 2008).

Jeans Theorem. – *Any steady-state solution of the collisionless Boltzmann equation depends on the phase-space coordinates only through integrals of motion in the given potential, and any function of the integrals yields a steady-state solution of the collisionless Boltzmann equation.*

It follows from this theorem that, if $I_1, \ldots, I_N$ is a set of N integrals of motion for a given collisionless and stationary system, its DF can be expressed as $f = f(I_1, \ldots, I_N)$, rather than as a general function of phase space. Nevertheless, before examining the conditions under which a closed-form solution to Equation (3.7) can be obtained, let us introduce the main moments of the DF, which are directly related to kinematic observables and will prove useful for this purpose.
The first- and second-order velocity moments of the distribution function f, the mean velocity and the velocity dispersion tensor, respectively, are defined as

$$\overline{v_i}(\mathbf{x}, t) \equiv \frac{1}{\rho(\mathbf{x}, t)} \int_{\mathbb{R}^6} d^3\mathbf{v}\, f v_i \tag{3.10}$$

and

$$\sigma_{ij}^2(\mathbf{x},t) \equiv \frac{1}{\rho(\mathbf{x},t)} \int_{\mathbb{R}^6} d^3\mathbf{v}\, f(v_i - \overline{v_i})(v_j - \overline{v_j}) = \overline{v_i v_j} - \overline{v_i}\,\overline{v_j}. \tag{3.11}$$

These quantities enter the so-called Jeans equations, which follow from taking the first- and second-order velocity moments of Equation (3.3). In Cartesian coordinates they are

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho \overline{v_i}}{\partial x_i} = 0 \tag{3.12}$$

and

$$\frac{\partial \rho \overline{v_i}}{\partial t} + \frac{\partial \rho \overline{v_i v_j}}{\partial x_j} = -\rho \frac{\partial \Phi}{\partial x_i} \tag{3.13}$$

respectively, where the Einstein summation convention over repeated indices is adopted.
Equations (3.12) and (3.13) generalize the mass continuity and momentum equations commonly used in fluid dynamics to collisionless, stellar systems. For a spherically symmetric system, Equation (3.13) can be written in the spherical coordinates system (r, θ, ϕ) as

$$\frac{d(\rho\sigma_r^2)}{dr} + 2\frac{\rho}{r}\beta\sigma_r^2 = -\rho\frac{d\Phi}{dr}, \tag{3.14}$$

where $\beta = 1 - \sigma_t^2/(2\sigma_r^2)$ quantifies the degree of velocity anisotropy, with positive and negative values corresponding to radial and tangential anisotropy, respectively, while $\sigma_t^2 = 2\sigma_\theta^2 = 2\sigma_\phi^2 \equiv \sigma^2$ (see, e.g., W. C. Saslaw 1985, Binney and Tremaine 2008).
A stellar system is defined to be isotropic if $\sigma_r^2 = \sigma_\theta^2 = \sigma_\phi^2$ and thus $\beta = 0$. It is now clear from Equation (3.14) that, for a spherically symmetric and isotropic system, with $\rho = \rho(r)$ and $d\Phi/dr = GM(<r)/r^2$, the 1D velocity dispersion formula, mentioned in § 2.3, is given by Equation (2.51), which we reproduce here for convenience:

$$\sigma^2(r) = \frac{1}{\rho(r)} \int_r^\infty ds\, \frac{GM(<s)}{s^2}\rho(s). \tag{3.15}$$

We can now turn to the analysis of Equation (3.7) and its solutions. To this end, we recall the main properties of an important subclass of equilibrium models, namely ergodic systems, which are the type of systems we aim to initialize in our numerical experiments.
By definition, a system with Hamiltonian H is described by an ergodic DF if the latter depends solely on the Hamiltonian, i.e. $f = f(H)$. It is worth noting that, according to Equation (3.9), all stationary and collisionless systems admit H as an integral of motion. Therefore, the Jeans theorem implies that the DF can, for these systems, be written as $f = f(H, I_2, \ldots, I_N)$, with the ergodic case representing a particular and highly restrictive subclass. A key property of ergodic systems follows from the first-order moments of the corresponding DF.
In the inertial Cartesian frame where stars move under the action of a steady-state potential $\Phi(\mathbf{x})$, the Hamiltonian of the system is an even function of $\mathbf{v}$, causing $\overline{v_x}(\mathbf{x}) = \overline{v_y}(\mathbf{x}) = \overline{v_z}(\mathbf{x})$ to vanish, owing to Equation (3.10). Similarly, the velocity-dispersion tensor can be shown to be isotropic, since an ergodic DF depends on the velocity only through v^2, that is

$$\sigma_{ij}^2(\mathbf{x}) = \sigma^2(\mathbf{x}) = \delta_{ij}\frac{4\pi}{3\rho(\mathbf{x})} \int_0^{+\infty} dv\, f\left(\frac{1}{2}v^2 + \Phi(\mathbf{x})\right) v^4. \tag{3.16}$$

Therefore, any system described by an ergodic DF is necessarily isotropic.
So far, we have focused on several theoretical aspects of N-body systems, with particular emphasis on their equilibrium properties, without yet discussing in detail how the DF is made. However, generating initial conditions (ICs) for equilibrium N-body models requires sampling the DF directly, rather than relying solely on its moments. For globally isotropic stellar systems,

the DF can be recovered from the density profile ρ through Eddington's inversion formula (Eddington 1916), which provides the corresponding PhS distribution in terms of the density and gravitational potential. It reads

$$f(\varepsilon) = \frac{1}{\sqrt{8}\pi^2}\frac{d}{d\varepsilon}\int_{\varepsilon_t}^{\varepsilon}\frac{d\rho}{d\Psi}\frac{d\Psi}{\sqrt{\varepsilon-\Psi}}, \tag{3.17}$$

with $\varepsilon = -H$ and $\Psi = -\Phi$ being the relative energy and potential, respectively, and ε_t the truncation energy of the system.
The theoretical framework developed above allows us to turn a prescribed equilibrium density profile $\rho(r)$ into a self-consistent set of N-body initial conditions. In the following Subsection, we describe how this procedure is implemented for spherically symmetric, isotropic stellar systems by means of the software OpOpGadget.

3.1.2 Generation of N-Body Initial Conditions with OpOpGadget

Having established the theoretical basis for describing dynamical equilibrium in stellar systems, we can now turn to the construction of N-body initial conditions. The goal is to generate a discrete realization of a given stellar system at equilibrium that consistently reproduces its features, while satisfying the equilibrium conditions discussed above. For a given stellar distribution, stellar positions and velocities are generated by means of `OpOpGadget` (see Iorio et al. 2019). This numerical code provides a range of tools for generating N-body initial conditions and analyzing the outputs of N-body simulations. It also includes an extensive library of analytical models for spherically symmetric stellar systems, allowing stellar systems with canonical profiles to be constructed from their specified structural parameters.
The first step in the workflow initialization is to specify the desired model, assign the values of the parameters required by the library model, and choose the number of particles to represent the stellar system. After that, the code proceeds with assigning a set of spherical coordinates (r, θ, ϕ) to each particle by randomly sampling the cumulative mass profile

$$M_*(< r) = 4\pi\int_0^r dy\, y^2\rho(y), \tag{3.18}$$

and stores the extracted mass in the variable $u = np.random.uniform(M_{min} = 0, M_{max} = M, N)$.
Since $M(< r)$ is a monotonically increasing function of radius, it can be inverted to obtain the corresponding radius, $r = M_*^{-1}(u)$, for the N sampled values. Yet, two additional independent random variables, uniformly distributed in the intervals $[-1, 1]$ and $[0, 2\pi]$ respectively, are used to sample the direction cosine and the azimuthal angle, thereby obtaining N sets of spherical coordinates $(r, \theta, \phi.)$, which are then transformed into Cartesian coordinates.
While particle positions are drawn from the cumulative mass distribution, the velocity components are obtained by sampling the DF, assuming the gravitational potential Φ to be generated by the mass distribution associated with the previously initialized particles. Therefore, three random numbers $\mathbf{v}$, lying in the range $[-1, 1]$, allow to temporary assign a normalized velocity to each particle with binding energy $\varepsilon = \Psi(1 - v^2)$. Then, the Von Neumann rejection method is implemented and the normalized velocity is accepted if

$$g \leq \frac{f(\varepsilon)}{f(\varepsilon = \Psi)}, \tag{3.19}$$

where g is another indipendent random number in $[0, 1]$. If the inequality in Equation (3.19) is satisfied, the final velocity vector is obtained by multiplying the normalized vector by the

factor $\sqrt{2\Psi}$. Once this procedure, which can be extended to multi-component stellar systems, is completed, each particle will have its own identification number (ID), mass $m = M_*/N$, position $\mathbf{x}$ and velocity $\mathbf{v}$ (see, e.g., Aarseth 2003).
All numerical experiments of merging between identical stellar systems with a central IMBH that will be presented in the next Sections involve Sersic stellar distribution as described by Prugniel and Simien (1997). However, the Sersic analytic model implemented within OpOpGadget library takes as a reference Lima Neto et al. (1999) de-projection

$$\rho(r) = \rho_0 \left(\frac{r}{a}\right)^{-p} e^{-(r/a)^\nu}, \tag{3.20}$$

where $\nu = n^{-1}$ denotes the inverse of the Sérsic index, while a is the Sérsic scale length, defined such that $\Sigma(a) = \Sigma_0$. These quantities are referred to as the primary parameters and the remaining structural parameters can be derived from them, so that the same density distribution may be equivalently described by different sets of parameters.
To convert from the scale length a to the effective radius R_e, we adopt the following analytical approximation (Lima Neto et al. 1999):

$$\ln R_e/a = \frac{0.6950 - \ln \nu}{\nu} - 0.1789. \tag{3.21}$$

Particular attention should be paid to the fact that the stellar particles to be initialized are required to be in dynamical equilibrium within the combined gravitational potential of both stars and central black hole. The presence of such a massive and compact object is expected to modify the equilibrium configuration of the stellar system compared to the isolated case. Therefore, the initial conditions must account for the gravitational contribution of all components. This is achieved by adopting a multi-component system comprising different components, which are at most in this thesis project: the stellar bulge, the nuclear star cluster and the central IMBH. Once the ICs of a reference, equilibrium system are generated, several methods and analysis tools with which OpOpGadget is equipped be applied. We list below the most relevant ones.
The positions, velocities, masses and IDs of each particle are stored within the object ***Nbody-Model:***, which can be then passed as argument of ***Analysis():***, provided of the following useful methods:

1. `Analysis.com()`: it returns Cartesian position and velocity of the center of mass for the generated N-body ICs.

2. `Analysis.softening_scale(kernel=Gadget)`: it computes the system's softening length ϵ[1], to be used in KETJU N-body simulations employing softened gravity.

3. `Analysis.tdyn(mq=50)`: It provides the dynamical time of the system as

$$T_{dyn} = 0.5\pi\sqrt{\frac{R_h^3}{GM_h}}, \tag{3.22}$$

with M_h being the mass inside the half-mass radius R_h. This estimate has been adjusted to match the definition of the dynamical time at half mass radius adopted here, which is

$$t_{dyn} = \frac{1}{\sqrt{G\bar{\rho}}}, \tag{3.23}$$

with $\bar{\rho}$ indicating the average mass density within R_h.

[1]The next subsection will examine in greater detail which softening length ϵ is computed with OpOpGadget and how its role in KETJU N-body experiments.

Lastly, the ***NbodyModel*** object can be passed to ***Profile(NbodyModel, xmin, xmax)*** to extract the radial profiles of relevant quantities over the range $[x_{\min}, x_{\max}]$. This object provides two methods for characterising the spatial distribution of the generated N-body realisation:

1. `Profile.dens()`: returns the radial grid and the corresponding density profile.

2. `Profile.mass()`: returns the radial grid and the corresponding cumulative mass profile.

As a reference case and standard procedure adopted throughout the numerical experiments presented in the following Sections, we now describe the generation and analysis of the N-body ICs for A NGC 205-like stellar distribution with a central IMBH, which is the same model considered in § 2.3.
First, NGC 205 observational data reported in Table 2.4 are used to derive the Sersic scale-lengths of the stellar bulge and NSC according to Equation (3.21). This yields $a_{bulge} = 1456$ pc and $a_{NSC} = 0.24$ pc, respectively. Then, following the choice adopted by F. M. Khan and Holley-Bockelmann (2021), we set the number of particles used to initialize the stellar bulge and NSC, N_{bulge} and N_{NSC}, such that the BH-to-stellar mass ratio is equal to 160. For an IMBH with mass $4 \times 10^4\, M_\odot$, this results in $N_{bulge} = 3888000$ and $N_{NSC} = 7200$ generated stellar particles. Once the parameters rquired by OpOpGadget library are given for each component, along with the corresponding number of particles to initialize, the N-body ICs are generated by assessing positions, velocities, masses, and particle IDs to all particles.
Figure 3.1 shows the spatial distribution of the stellar particles and the IMBH projected onto the x-y plane, on both the scale of the stellar bulge and that of the NSC.

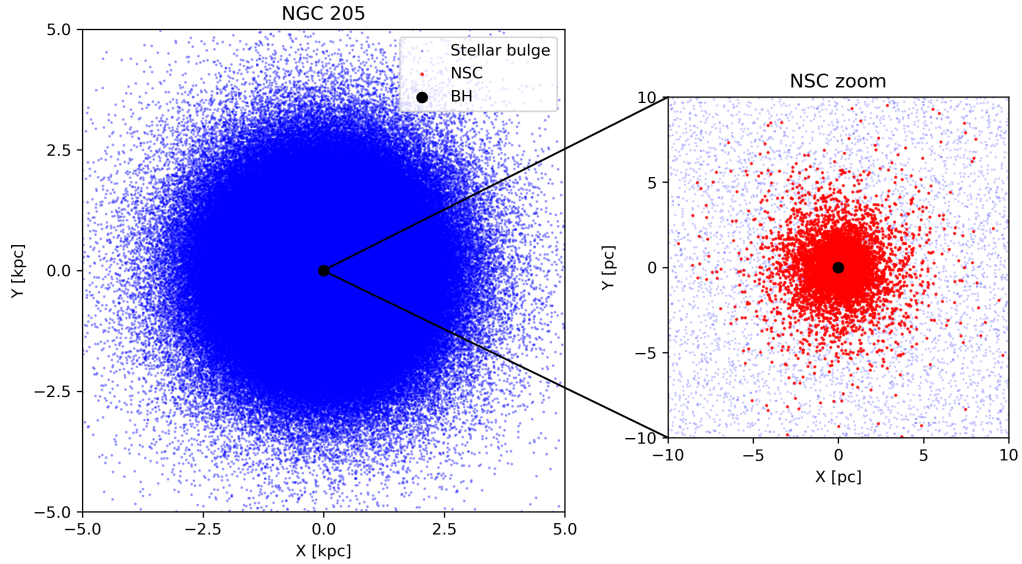


Figure 3.1: Positions of the N-body initial conditions for NGC 205 stellar bulge, NSC, and central IMBH (blue, red, and black dots, respectively), projected onto the x-y plane. The main panel shows the system on the stellar-bulge scale, while the inset provides a zoomed-in view on the scale of the NSC.

Another way of visualizing the generated N-body ICs is provided in Figure 3.2, which shows the three Cartesian components of the position and velocity of the particles in six histograms. As expected, all distributions are centred around 0.

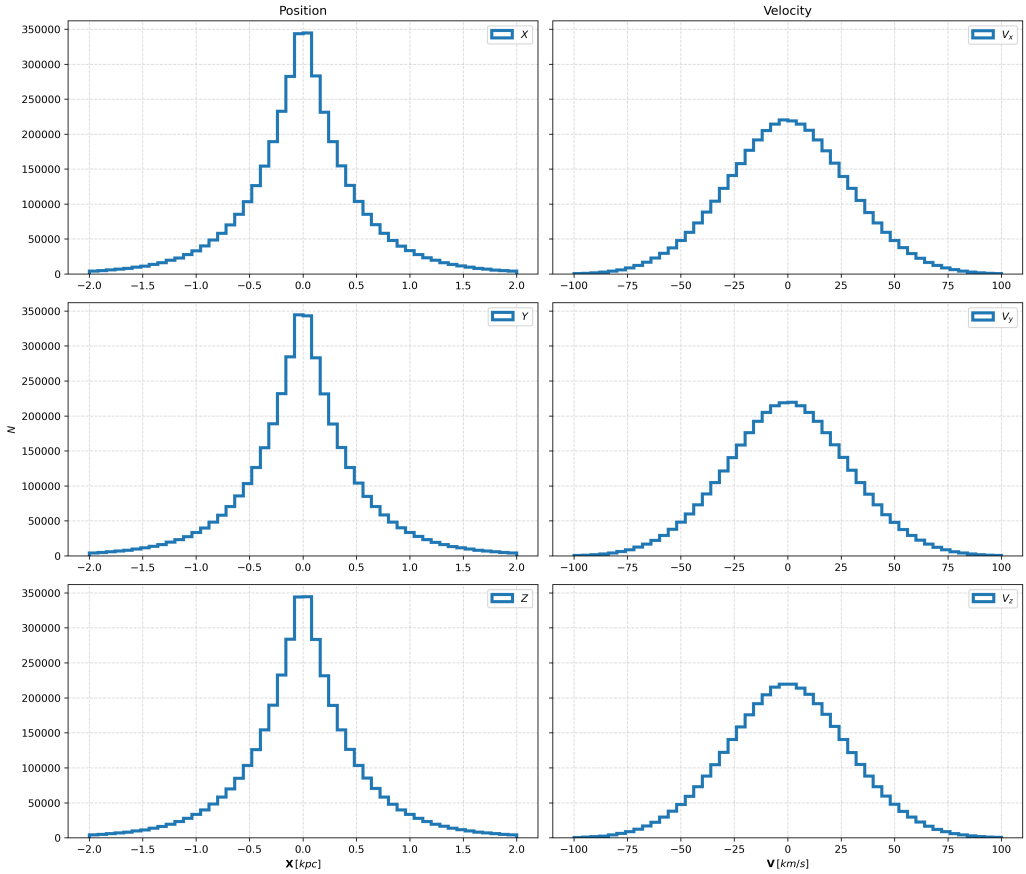
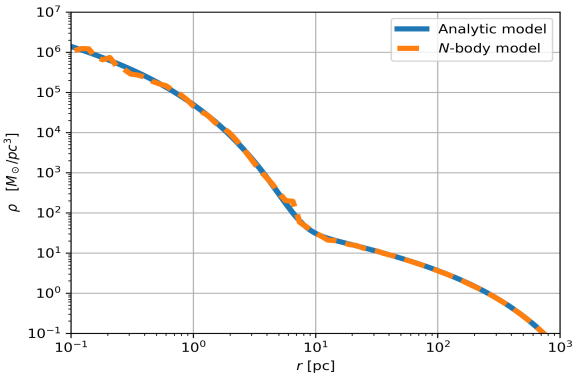
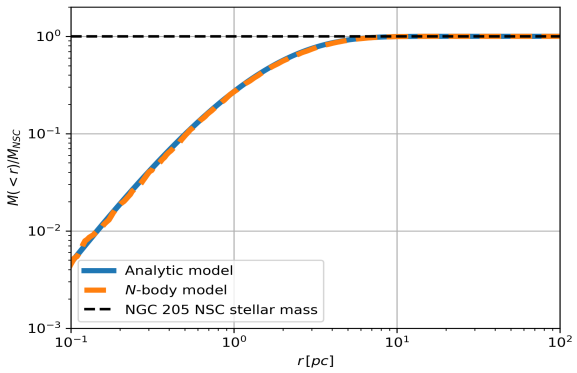


Figure 3.2: Histograms of the position and velocity distributions of the N-body realization of NGC 205 stellar bulge and NSC. Each row shows, on the left, the distribution of the i-th Cartesian component of the stellar positions and, on the right, the corresponding distribution of the i-th component of the stellar velocities. The position histograms span $[-2, 2]\,kpc$ and are divided into 50 bins, each with a width of $\Delta x_i = 80\,pc$, while the velocity histograms span $[-100, 100]\,km/s$ and are divided into 50 bins, each with a width of $\Delta v_i = 4\,km/s$.

The 0-th order moments of these six distributions, namely the system centre-of-mass position x_{CM} and velocity v_{CM}, are obtained using the OpOpGadget `Analysis.com()` method, while the corresponding first-order moments, namely the position and velocity dispersions, are computed through Python coding. The analysis yields $x_{CM} = [0.34 \pm 626,\ -0.37 \pm 627,\ 0.10 \pm 627]$ pc and $v_{CM} = [-0.0086 \pm 28.9281,\ -0.0065 \pm 28.9258,\ 0.0243 \pm 28.8984]$ km/s, compatible with a spherically symmetric and isotropic stellar distribution whose centre of mass is located at the origin of the distribution's inertial frame S_{CM}.
Lastly, applying the `Profile.dens()` and `Profile.mass()` methods to the generated ICs provides the N-body representations of the stellar density and cumulative mass profiles of NGC 205. These profiles are shown in panels (a) and (b) of Figure 3.3, respectively, together with the corresponding analytical models.



(a) Comparison between the analytic stellar mass density profile of NGC 205 (blue solid line), also shown in Figure 2.7a, and the corresponding profile reconstructed from the generated N-body initial conditions (orange dashed line) using `Profile.dens()`.



(b) Comparison between the analytic stellar cumulative mass profile of NGC 205 (blue solid line), also shown in Figure 2.7b, and the corresponding profile reconstructed from the generated N-body initial conditions (orange solid line) using `Profile.mass()`.

Figure 3.3: Comparison between the analytic and reconstructed stellar density (left) and cumulative mass (right) profiles of NGC 205 from the N-body ICs generated with OpOpGadget.

Before examining the results of our suite of numerical experiments, we conclude this Section with a Subsection devoted to reviewing the main features of the numerical code KETJU, employed to perform the simulations, and to assessing whether the initially equilibrium models remain in equilibrium when evolved with KETJU over several dynamical timescales, t_{dyn}.

3.1.3 Testing the Dynamical Equilibrium of OpOpGadget Initial Conditions with KETJU

As anticipated, the numerical code employed in this thesis to simulate galaxy mergers and, consequently, the dynamical evolution of IMBH binaries is KETJU. Originally developed as an extension of GADGET-3, here we adopt KETJU publicly available version implemented in the GADGET-4 code (see, e.g., Springel et al. 2021; Mannerkoski, Rawlings, et al. 2023). Before assessing the equilibrium N-body initial conditions when evolved with KETJU, we briefly review the main features of GADGET-4 code and KETJU module, and describe the numerical setup adopted for our simulations.
GADGET-4 is a highly parallelized N-body and hydrodynamical code that provides two methods for computing gravitational forces: the one-sided, tree-based, multipole expansion and the Fast Multipole Method (FMM). Unlike the tree-based approach, the FMM computes gravitational interactions symmetrically, thereby ensuring momentum conservation when combined with a suitable time-integration scheme. In this regard, GADGET-4 provides two time-integration schemes: the leapfrog integrator and a hierarchical time-integration scheme based on a hierarchy of timebins. The former assigns to each particle with acceleration $\mathbf{a}$ an adaptive time step (Springel 2005),

$$\Delta t = \sqrt{\frac{2\eta\epsilon}{|\mathbf{a}|}}, \tag{3.24}$$

where η is a user-specified accuracy parameter. The latter instead relies on a decomposition of the Hamiltonian into fast and slow components, optimizing the time integration by identifying subsystems that evolve slowly or rapidly and integrating each of them on the appropriate timescale.

The main improvement introduced by KETJU is the regularized MSTAR integrator (Rantala et al. 2020). By employing algorithmic regularization, this integrator treats BH–star interactions without gravitational softening when a star enters the corresponding regularized region. This allows KETJU to accurately resolve close encounters and, in particular, the dynamical interactions between BHs and surrounding stars at small separations (see, e.g., Mikkola and Tanikawa 1999; Mikkola and Tanikawa 1999). The regularized region surrounding each BH-like particle is required to extend beyond the support radius $h \equiv 2.8\epsilon$ of Gadget-2 spline kernel (Springel 2005), beyond which the gravitational potential reverts to the standard Newtonian form. This softening procedure replaces the mass density of a point-like particle, $\rho(r) = m\delta(r)$, with the normalized spline kernel

$$W(r,h) = \frac{8}{\pi h^3} \begin{cases} 1 - 6q^2 + 6q^3, & 0 \leq q \leq \frac{1}{2}, \\ 2(1-q)^3, & \frac{1}{2} < q \leq 1, \\ 0, & q > 1, \end{cases} \tag{3.25}$$

with $q \equiv r/h$.
The quantity ϵ, whose value depends on the system we deal with and appearing in Equation (3.25) through q, is the same provided by OpOpGadget through the method `Analysis.softening_scale(k` mentioned above.
The lower limit to the size of KETJU regularized regions ensures that BH–star interactions are exactly Newtonian when the star is outside the regularized region. In all simulations presented in the following Sections, the physical radius of the regularized region is set to 3ϵ.
When two regularized regions overlap, KETJU can additionally account for relativistic corrections by including Post-Newtonian (PN) terms up to 3.5PN order (see, e.g., Thorne and Hartle 1985; Blanchet 2014). This treatment allows the evolution of a BH binary to be followed from the hardening phase to its final coalescence, defined by default to happen when the separation is $12R_s$, whith R_s being the Schwarzschild radius corresponding to the total binary mass.
Finally, we summarize the main numerical setup adopted throughout the experiments presented in the following Sections. The simulations are controlled through two sets of options: compile-time options, specified in the ***Configuration file***, and run-time options, specified in the ***Parameter file***. The main options adopted in each case are reported below.
The main options enabled in the ***Configuration file*** are:

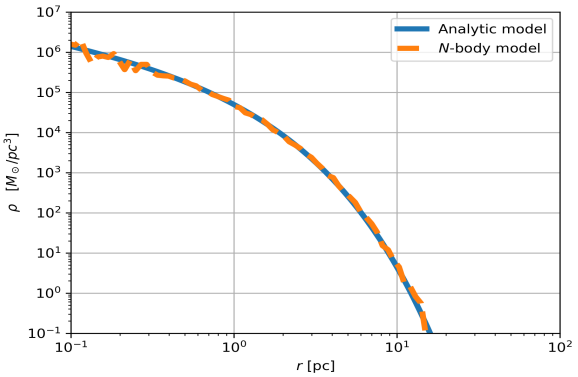
- *HIERARCHICAL_GRAVITY*: enables hierarchical time integration by splitting the Hamiltonian into fast and slow components. This scheme ensures momentum-conserving time integration.
- *FMM*: enables the Fast Multipole Method (FMM) for the computation of gravitational forces.
- *KETJU*: activates the KETJU functionality, including the MSTAR integration within the regularized regions.

The main options specified in the ***Parameter file*** are grouped in:

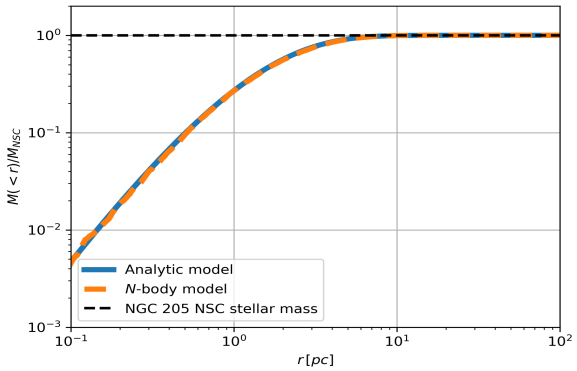
- *System of units*: the internal units are defined through `UnitLength_in_cm`, `UnitMass_in_g`, and `UnitVelocity_in_cm_per_s`, corresponding to $1\,\mathrm{pc}/h$, $1\,M_\odot/h$, and $1\,\mathrm{km\,s^{-1}}$, respectively. The gravitational constant is set to `GravityConstantInternal = 0`, so that G is expressed in internal units. With this choice, the corresponding internal time unit is approximately $9.8 \times 10^5\,\mathrm{yr}/h$.

- *Simulation time*: the beginning and end of the simulation are specified through `TimeBegin` and `TimeMax`, respectively, in code time unit.
- *KETJU regularization region*: the physical radius of the KETJU region is specified through `KetjuRegionPhysicalRadius`, set to three times the softening length of the corresponding particle class.`KetjuPNTerms` is set to 3.5.
- *Integration accuracy*: the relative tolerance of the KETJU integration is controlled by `KetjuIntegrationRelativeTolerance`, set to 10^{-7}. The relative tolerance used for the KETJU output is specified by `KetjuOutputTimeRelativeTolerance` and set to 10^{-6}.

Once the main KETJU features and the setup adopted for our numerical experiments have been described, we can assess whether an isolated N-body system generated in equilibrium remains in this configuration after several orbital periods of a typical star. Since the following sections will focus on high-resolution numerical experiments of merging stellar hosts, here we assess the equilibrium of anisolated, N-body system composed solely of NGC 205-like NSC and IMBH, as described in § 2.3. To this end, we let the system evolve for $50 \times t_{dyn}$ with the *KETJU* functionality enabled, and monitor the evolution of the stellar mass density and cumulative mass profiles, as well as the virial ratio Q, over this timescale.
Keeping the same BH-to-stellar mass ratio of 160 adopted in the previous Subsection, we initialize $N_{NSC} = 7200$ NSC stars, whose density and cumulative mass profiles are shown in Figure 3.4 and compared with their corresponding analytical models.



(a) Comparison between the analytic stellar mass density profile of NGC 205 NSC (blue solid line) and the corresponding profile reconstructed from the generated N-body initial conditions (orange dashed line) using `Profile.dens()`

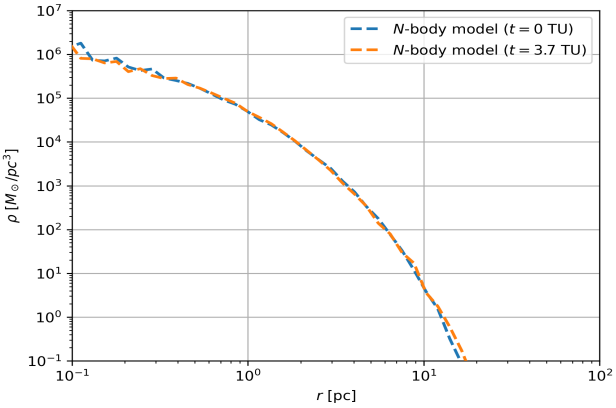


(b) Comparison between the analytic cumulative mass profile of NGC 205 NSC (blue solid line) and the corresponding profile reconstructed from the generated N-body initial conditions (orange dashed line) using `Profile.mass()`

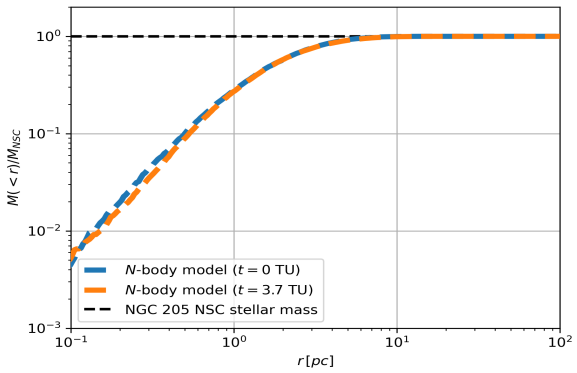
Figure 3.4: Comparison between the analytic and reconstructed stellar density (left) and cumulative mass (right) profiles of NGC 205 NSC from the N-body ICs generated by OpOpGadget.

As already discussed, although OpOpGadget constructs the system such that particle coordinates are defined in the inertial frame of the centre of mass, S_{CM}, a non-vanishing centre-of-mass velocity can arise from the discrete realization of the N-body system. This contribution must be accounted for when computing the virial ratio, by subtracting the center of mass velocity to each particle velocity. Therefore, at the beginning of the simulation, we find $Q \approx 1$, clearly indicating a dynamically equilibrated configuration, as widely discussed in § 1.1.
The dynamical time evaluated at the half-mass radius of the initialized system is found to be $t_{dyn} = 0.072$ Myr. The system is therefore evolved over the interval $[0, \texttt{TimeMax}]$, with $\texttt{TimeMax} \approx 50 \times t_{dyn} = 3.7\,TU$. Finally, analyzing the KETJU output allows us to reconstruct

the same distributions shown in Figure 3.4, compare them with the corresponding initial profiles, and recompute the virial ratio. Applying the same procedure at the end of the simulation, we obtain $Q \approx 1$. The evolution of the system is further assessed in Figure 3.5, where the initial and final mass density and cumulative mass profiles are compared.



(a) Comparison between the reconstructed mass density profiles of the NGC 205 NSC at the beginning (blue dashed line) and at the end (orange dashed line) of the simulation.



(b) Comparison between the reconstructed cumulative mass profiles of the NGC 205 NSC at the beginning (blue dashed line) and at the end (orange dashed line) of the simulation.

Figure 3.5: Comparison of the NGC 205 NSC N-body realization at the beginning and end of the simulation: mass density profiles (left) and cumulative mass profiles (right).

In conclusion, as demonstrated by Mannerkoski, Rawlings, et al. (2023), activating the KETJU module preserves the ability of GADGET-4 to evolve initially at equilibrium systems without introducing spurious dynamical evolution of numerical origin. The analysis presented in this Section has allowed us to review the N-body theory, the main features of the numerical tools adopted in this thesis, establish the standard procedure used to generate and analyse the N-body initial conditions, and verify that the KETJU implementation is well suited to our purposes. In particular, the stability of the isolated system over several dynamical times provides a necessary validation of the high-resolution merger simulations, involving pairs of stellar host and IMBH, that will be presented in the following Sections.

3.2 Evolution of an IMBH Binary in an N-body Simulation of Merging NSCs

In Chapter 2, we presented the theoretical foundations of the self-consistent hybrid model developed by Quinlan (1996) and Sesana, Haardt, et al. (2006), together with a series of numerical experiments investigating the evolution of IMBH binaries within this framework. Although the model was originally formulated for a fixed, homogeneous, and isotropic stellar background, we also described the procedure introduced by Sesana and F. M. Khan (2015) to extend its applicability to arbitrary mass density distributions. Its versatility and its ability to reproduce evolutionary tracks in good agreement with those obtained from analogous N-body simulations allowed us to obtain the first predictions on IMBH binaries separation and eccentricty evolution, while studying the role of the central stellar density, eccentricity at pairing and cosmic size evolution effects on the original stellar host. Nevertheless, the hybrid model remains subject to several limitations arising from its underlying assumptions. We now identify the most relevant ones, which we aim to overcome through the use of KETJU N-body simulations:

- **Artificial BH binary initialization.** The secondary BH is artificially introduced inside the stellar background of the primary BH:

  1. **Merging stellar systems not included.** The model does not account for the fact that massive BH binaries are expected to form following the merger of their stellar systems. As a consequence, the structural and dynamical properties of the stellar background, i.e. the merger remnant, may differ significantly from those of either progenitor.
  2. **Unknown pre-binary evolution.**The model does not describe the dynamical evolution preceding the formation of the BH binary. As a result, the binary initial conditions (a_0, e_0) must be prescribed as input parameters rather than being determined self-consistently from the preceding dynamical evolution. While this represents an advantage of the model, as it enables an efficient exploration of the (a_0, e_0) parameter space, it also constitutes a limitation when the goal is to investigate how the binary initial configuration emerges from the dynamical processes leading to its formation.
  3. **Fixed binary center of mass.** The center of mass of the binary is placed at the center of symmetry of the stellar distribution, preventing the model from describing possible displacements of the binary from the galactic center.

- **Arbitrary loss-cone refilling parameter.** Besides the initial binary conditions (a_0, e_0), the only free parameter controlling the evolution in the hybrid model is the loss-cone refilling parameter LCN, mentioned § 1.4. However, its value must be prescribed a priori, introducing an additional degree of arbitrariness into the model.

- **Assumption of isotropy.** By construction, the model assumes an isotropic stellar background and is therefore not suited to describe systems with significant net rotation or velocity anisotropy, which are instead expected for merger remnants.

- **Fixed stellar background.** Although Equation (2.3) provides a prescription for computing the stellar mass ejection rate, the binary evolution is still reconstructed within a fixed stellar background. The model therefore does not self-consistently account for the dynamical response of the stellar system to the binary's evolution.

These assumptions, along with the limitations arising from them, can be overcome by adopting a fully self-consistent N-body treatment. We therefore present here a KETJU N-body experiment designed to follow the evolution of a head-on, parabolic encounter between two N-body realizations of the NGC 205-like NSC (see Table 2.4), each hosting an IMBH of mass $m_1 = m_2 = 4 \times 10^4, M_\odot$ at its center[2].
For this numerical experiment we adopted a BH-to-stellar mass ratio of 160 (see F. M. Khan and Holley-Bockelmann 2021), $N_{NSC} = 7200$ stellar particles for each realization of NGC 205-like NSC, for a total of $N_{tot} = 14400 + 2$ particles. Each model of NSC and IMBH has been generated by employing OpOpGadget, which assigns particles coordinates by assuming the center of symmetry to be placed at the origin with null velocity. The IMBHs are placed at coordinates $(\mathbf{x}, \mathbf{v}) = (\mathbf{0}, \mathbf{0})$.
The initial conditions adopted for this merger follow the approach of C. Nipoti et al. (2003). The two NSCs, namely $NSC\,1$ and $NSC\,2$, are initially positioned such that the spheres enclosing 90 % of their stellar mass, with radius $r_{90} = 5.07\,\mathrm{pc}$, are tangent to each other, neglecting distortions induced by tidal forces. They are then assigned a relative velocity equal to that

[2]NSCs reside at galaxy centres, so more realistic simulations should also include the stellar bulge and dark matter. However, our simplified approach is justified when galaxy interactions bring the two NSCs into close proximity.

which two point masses, each with the corresponding NSC mass, would possess at a separation of $2r_{90}$ if they were initially at rest at infinite separation. To this end, the two particle sets are displaced in position by $\mp(r_{90}, 0, 0) = \mp(5.07, 0, 0)\,\mathrm{pc}$ and assigned opposite velocity offsets, $\pm(v_{90}, 0, 0) = \pm(27.94, 0, 0)\,km/s$, with v_{90} defined by

$$v_{90} = \sqrt{\frac{G\,(M_{NSC} + m_1)}{2r_{90}}}. \tag{3.26}$$

Figure 3.6 shows the initial configuration of the numerical experiment in its center-of-mass reference frame, S_{CM}. The black arrows indicate the bulk velocities of the two systems, with their lengths representing the distance each system would travel over a timescale $t_{dyn} \approx 0.07\,\mathrm{Myr}$ if it continued to move at the initial speed v_{90}.

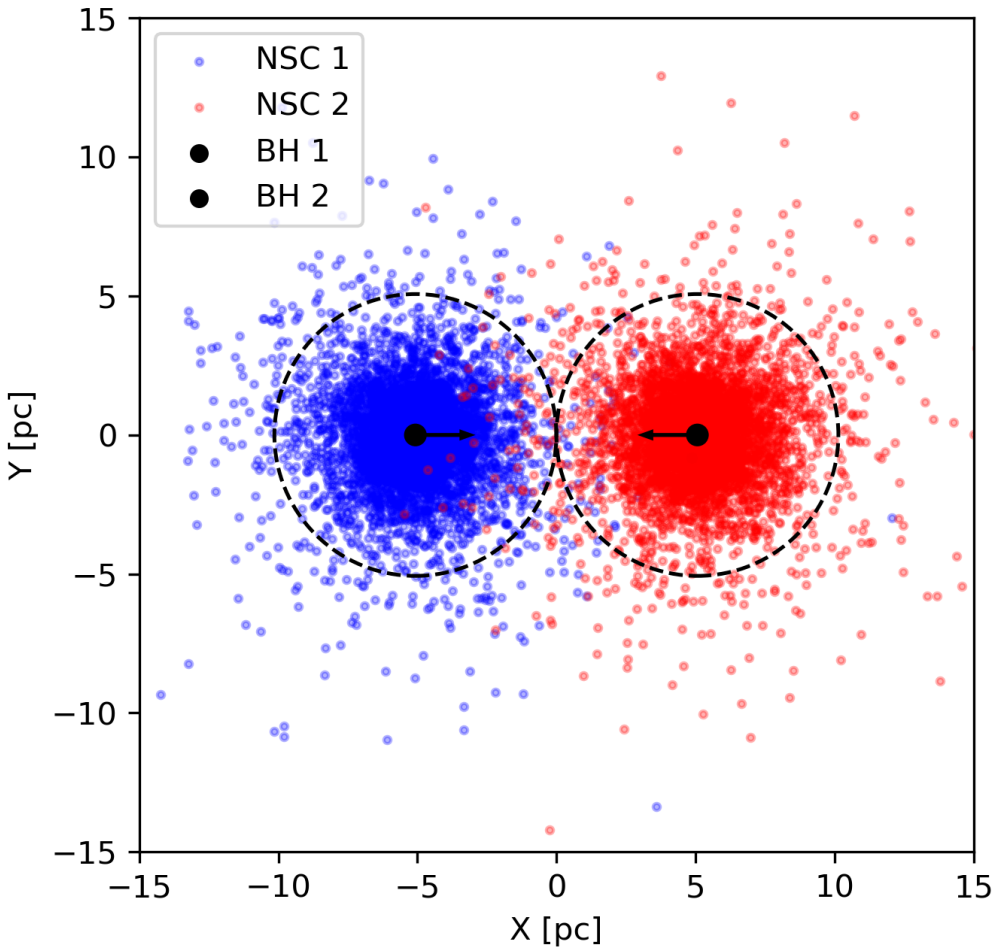


Figure 3.6: Positions of the initialized NSC stars (blue and red dots) and IMBHs (black dots) in the center-of-mass frame S_{CM}. The black arrows indicate the initial bulk velocities of the two NSCs, with lengths corresponding to the displacement over their average half-mass dynamical time. The dashed circles mark the spheres containing 90 % of the total mass for each NSC.

Following the same analysis of the IC generation performed in § 3.1, we examine the stellar spatial and velocity distributions of the two NSCs, shown in Figures 3.7 and 3.8, respectively. It is worth noting that, unlike in the analysis presented in the previous Section, the spatial distributions are centered at $\mp(r_{90}, 0, 0)$, while the corresponding velocity distributions are centered at $\mp(v_{90}, 0, 0)$ for the first and second NSCs, respectively.

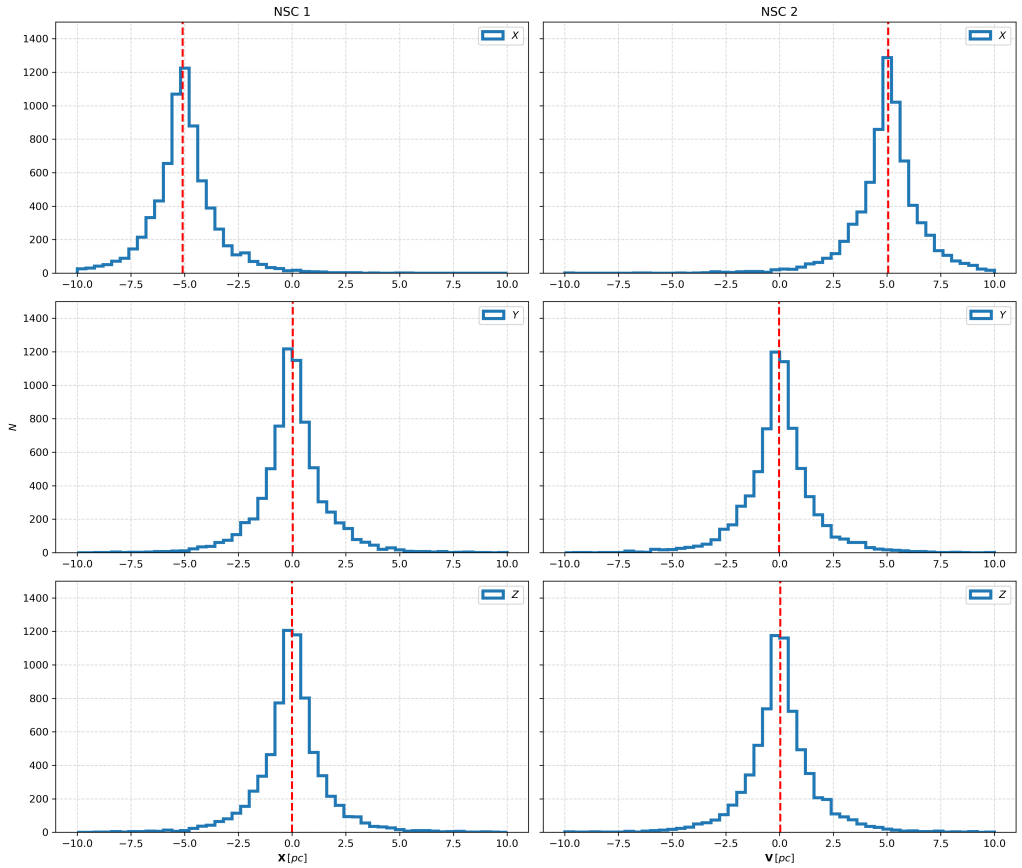


Figure 3.7: Histograms of the position distributions of both, N-body generated, NSCs. Each row shows the distribution of the i-th Cartesian component of the stellar positions for $NSC\,1$ (left) and $NSC\,2$ (right). All histograms span $[-10, 10]\,\mathrm{kpc}$ and are divided into 50 bins, each with a width of $\Delta x_i = 400\,\mathrm{pc}$. The first column distributions peak around $-(r_{90}, 0, 0)$, while the second column around $+(r_{90}, 0, 0)$.

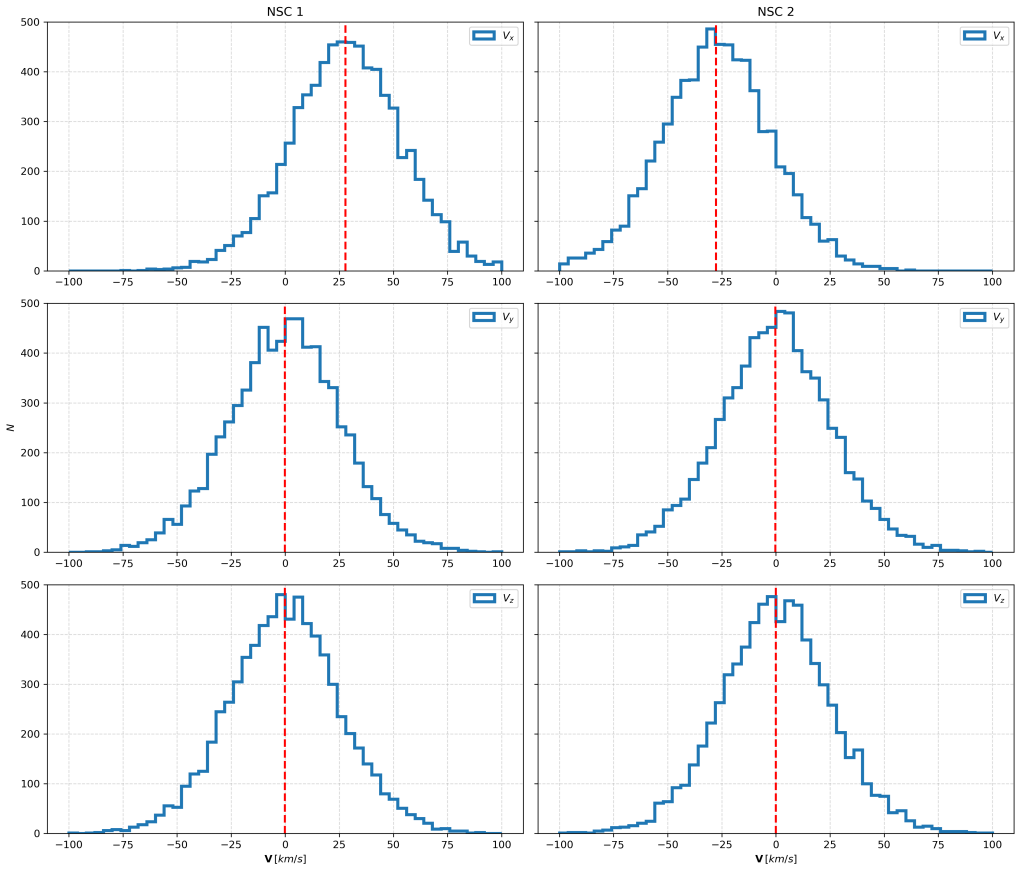


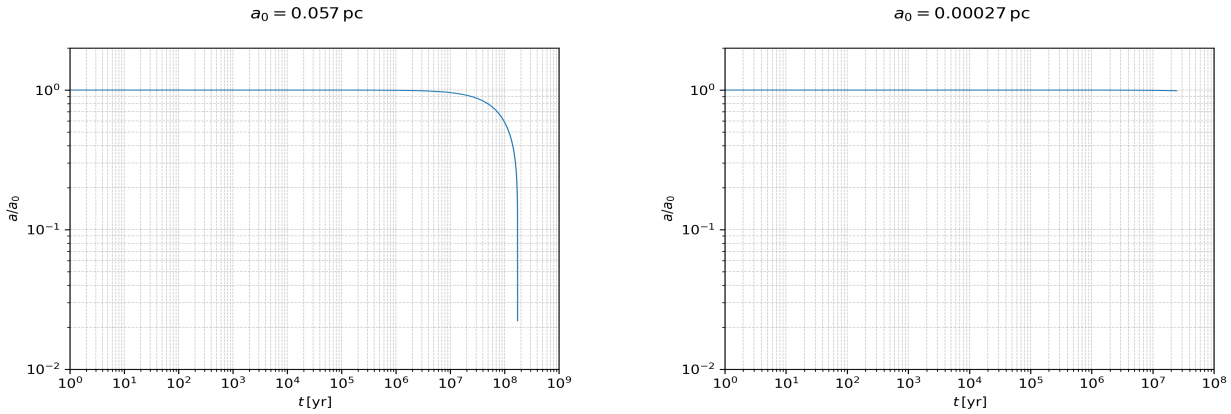
Figure 3.8: Histograms of the velocity distributions of both, N-body generated, NSCs. Each row shows the distribution of the i-th Cartesian component of the stellar velocities for $NSC\,1$ (left) and $NSC\,2$ (right). All histograms span $[-100, 100]\,\mathrm{km/s}$ and are divided into 50 bins, each with a width of $\Delta v_i = 4\,\mathrm{km/s}$. The first column distributions peak around $(v_{90}, 0, 0)$, while the second column around $-(v_{90}, 0, 0)$.

One of the main challenges encountered in the KETJU simulations is the unexpectedly long integration time required to evolve the systems. In all simulations considered in this thesis, including those presented in the following Section, we identified a numerical bottleneck associated with excessively short integration time steps Δt compared with the timescale of the binary evolution. This behavior becomes particularly pronounced around the transition scale a_{GW}, defined in Equation (2.34), below which GW emission dominates over stellar-scattering-driven hardening and drives the subsequent orbital decay.
A physical interpretation of this behavior can be obtained, at least at the order-of-magnitude level, by comparing the binary coalescence timescale $\tau_0 = \tau(a_0)$, as given by Equation (1.142), with the initial orbital period T_0. For a BH binary with total mass M_{12} and initial separation a_0, the coalescence timescale due to gravitational-wave emission scales as $\tau(a_0) \propto a_0^4/M_{12}^3$, whereas the orbital period scales as $T_0 \propto a_0^{3/2}/M_{12}^{1/2}$. Their ratio therefore scales as

$$\frac{\tau_0}{T_0} \propto \frac{a_0^4}{M_{12}^3}\left(\frac{a_0^{3/2}}{M_{12}^{1/2}}\right)^{-1} \propto \left(a_0 M_{12}^{-1}\right)^{5/2}. \tag{3.27}$$

When the initial separation is identified with the GW transition scale, $a_0 = a_{\mathrm{GW}}$, this relation implies that the ratio $\tau(a_{\mathrm{GW}})/T_0$ decreases with increasing binary mass, scaling as $M_{12}^{-5/8}$. The numerical bottleneck is therefore not primarily associated with the overall dynamical evolution of the stellar system, whose characteristic timescale remains of order t_{dyn}, but rather with the short orbital period of the binary compared with the remaining coalescence timescale.

This trend is further supported by two independent numerical experiments following the dynamical evolution of two symmetric BH binaries with different total masses, $M_{12} = 2 \times 10^8\, M_\odot$ and $M_{12} = 2 \times 4 \times 10^4\, M_\odot$, and initial separations scaling with $M_{12}^{-5/8}$. While the SMBH binary reaches coalescence within approximately two hours of computational time, the IMBH binary, over the same runtime, remains far from exhibiting the characteristic drop of separation due to efficient GW emission, as shown in Figure 3.9.

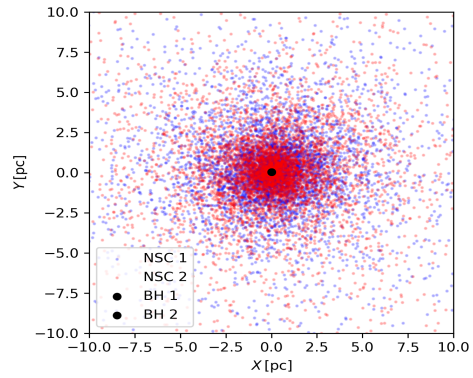


(a) Evolution of the semimajor axis of the SMBH binary with total mass $M_{12} = 2 \times 10^8\, M_\odot$, normalized to the initial separation a_0. The track was reconstructed from a 2-hour simulation run on a single CPU.

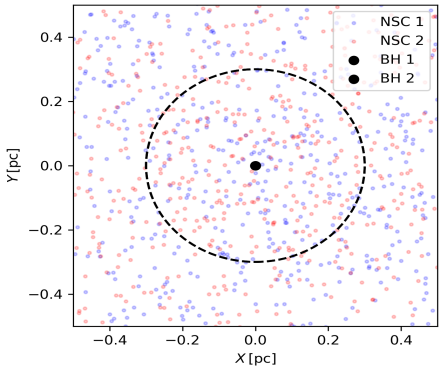
(b) Evolution of the semimajor axis of the IMBH binary with total mass $M_{12} = 2 \times 4 \times 10^4\, M_\odot$, normalized to the initial separation a_0. The track was reconstructed from a 2-hour simulation run on a single CPU.

Figure 3.9: Comparison of the semimajor axis evolution of SMBH (left) and IMBH (right) binaries, normalized to the respective initial separations a_0. Both evolutionary tracks were reconstructed from 2-hour simulation runs on a single CPU.

To follow the complete dynamical evolution of the IMBH binary, from the merger of the stellar hosts to the final coalescence, we combine the N-body and hybrid approaches. During the numerical integration, KETJU periodically outputs a series of snapshots at fixed time intervals Δt_{snap}, containing the positions, velocities, masses, and IDs of all particles generated with OpOpGadget. In addition, it produces a separate series of files containing only the BBH information, e.g. $(t, a(t), e(t))$,which can be analysed using the KETJU-Utils analysis toolkit. The final snapshot can then be read with OpOpGadget, allowing us to perform the same analysis described at the end of § 3.1. This procedure provides a physically motivated initial condition (a_0, e_0) for the subsequent binary evolution, directly determined from the final KETJU state. Moreover, it allows us to derive the main quantities at the influence radius required by the hybrid model from the actual stellar background of the merger remnant. Overall, this approach combines the complementary strengths of the two methods: KETJU captures the dynamics of the galaxy merger and the subsequent hardening of the binary through stellar scattering, while the hybrid model can then be used to evolve the binary from a physically constrained stellar background and a initial condition.
The numerical integration is stopped at the physical time $t_{snap} = 18.75\,\mathrm{TU} \simeq 18\,\mathrm{Myr}$. Figure 3.10a shows the state of the system in its centre-of-mass reference frame S_{CM} at the time of the final snapshot.



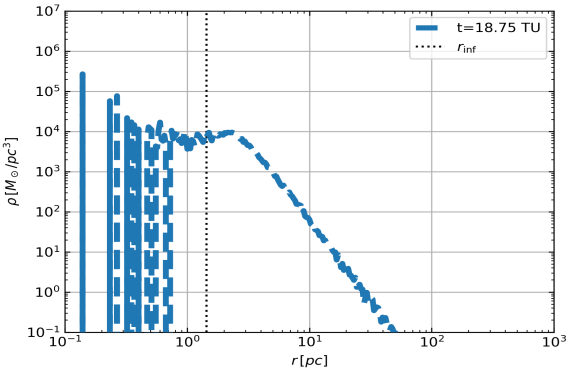
(a) Positions of the initialized NSC stars (blue and red dots) and IMBHs (black dots) in the center-of-mass frame S_{CM} at time $t_{snap} = 18.75\,\mathrm{TU}$.



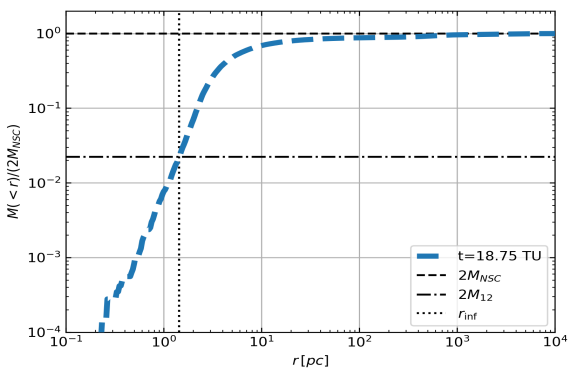
(b) Zoom-in on the IMBH binary shown in Figure 3.10a. The dashed circle marks the KETJU regularized region at time $t_{snap} = 18.75\,\mathrm{TU}$.

Figure 3.10: Positions of the two NSCs and their IMBH binary at $t_{snap} = 18.75\,\mathrm{TU}$ in the center-of-mass frame S_{CM}, with a zoom-in on the KETJU regularized region shown on the right.

The two original stellar components appear to be well mixed, forming a more compact central core surrounded by diffuse outskirts, while the IMBH binary is located close to the centre of mass of the stellar distribution.
From this configuration, the radial, i.e. angle-averaged, stellar density and cumulative mass profiles are derived using OpOpGadget, as shown in Figure 3.11(a) and (b), respectively.



(a) Reconstructed radial stellar density profile of the remnant stellar system at $t_{snap} = 18.75\,\mathrm{TU}$. The centre of the distribution was determined using the shrinking-spheres method (see, e.g., Power et al. 2003). The black vertical dotted line marks the binary influence radius, r_{inf}.



(b) Reconstructed cumulative mass profile of the remnant stellar system at $t_{snap} = 18.75\,\mathrm{TU}$. The black horizontal dashed and dot-dashed lines mark twice the stellar mass and the binary mass, respectively, while the black vertical dotted line marks the binary influence radius, r_{inf}.

Figure 3.11: Reconstructed radial stellar density (left) and cumulative mass (right) profiles of the remnant stellar system at $t_{snap} = 18.75\,\mathrm{TU}$.

Yet, imposing $M_*(< r_{inf}) = 2M_{12}$ yields an influence radius of $r_{inf} = 1.44\,\mathrm{pc}$ for the system at the time of the snapshot. This value is then used to compute the corresponding stellar mass density, which is given by $\rho_{inf} = 7.11 \times 10^4\, M_\odot/\mathrm{pc}^3$.
Another functionality of OpOpGadget, not introduced so far but particularly useful at this stage, is the `Profile.vdisp3d()` module. This routine is used to construct the radial velocity-

dispersion profile, shown in Figure 3.12, and to determine its value at the binary influence radius, $\sigma_{inf} = 23.3, \mathrm{km/s}$. Finally, substituting σ_{inf} into Equation (2.50) gives $a_h = 0.16, \mathrm{pc}$.

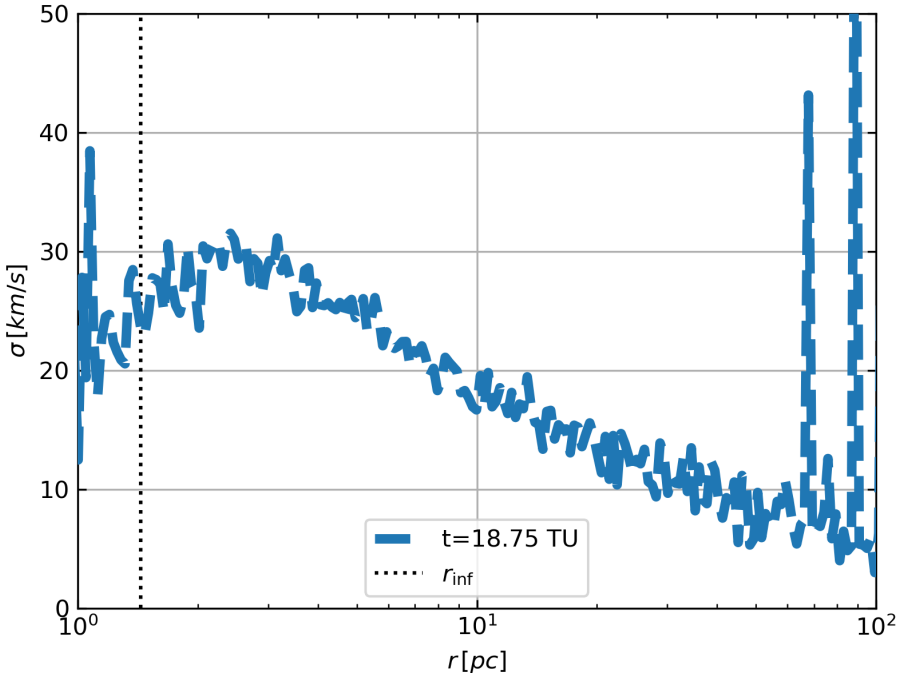


Figure 3.12: Reconstructed radial velocity dispersion profile of the remnant stellar system at $t_{snap} = 18.75\,\mathrm{TU}$. The black vertical dotted line marks the binary influence radius, r_{inf}.

The last quantity required by the hybrid model before starting the numerical integration with the RK4 scheme is the dimensionless factor LCN, which enters Equations (2.20) by replacing ρ_{inf} with $LCN \cdot \rho_{inf}$. This normalization factor, which was always set to unity in Chapter 2, represents the only fine-tuning parameter of the model and accounts for the efficiency of the loss-cone replenishment process. Its value is therefore crucial for an accurate description of the binary evolution, but it is also difficult to determine a priori. This is because the assumptions underlying the hybrid model neglect several physical effects, such as velocity-dispersion anisotropy and the departure from spherical symmetry induced by the merger of the stellar hosts, which can significantly affect the rate at which the loss cone is replenished. However, unlike the purely hybrid-model experiments, we have directly constrained the evolution of the binary separation and eccentricity over the time interval covered by t_{snap}. In particular, the KETJU output $(t, a(t))$ can be used to estimate the status of the loss cone when transitioning to the hybrid model. To this end, the data points satisfying $a < a_{in} \equiv 0.1 a_h$ are fitted with the curve

$$a(t) = \left[\frac{1}{a_{in}} + \frac{G \rho_{inf} H(a_{in})}{\sigma_{inf}} LCN (t - t_{in}) \right]^{-1} \tag{3.28}$$

obtained by integrating Equation (2.1), under the assumption that H_u remains approximately constant in this regime, as discussed in § 2.1. This operation, shown in Figure 3.13, yields $LCN = 14.00 \pm 0.03$.

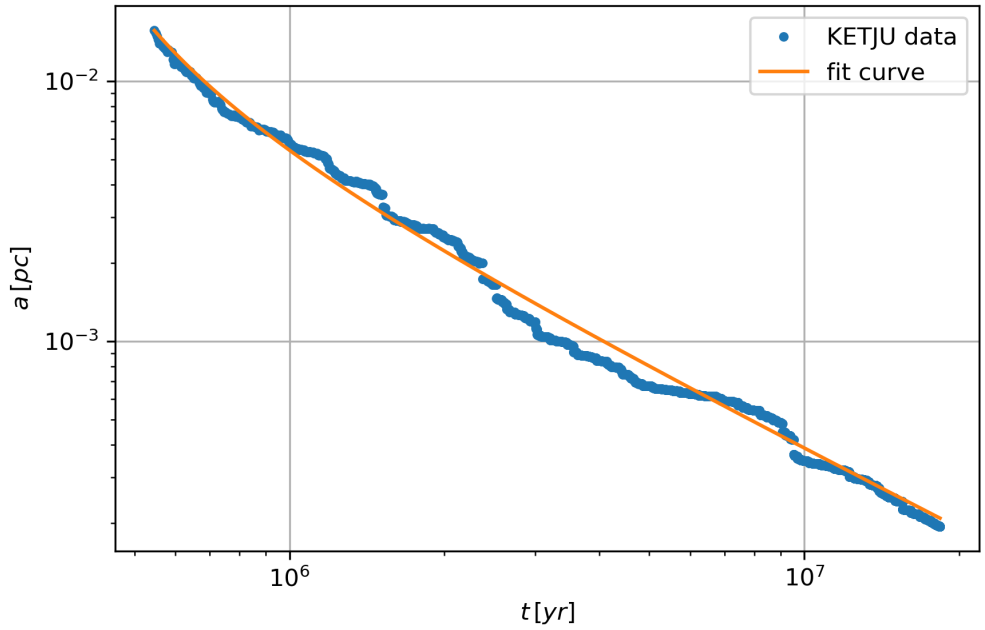
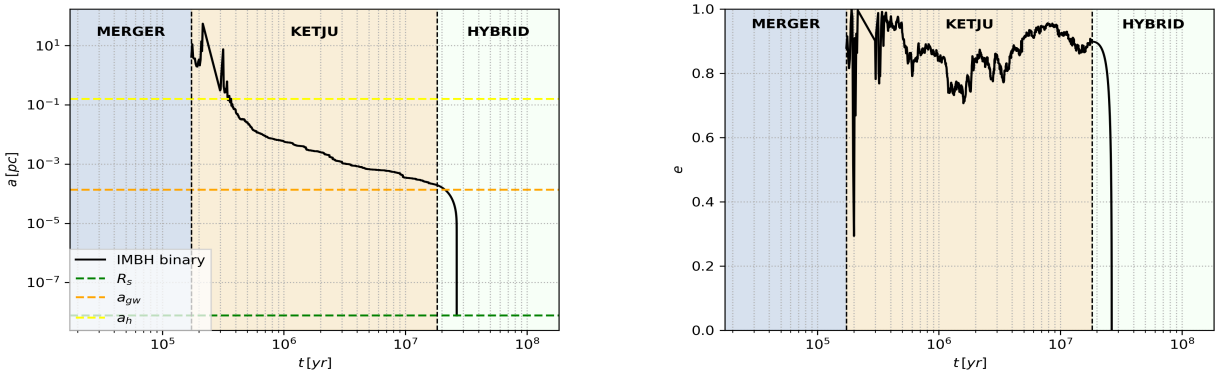


Figure 3.13: Best-fit curve (orange solid line), defined by Equation 3.28, to the KETJU output (blue dots) over the range from $0.1a_h$ to the IMBH binary separation at the end of the KETJU numerical simulation.

Once all the required quantities have been determined, the hybrid-model differential Equations (2.20), with ρ_{inf} replaced by $LCN \cdot \rho_{inf}$, are integrated starting from the IC $(t_0, a_0, e_0) = (18.35\,\text{Myr},\ 1.94 \times 10^{-4}\,\text{pc},\ 0.90)$, corresponding to the final KETJU binary output. This allows us to follow the IMBH binary evolution down to a separation equal to its Schwarzschild radius R_s, as shown in Figure 3.14(a)-(b), where the KETJU and hybrid-model pieces of evolutionary tracks are shown together.



(a) IMBH binary semimajor axis time evolution.

(b) IMBH binary eccentricity time evolution.

Figure 3.14: IMBH binary evolutionary tracks reconstructed by combining KETJU numerical simulation and the Quinlan-Sesana hybrid model. Both plots are divided into three regions: "MERGER" denotes the phase, followed with KETJU, during which the IMBHs are still unbound and no output is provided by KETJU; "KETJU" denotes the subsequent binary dynamical evolution directly reconstructed from the KETJU simulation; and "HYBRID" denotes the remaining evolutionary track obtained with the hybrid model.

Figure 3.14(a) and (b) warrant further discussion. First, the label "MERGER", shown on

the left-hand side of both panels, has been added to highlight the phase in which the two IMBHs are still unbound and the system is being numerically evolved with KETJU. During this stage, KETJU does not provide the usual binary output, as indicated by the absence of any curve to the left of the "KETJU" regions. Then, although the eccentricity evolution reconstructed from KETJU is relatively noisy, it still exhibits an increasing trend during the stellar-scattering phase, as discussed in § 2.3. By contrast, the binary separation is less affected by numerical fluctuations and displays a smooth evolution, in good agreement with the evolutionary tracks presented in § 2.3.
The resulting coalescence timescale, here defined as the time elapsed from the beginning of the merger simulated with KETJU to the final coalescence reached with the hybrid model, is $\tau_{coal} = 26.45\,\mathrm{Myr}$. Most importantly, this timescale is of the same order of magnitude as those obtained in § 2.3 for different initial eccentricities e_0, providing further support for the hybrid-model approach. It is worth noting that, although, in this chapter, the coalescence time is measured from an earlier stage of the binary evolution than in the definition adopted in Chapter 2, the value $LCN > 1$ indicates that stellar-scattering hardening is more efficient in the merger remnant than assumed in the previous calculations, where $LCN = 1$, promoting a faster orbital decay.
We conclude this chapter by presenting the same analysis for different merger configurations involving a high-redshift, stellar clumps.

3.3 KETJU IMBH Binary evolution from Parabolic Encounters of Stellar Clumps

So far, we have investigated the dynamical evolution of IMBH binaries through a series of numerical experiments designed to characterise the mechanisms governing their orbital decay and coalescence. In Chapter 2, we addressed this problem using the Quinlan–Sesana hybrid model, reconstructing the time evolution of the binary orbital parameters by treating the central BH of the stellar system as the primary component and introducing a secondary IMBH as an intruder initially placed in the central region of the host galaxy. In § 3.2, we instead investigated the binary evolution with high-resolution KETJU N-body simulations of mergers between identical nucleated stellar systems, each hosting a central IMBH.
Recent, observational evidences (see, e.g., (Nakane et al. 2026)) suggest that galaxies at high redshift may have undergone a clumpy phase, with present-day globular clusters potentially representing the surviving remnants of these early stellar clumps. The presence of IMBHs within such systems may therefore provide a common link between IMBH formation and the assembly of nuclear star clusters. Through dynamical friction, these compact stellar systems can lose orbital energy and migrate towards the galactic centre, where their subsequent infall and merging may contribute to the formation of dense nuclear environments (see, e.g., I. Khan et al. 2026). The resulting high stellar densities may in turn favour runaway stellar collisions, providing favourable conditions for the formation and subsequent growth of IMBHs.
Motivated by this broader evolutionary picture, we conclude Chapter 3 with a suite of four N-body experiments, namely Model 1-2-3-4, aimed at:

1. Reconstructing the dynamical history of IMBH binaries formed as a result of the merger between observationally motivated stellar clumps models.

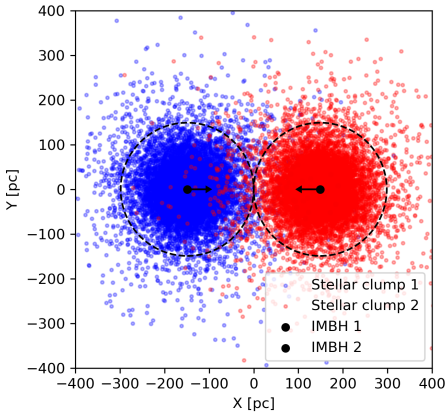
2. Exploring different initial conditions of the merger, thus considerning parabolic encounters with non-vanishing angular momentum.

As shown in Figure 2, the stellar clumps resolved in the lensed, high-redshift Misty Moons have stellar masses of the order of $\sim 10^7\,M_\odot$ and effective radii ranging from ~ 10 pc to $\sim 70\,$pc.

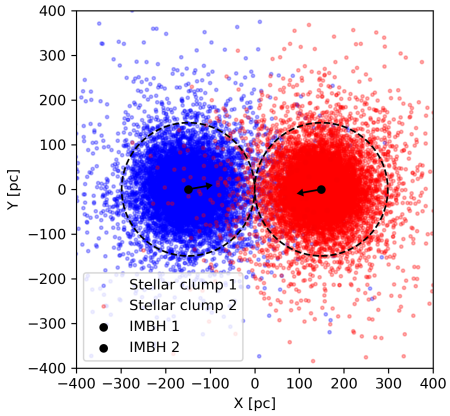
Motivated by these observational constraints, we used OpOpGadget to generate four pairs of identical, spherical and isotropic equilibrium systems, each consisting of a stellar clump-like system and a central IMBH. The stellar component was modelled with a Sérsic profile, as defined in Equation (2.37), with a total stellar mass of $M_* = 10^7\,M_\odot$, a Sérsic index of $n = 2$, and an effective radius of $R_e \approx 35\,\mathrm{pc}$. Although there is currently no observational evidence for IMBHs residing in the stellar clumps of the Misty Moons, we included a central IMBH with a mass of $10^5\,M_\odot$ in each system. This choice is motivated by the increasing BH-to-stellar mass ratio towards higher redshifts, as illustrated in Figure 2. All stellar clumps are generated with a fixed BH-to-stellar mass ratio of $1:200$, corresponding to 20000 stellar particles per clump. The four merger simulations, namely Model 1-2-3-4, were performed with KETJU, adopting the same initial separation of $2r_{90} = 2 \times 149.1\,\mathrm{pc}$ between the two IMBHs. The initial relative velocity was chosen to yield a parabolic encounter, with a magnitude determined by the total mass enclosed within r_{90}. To investigate the impact of orbital angular momentum on the merger dynamics, we varied the orientation of the initial velocity vectors across the four simulations. Specifically, the velocity vectors of the two stellar systems, both having modulus $v_{90} = 11.3\,\mathrm{km/s}$, were symmetrically rotated by $\theta = 0°$, $10°$, $20°$, and $30°$ with respect to the initial line connecting their centres.
For a rotation angle θ, the Cartesian components of the velocity of each system centre of mass were then assigned according to (see, e.g. Carlo Nipoti et al. 2007)

$$v_x = \frac{v_{90}}{\sqrt{1+\tan^2\theta}}, \qquad v_y = v_x \tan\theta, \tag{3.29}$$

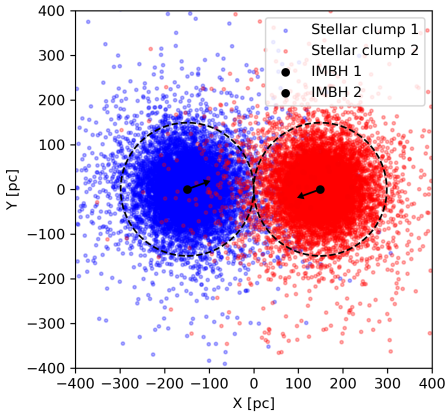
with opposite signs for the two systems to ensure their prescribed relative motion. This setup preserves the magnitude of the initial relative velocity, and hence the parabolic nature of the encounter, while introducing a non-zero tangential component for $\theta > 0°$. The resulting sequence of simulations therefore provides a controlled test of how increasing orbital angular momentum affects the dynamical evolution of IMBH pairs during stellar-clump mergers.
Model 1, Model 2, Model 3 and Model 4 generated initial conditions are shown in Figures 3.15a, 3.15b, 3.15c and 3.15d, respectively.



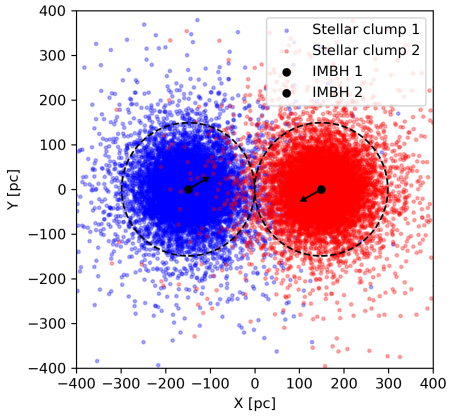
(a) Model 1 ($\theta = 0°$): Positions of the initialized stellar clumps stars (blue and red dots) and IMBHs (black dots) in the center-of-mass frame S_{CM}. The black arrows indicate the initial bulk velocities of the two stellar clumps, with lengths corresponding to the displacement over their average half-mass dynamical time. The dashed circles mark the spheres containing 90 % of the total mass for each NSC.



(b) Model 2 ($\theta = 10°$): Positions of the initialized stellar clumps stars (blue and red dots) and IMBHs (black dots) in the center-of-mass frame S_{CM}. The black arrows indicate the initial bulk velocities of the two stellar clumps, with lengths corresponding to the displacement over their average half-mass dynamical time. The dashed circles mark the spheres containing 90 % of the total mass for each NSC.



(c) Model 3 ($\theta = 20°$): Positions of the initialized stellar clumps stars (blue and red dots) and IMBHs (black dots) in the center-of-mass frame S_{CM}. The black arrows indicate the initial bulk velocities of the two stellar clumps, with lengths corresponding to the displacement over their average half-mass dynamical time. The dashed circles mark the spheres containing 90 % of the total mass for each NSC.



(d) Model 4 ($\theta = 30°$): Positions of the initialized stellar clumps stars (blue and red dots) and IMBHs (black dots) in the center-of-mass frame S_{CM}. The black arrows indicate the initial bulk velocities of the two stellar clumps, with lengths corresponding to the displacement over their average half-mass dynamical time. The dashed circles mark the spheres containing 90 % of the total mass for each NSC.

Figure 3.15: Spatial distribution of the stellar particles and IMBHs for the four initial configurations of Model 4, corresponding to different inclination angles.

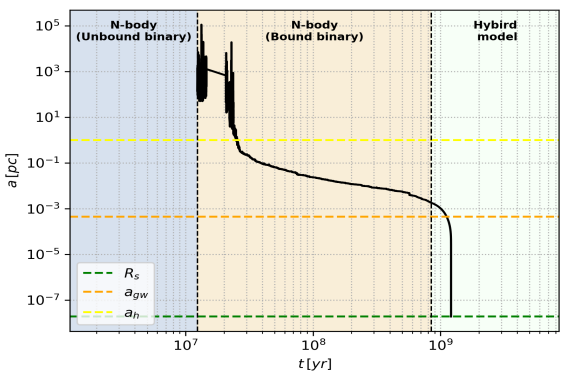
All four numerical experiments encountered the same bottleneck towards the end of the stellar-scattering phase, preventing the simulations from following the subsequent binary evolution to coalescence. We therefore resumed the evolution using the Quinlan–Sesana hybrid model, following the procedure described in § 3.2. Table 3.1 summarises the binary orbital

parameters at the end of each KETJU simulation, at time $t_{\rm snap}$, which serve as initial conditions for the hybrid model, together with the quantities required to integrate the subsequent evolution.

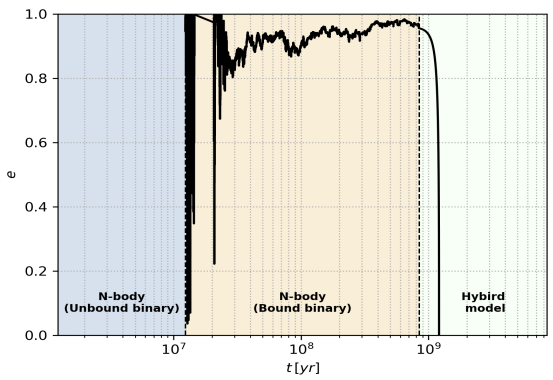
Model	$t_{\rm snap}$ [Myr]	$a_{\rm snap}$ [pc]	$e_{\rm snap}$	LCN	$r_{\rm inf}$ [pc]	$\rho_{\rm inf}$ [$M_\odot$/pc^3]	$\sigma_{\rm inf}$ [km/s]	a_h [pc]
1	848.6	0.0018	0.96	3.79	11.2	30.1	14.7	1.00
2	488.8	0.0026	0.98	3.08	8.96	61.1	17.6	0.70
3	783.6	0.0031	0.51	4.17	11.9	30.5	14.7	1.00
4	430.0	0.0061	0.71	1.29	7.65	97.6	14.0	1.10

Table 3.1: IMBH binary status at the last KETJU output (left of the double vertical line) and hybrid-model parameters (right of the double vertical line) at t_{snap} of Model 1-2-3-4. Column 1: Model name. Column 2: Time of last KETJU output. Column 3: IMBH binary at t_{snap}. Column 4: IMBH binary eccentricity at t_{snap}. Column 5: Loss-cone normalization from fitting KETJU outupt. Column 6: IMBH binary influence radius. Column 7: Stellar mass density at binary influence radius. Column 8: Stellar velocity dispersion at binary influence radius. Column 8: Binary hardening radius.

Finally, the evolutionary tracks $(t, a(t))$ and $(t, e(t))$ of Model 1-2-3-4 are represented in Figure 3.16a-3.17a-3.18a-3.19a, respectively.

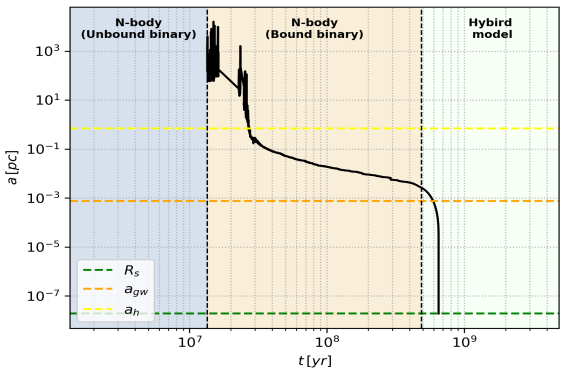


(a) IMBH binary semimajor axis time evolution reconstructed with KETJU (N-body regions) and Quinlan-Sesana model (Hybrid model region).

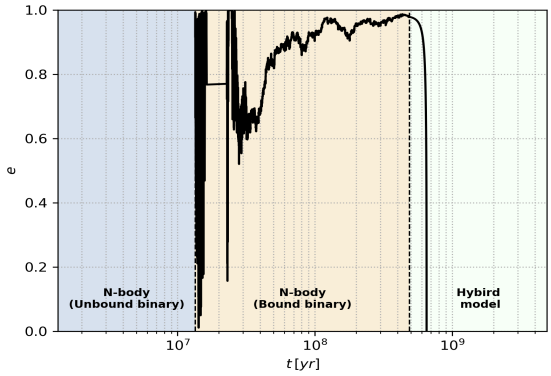


(b) IMBH binary eccentricity time evolution reconstructed with KETJU (N-body regions) and Quinlan-Sesana model (Hybrid model region)..

Figure 3.16: Model 1 evolutionary tracks $(t, a(t))$ (on left) and $(t, e(t))$ (on right).

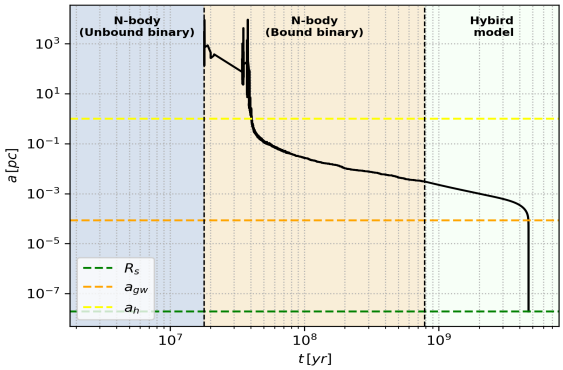


(a) IMBH binary semimajor axis time evolution reconstructed with KETJU (N-body regions) and Quinlan-Sesana model (Hybrid model region)..

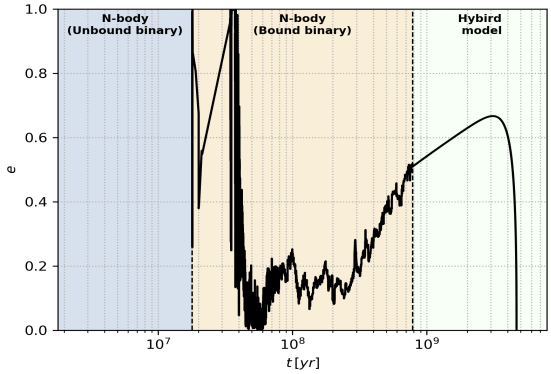


(b) IMBH binary eccentricity time evolution reconstructed with KETJU (N-body regions) and Quinlan-Sesana model (Hybrid model region)..

Figure 3.17: Model 2 evolutionary tracks $(t, a(t))$ (on left) and $(t, e(t))$ (on right).

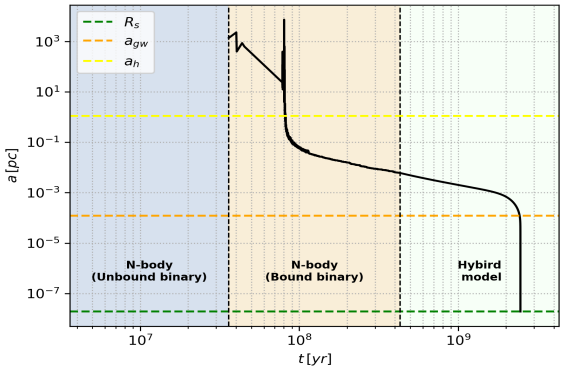


(a) IMBH binary semimajor axis time evolution reconstructed with KETJU (N-body regions) and Quinlan-Sesana model (Hybrid model region)..

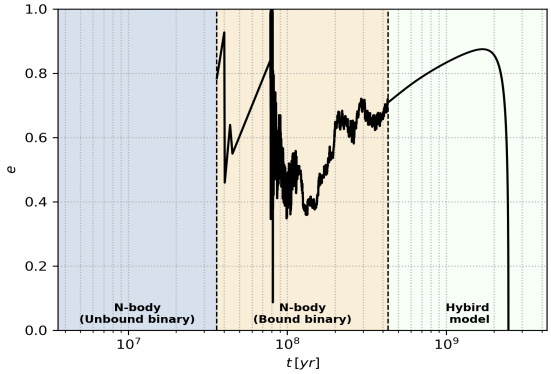


(b) IMBH binary eccentricity time evolution reconstructed with KETJU (N-body regions) and Quinlan-Sesana model (Hybrid model region)..

Figure 3.18: Model 3 evolutionary tracks $(t, a(t))$ (on left) and $(t, e(t))$ (on right).



(a) IMBH binary semimajor axis time evolution reconstructed with KETJU (N-body regions) and Quinlan-Sesana model (Hybrid model region)..



(b) IMBH binary eccentricity time evolution reconstructed with KETJU (N-body regions) and Quinlan-Sesana model (Hybrid model region)..

Figure 3.19: Model 4 evolutionary tracks $(t, a(t))$ (on left) and $(t, e(t))$ (on right).

The resulting coalescence timescales are $\tau_{\rm coal} = ()$ for Models 1–4, respectively. These results suggest that, for the adopted initial conditions, stellar clumps similar to those observed in the Misty Moons may provide favourable environments for IMBH binary coalescence. Assuming that these systems formed at the same redshift as the Misty Moons ($z = 11$), the inferred timescales indicate that the binaries could coalesce by the present day.

Discussion and Conclusions

This thesis investigated the dynamical evolution of intermediate-mass black hole (IMBH) binaries embedded in various stellar environments. The main objectives were to reconstruct their evolutionary tracks, namely the time evolution of IMBH binaries separation and eccentricity, and to estimate their coalescence times across a representative range of the parameter space. To this end, two complementary approaches were employed: the self-consistent hybrid model developed by Quinlan (1996) and Sesana (2010), presented in Chapter 2, and direct N-body simulations performed with KETJU, presented in Chapter 3.
The computational efficiency of the hybrid approach allowed an extensive exploration of the parameter space for IMBH binaries embedded in fixed, isotropic stellar backgrounds. In § 2.2, the evolution of the same IMBH binary was studied within stellar cusps characterized by different density slopes, while keeping the initial conditions fixed. The results confirmed that a higher central stellar density enhances the efficiency of stellar-scattering-driven binary hardening, thereby shortening the timescale to coalescence (Sesana 2010).
Motivated by these results, § 2.3 focused on an IMBH binary evolving within a present-day, nucleated dwarf galaxy, built taking as a reference model NGC 205. Concurrently, we assessed the extent to which the initial orbital eccentricity influences the subsequent hardening and coalescence of the binary.
The main results can be summarized as follows:

1. Binaries entering the stellar-scattering hardening phase with higher eccentricities tend to further increase their eccentricity, thereby reaching the GW-dominated regime at larger separations and coalescing on shorter timescales (Sesana 2010).

2. The exceptionally high central stellar density of an NSC can sustain efficient binary hardening and lead to coalescence timescales of only a few tens of Myr (F. M. Khan and Holley-Bockelmann 2021). This result indicates that GW signals from IMBH binaries hosted by NSCs could, in principle, be observable in the present Universe even if the formation of a bound binary occurred relatively recently.

In § 2.4, the same experiment was performed using the bulge of NGC 205 as a reference model for a present-day non-nucleated dwarf galaxy. The main outcome is that the low stellar density leads to coalescence timescales exceeding the Hubble time, thereby hindering the detection of the binary inspiral GW signal within the present age of the Universe.
The final part of Chapter 2, § 2.5, extended the analysis to non-nucleated, high-redshift galaxies through a series of numerical experiments. Since galaxies at fixed stellar mass are expected to become progressively more compact towards higher redshifts, a sequence of high-z galaxy models was constructed by applying the empirical cosmic size evolution formula, provided by Ono et al. (2023), to the stellar bulge of NGC 205, adopted as the present-day ($z = 0$) endpoint of the sequence of non-nucleated galaxies all sharing the same mass. For each galaxy model, corresponding to redshifts of IMBH binary pairing, and ranging from $z = 0$ to $z = 12$ in steps of $\Delta z = 0.5$, and for a range of initial binary eccentricities, $e_0 = (0.1, 0.3, 0.5, 0.7, 0.9)$,

the corresponding evolutionary tracks were reconstructed. This allowed us to address which combinations of galaxy redshift, and hence look-back time, and initial eccentricity enable the same IMBH binary to coalesce within the present age of the Universe.
The main conclusions are:

1. Typical low-redshift, non-nucleated dwarf galaxies do not provide sufficiently dense stellar environments to sustain efficient loss-cone replenishment. As a result, IMBH binaries can undergo prolonged stalling, with coalescence timescales exceeding the Hubble time.

2. High-redshift, non-nucleated galaxies, owing to their enhanced compactness, may host IMBH binaries capable of producing GW signals detectable by future observatories such as LISA and ET, provided that the binaries form sufficiently early in cosmic history.

In light of these results, the same astrophysical problem is investigated in Chapter 3 through a suite of direct, high-resolution N-body simulations performed with KETJU (Mannerkoski, Rawlings, et al. 2023). The algorithmically regularized integrator `mstar` enables the accurate treatment of close BH-star interactions without using any gravitational softening. However, its limited parallelization makes these simulations computationally demanding, preventing the direct N-body calculation from covering the entire IMBH binary evolution. The final stages of the evolution were therefore completed using the hybrid model presented in chapter 2.
First, in § 3.2, a head-on, parabolic merger between two identical N-body realizations of NGC 205 NSC, each hosting an IMBH, was simulated. The main results can be summarized as follows:

1. The merger breaks the spherical symmetry assumed by the hybrid model, enhancing loss-cone replenishment, resulting in a shortening of the stellar-scattering-driven phase.

2. The resulting coalescence timescale is comparable to that obtained in the numerical experiments presented in § 2.3, amounting to a few tens of Myr and supporting the consistency between the two approaches.

Lastly, § 3.3 presented a suite of KETJU simulations addressing whether massive stellar clumps in high-redshift galaxies can host GW sources if they harbour central IMBHs. To this end, were simulated parabolic encounters between observationally motivated stellar systems, each containing an IMBH, with different initial angular momenta (Nakane et al. 2026). The results obtained show that:

1. Increasing the initial angular momentum leads to a lower eccentricity of the IMBH binary at the moment of pairing.

2. Increasing the initial angular momentum gives rise to longer coalescence timescales

3. At sufficiently high redshift, these stellar environments can provide a viable evolutionary pathway towards IMBH coalescence within $z = 0$, making them potential sites for the formation of gravitational-wave sources detectable by LISA or ET.

Future perspectives. The results presented in this thesis open several directions for future work. These can be broadly divided into two complementary lines. The first concerns extensions of the present study, aimed at relaxing some of its assumptions and exploring additional physical processes and initial configurations. The second involves related research topics that build upon the results obtained here, extending the analysis towards other aspects of the formation, evolution, and observational signatures of IMBH binaries. The main prospects along these two lines are outlined below.

Exploring the parameter space. The parameter space relevant to this astrophysical problem remains far from being comprehensively explored. Further numerical experiments, building upon and complementing those presented in this thesis, could therefore investigate a broader range of initial configurations and physical processes not considered in the present work. The most relevant directions include:

- Exploring the evolution of IMBH binaries across a broader range of IMBH masses and mass ratios.
- Extending the N-body simulations to minor merger events and assessing their impact on IMBH binary evolution.
- Including gas dynamics to model wet mergers and quantify the role of ISM drag, as well as the effects of IMBH accretion disks, on the binary evolutionary tracks.
- Accounting for possible galaxy rotation and studying different merger configurations

Reconstruction of the GW signal. A natural extension of the present work would be to connect the dynamical evolution of the IMBH binaries to their gravitational-wave signatures. The binary evolution obtained from the simulations could be used to reconstruct the subsequent GW-driven inspiral and characterize the resulting signal, following the approaches adopted in Mannerkoski, Johansson, et al. 2019 and Keitaanranta et al. 2026, outlined as:

- Reconstructing the time evolution of the binary semimajor axis $a(t)$ and eccentricity $e(t)$.
- Determining the coalescence time of the binary from the onset of the pairing phase.
- Deriving the spectral energy distribution (SED) of the emitted gravitational radiation from the predicted binary orbital evolution.
- Generating the corresponding gravitational-wave waveform and assessing its detectability with current and future observatories.

NSC and IMBH formation via collision runaway. Another complementary direction would be to investigate the earlier evolutionary stages leading to the formation of the systems considered in this thesis. In particular, the interplay between the assembly of nuclear star clusters through the dynamical-friction-driven infall of globular clusters at high redshift (I. Khan et al. 2026) and the formation of a central IMBH through stellar collision runaway could be explored within a unified numerical framework. This would make it possible to estimate the timescales required for both the NSC and the IMBH to form, and to assess whether the resulting systems can emerge early enough to subsequently undergo the merger events analysed in this work.

From galaxy mergers to the IMBH merger rate. The growing capabilities of current and forthcoming observational facilities may provide increasingly accurate constraints on the galaxy merger rate across cosmic time, including at high redshift (see, e.g., Duncan et al. 2019; Pacucci et al. 2023). Combining these observational constraints with the dynamical evolution of IMBH binaries studied in this thesis would allow the expected IMBH merger rate to be estimated as a function of redshift. This, in turn, would provide a population-level prediction for the gravitational-wave signal produced by these systems and its evolution across cosmic time, offering a direct connection between the observed galaxy merger population and the expected IMBH contribution to the GW sky (Marta Volonteri et al. 2020).

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